

ARM-450 rev I -- final design

ARM-450 rev I -- release pass (all stages 0)

J1	0
J2	0
J3	0
J4	0
WRIST	0
HARDWARE (bearings + bolts as solids)	0
LINK LOCK (J2/J3 set screws, centring rings)	0
FASTENERS (every screw, insert, nut)	0
PARAMETER AUDIT	0
TOOL ACCESS (hex key to every screw at its build step)	0
RULE 4 (mate, never overlap; pathways; attached)	0
SPRINGS	0
SWEEP	0
6-DOF (axes two ways, Jacobian rank)	0
OFFICIAL ST3215 (manufacturer's model, rear idler horn)	0
DISASSEMBLY PATHS (our servo model)	0
DISASSEMBLY PATHS (official servo)	0
AUDIT GAPS (joints together, tipping, walls, bed)	0
GATES (each part in its chosen PRINT orientation)	0
PAIRS (static all-pairs)	0
WIRING (both bus plugs in every servo, way out)	0
HEAD CLEARANCE (screw heads vs turning parts, contact angle)	0
CABLE ROUTE (one joint per cable, path change, loops)	0
PLUG INSERTION (bus plugs in and out with the arm assembled)	0
UNION JOINTS (every glued feature joined over its full section)	0
SECTION MEASURE (manual check by slicing)	0
PRINT SLICER on the print-ready files	0

VERIFICATION AUDIT 2026-09-24 (Rules 3, 4, 6, 8) -- ARM450_AUDIT_REPORT.pdf:

- * every servo SCREWED by its own back holes (20 self-tappers) and seated on 4 supports
- * rear idler horn of the real ST3215 cleared at J1/J4/J6; J5 axle assemblies; table holes
- * thin walls fixed; fork-floor bolt heads sunk; wire windows fitted to the plugs
- * new checks: manufacturer's servo, disassembly, wiring, head clearance, cable route

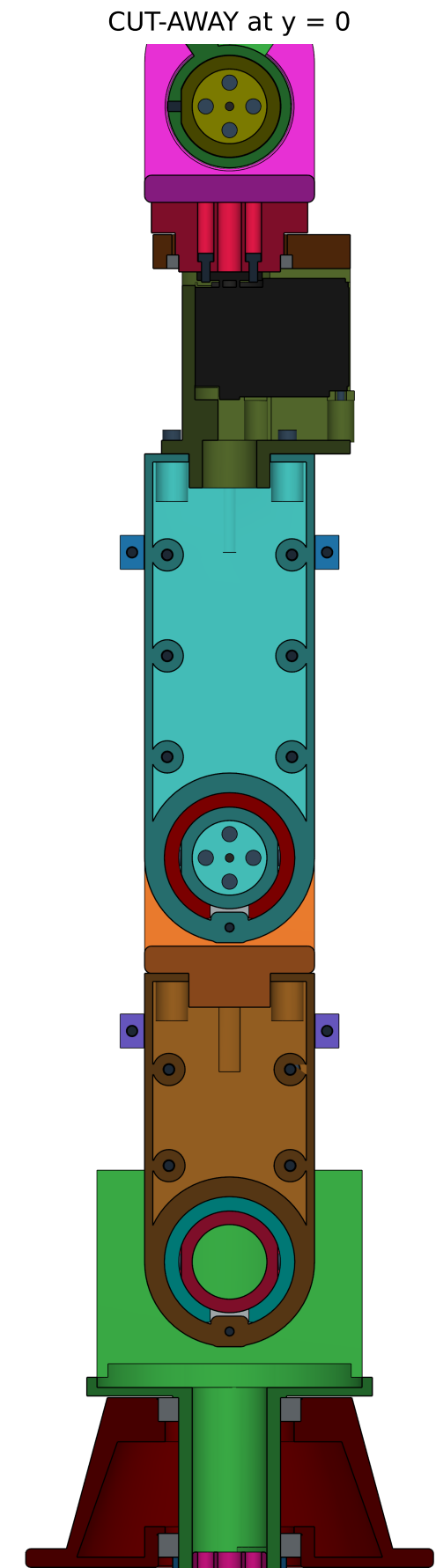
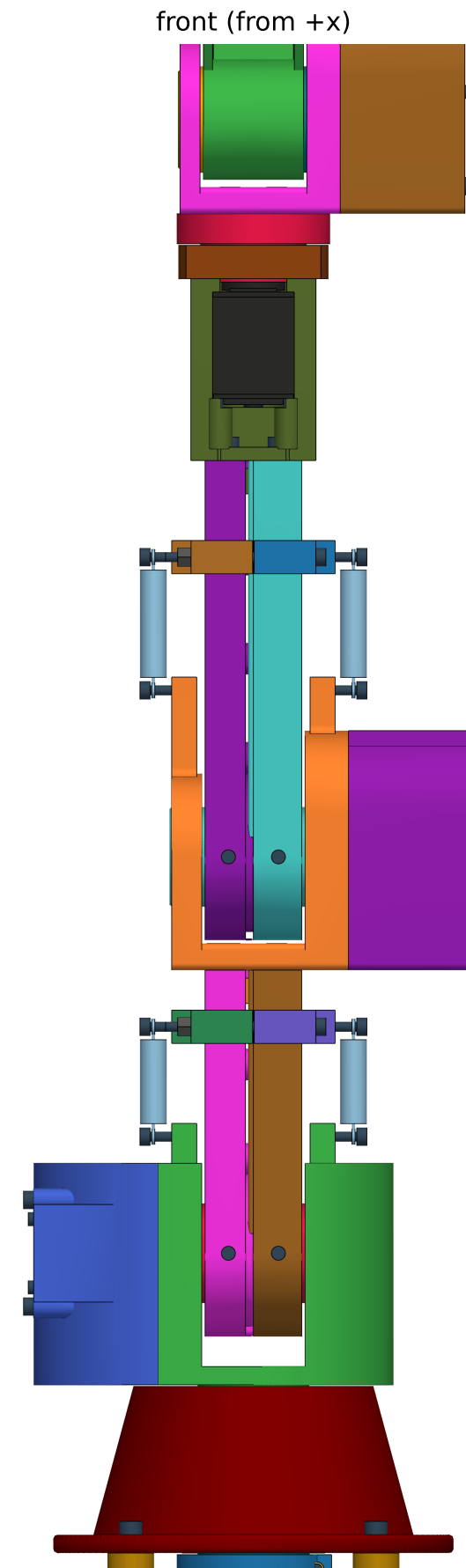
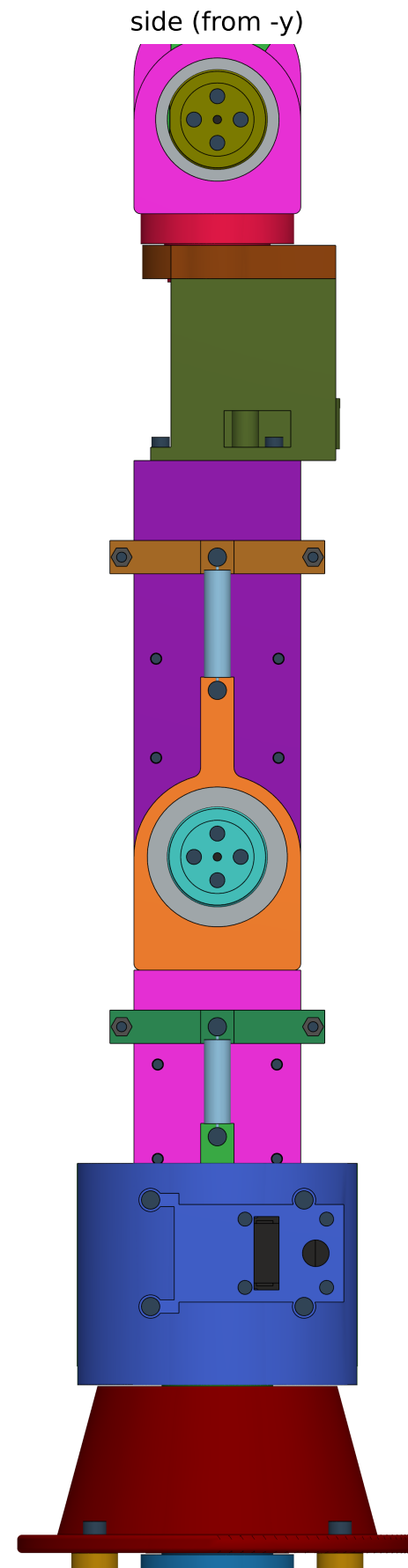
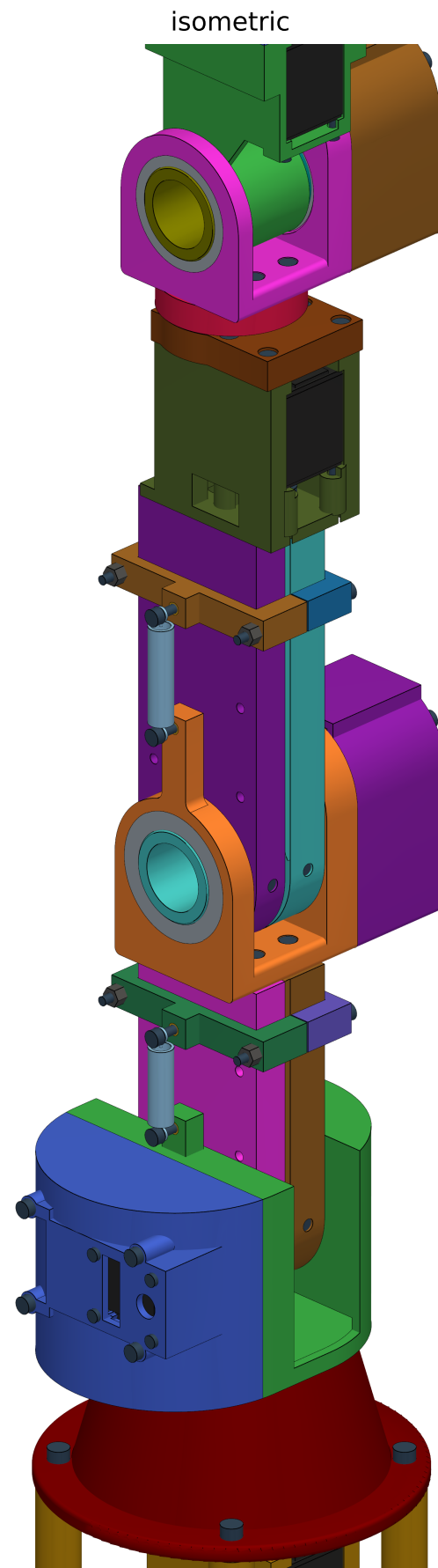
RULE 4 (mate, never overlap; interact only through proper pathways) found and fixed:

- * J1: hub D-plug in the spigot on 0.1/0.2 mm clearance (+-1.9 deg yaw play) -> spigot end slit into a COLLET, closed by the spigot collar (slot 2.0: 1.10 needed)
- * J2/J3: links hung on +-0.2 mm clearances (+-1 deg) -> 2 x M5 set screws per link half, self-tapped in the link's own 04.2 side holes, on double-D shaft flats
- * J1: turret land sat on the base and the 6806 OUTER ring -> land on the inner ring only
- * J4/J6: stationary lips on the 6706's turning INNER ring -> 0.5 mm relief
- * J4/J6 caps overlapped the bearings 0.01 mm; J5 spacers floated 0.05 per face
- * every set screw / grub modelled TIGHTENED; split clamps checked for closing travel

Earlier in this pass (each measured, then re-checked):

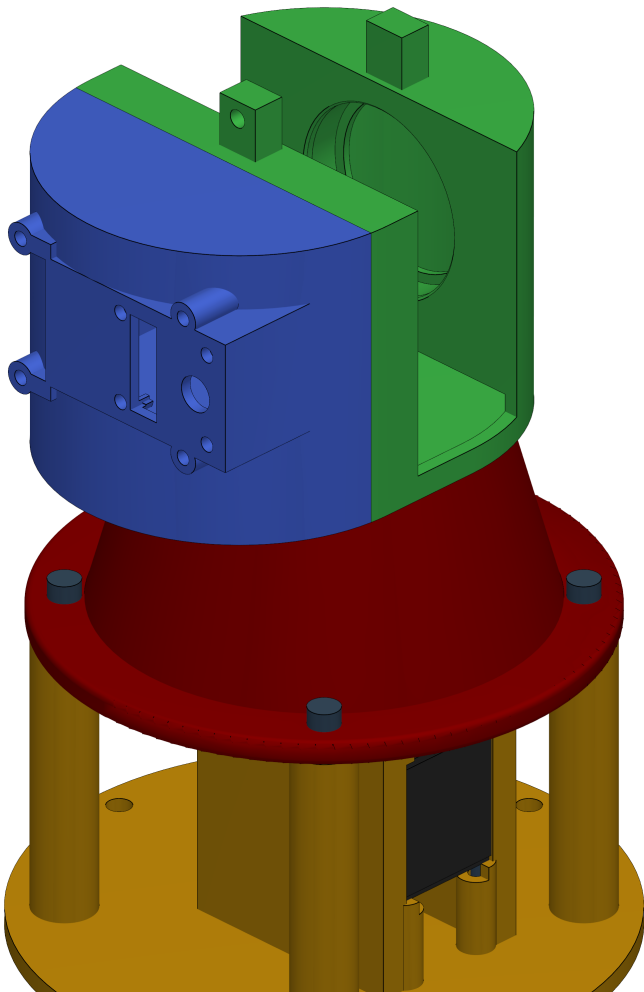
- * shaft clamps: the released 038 ring could not enter the links' bores (seam ear);

ARM-450 rev I -- whole arm, every part and every screw, opaque

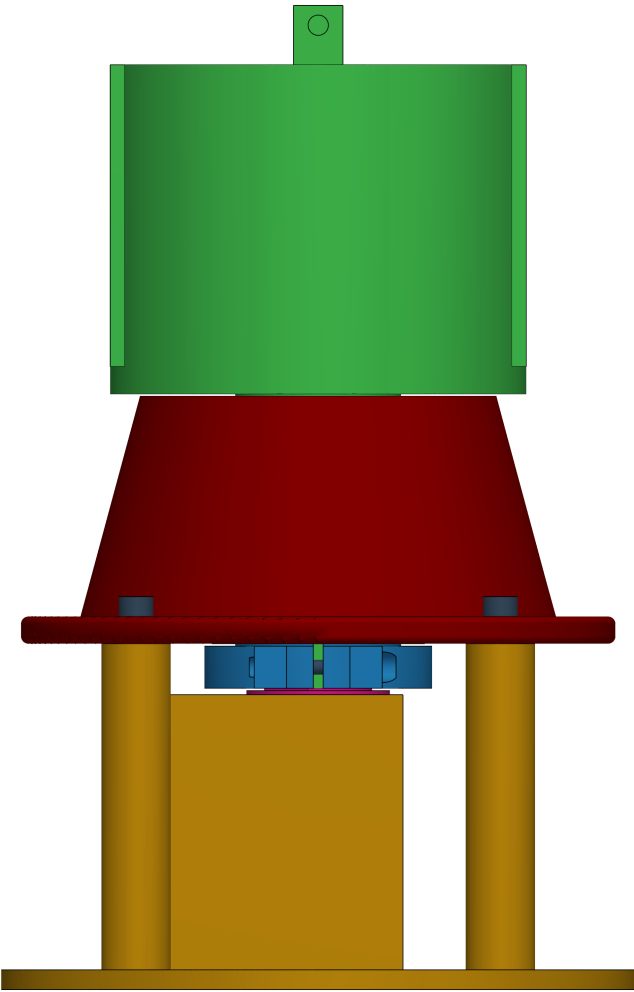


J1 base yaw -- every part opaque in its own colour, black edges (3D, exact solids)

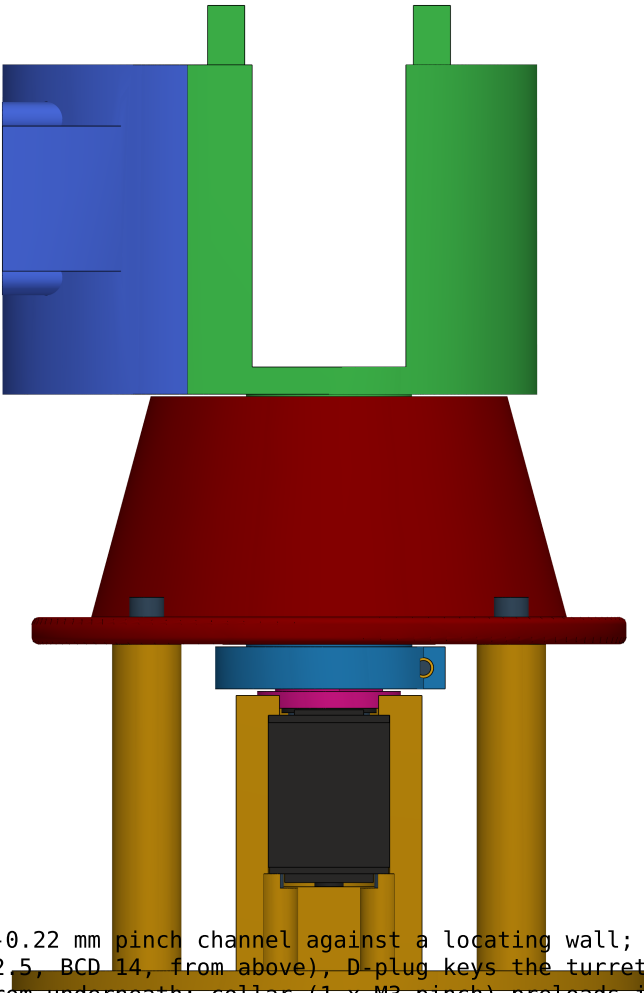
isometric



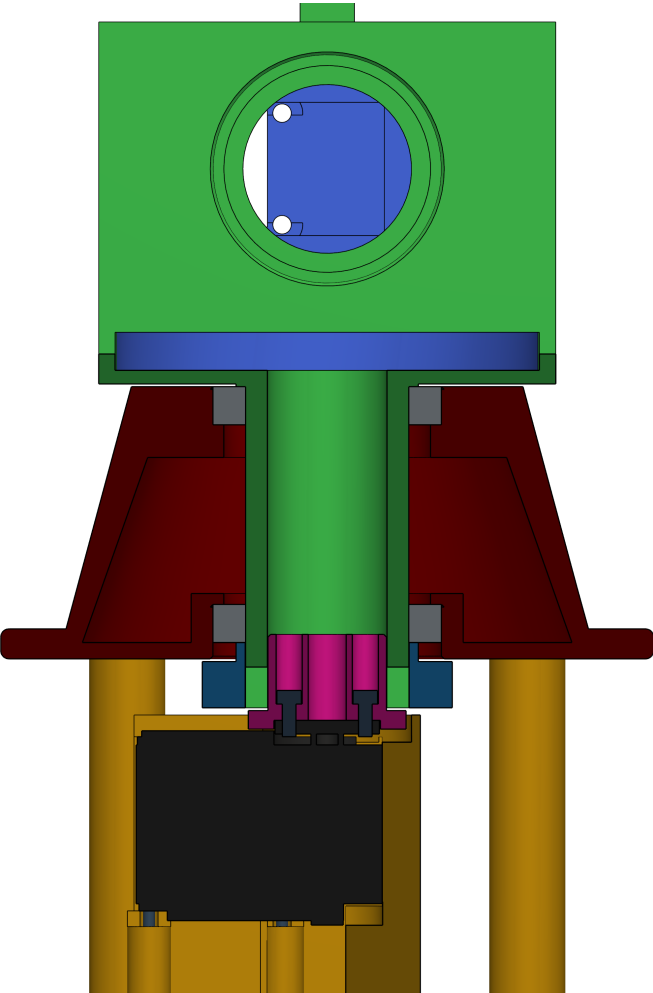
side (along the cut normal)



front

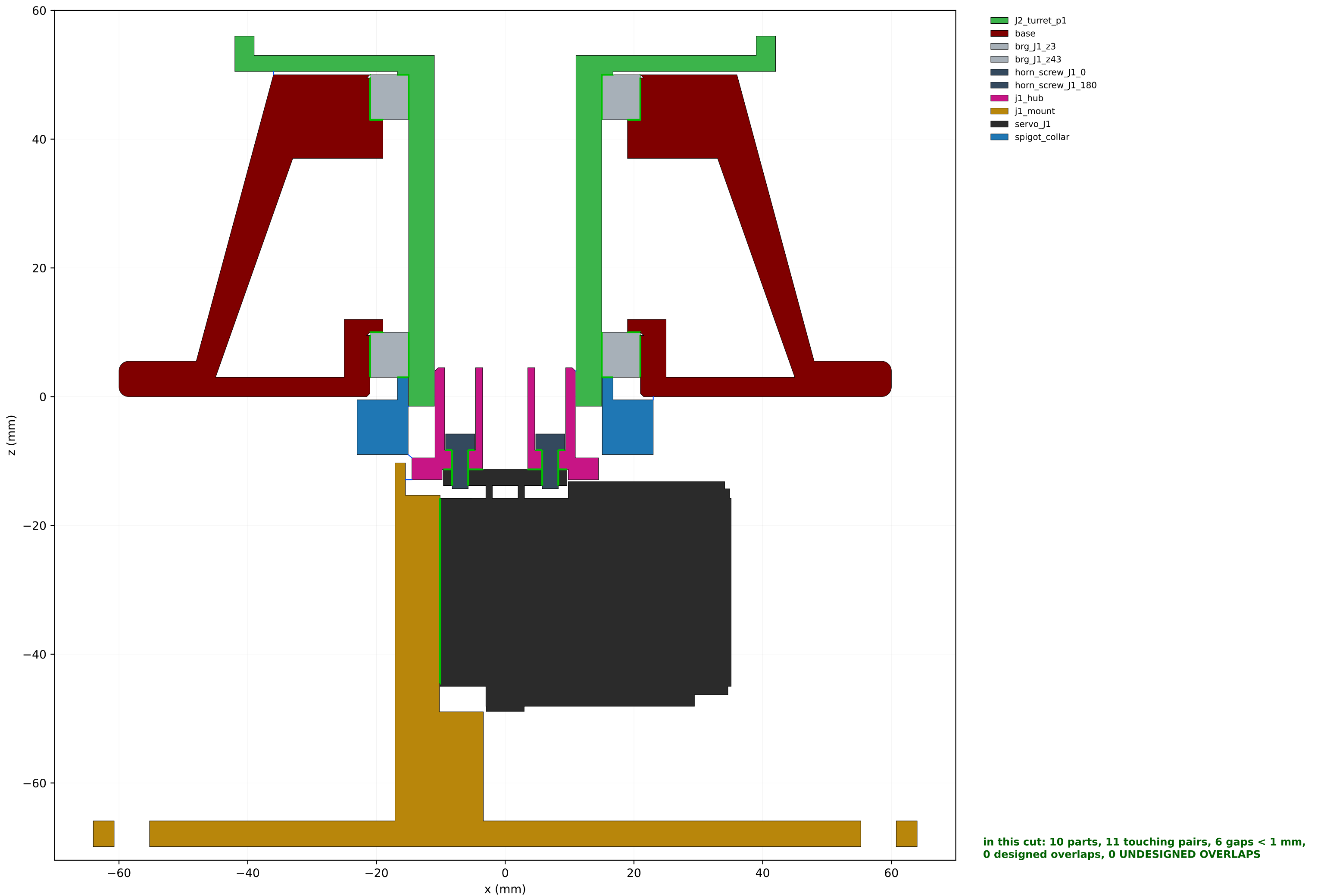


CUT-AWAY through the joint plane

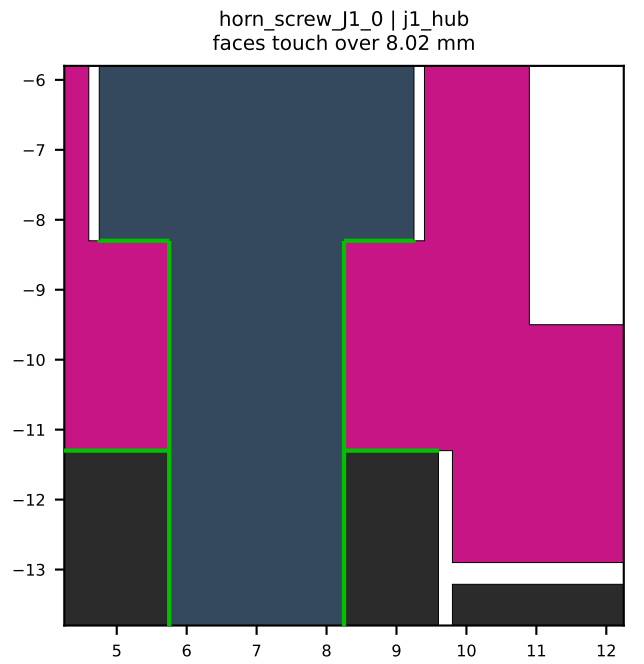
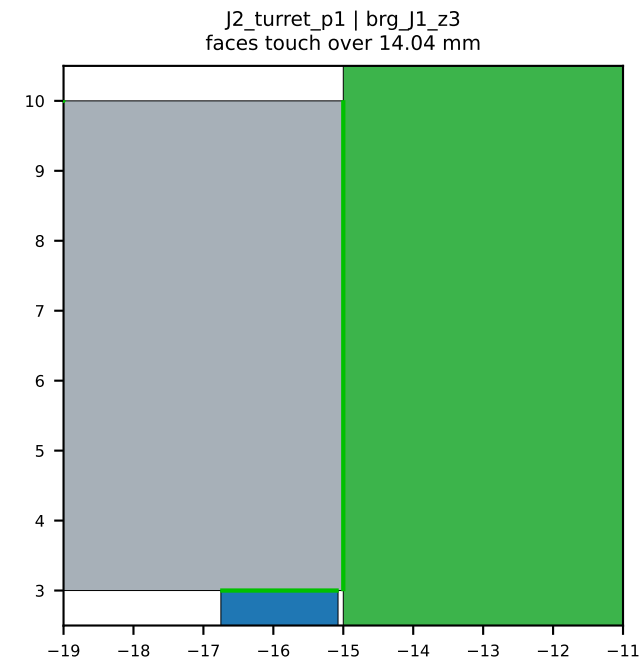
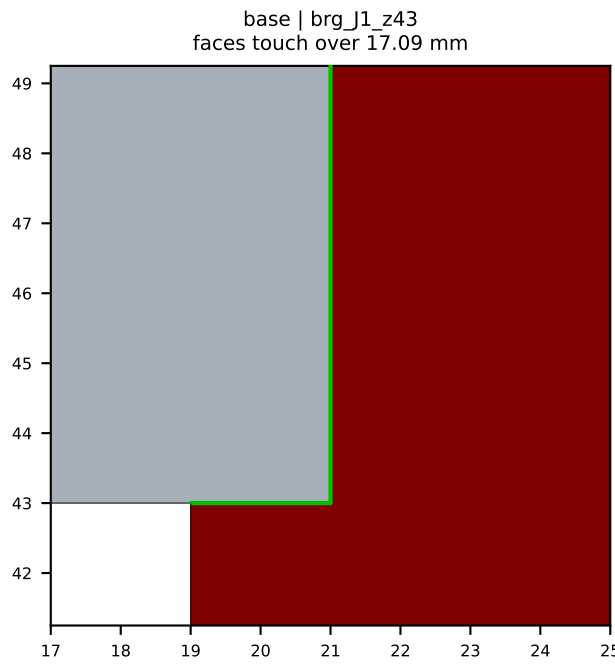
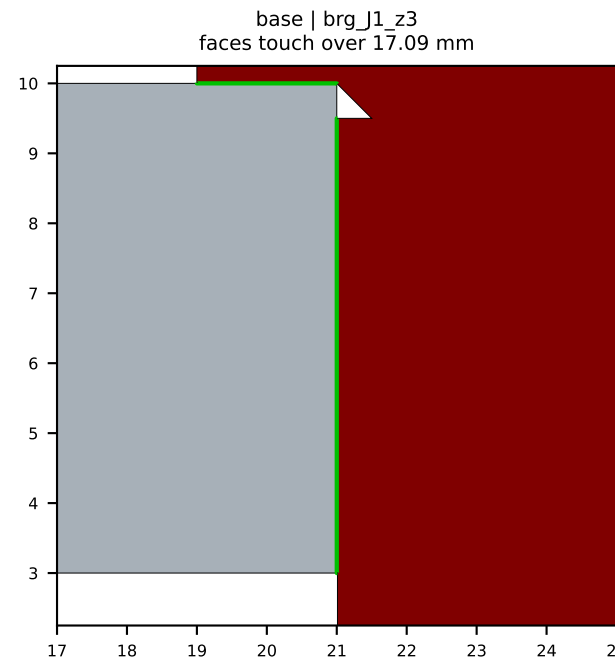
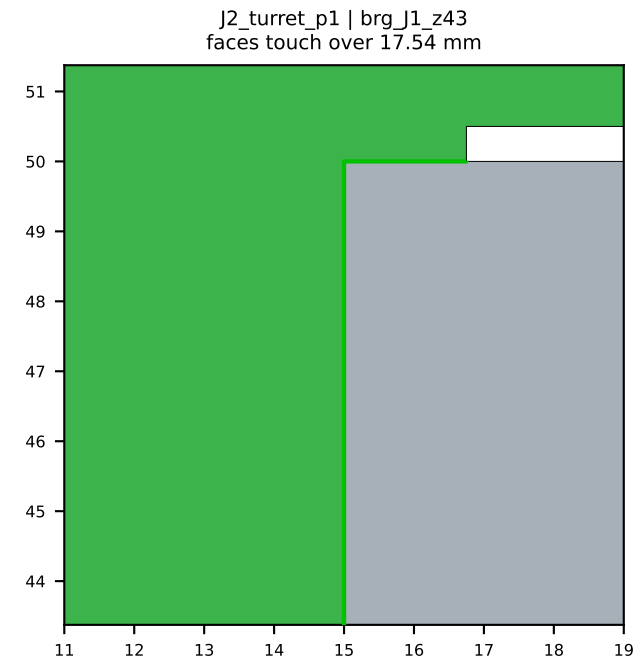
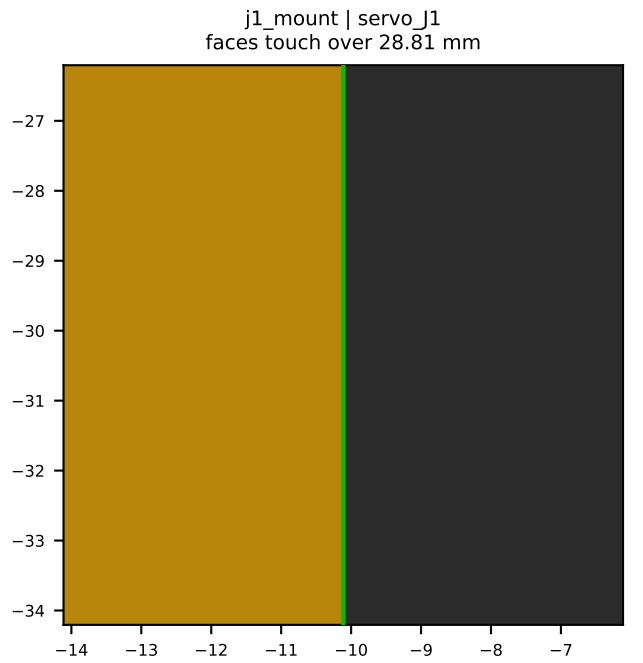
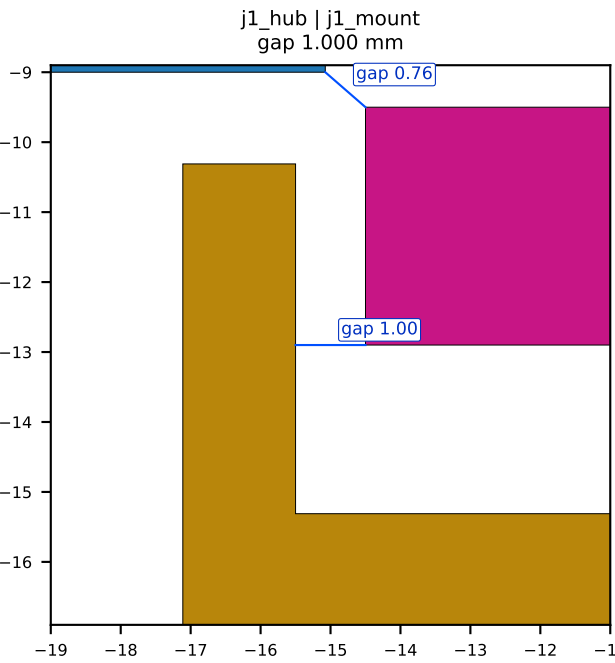
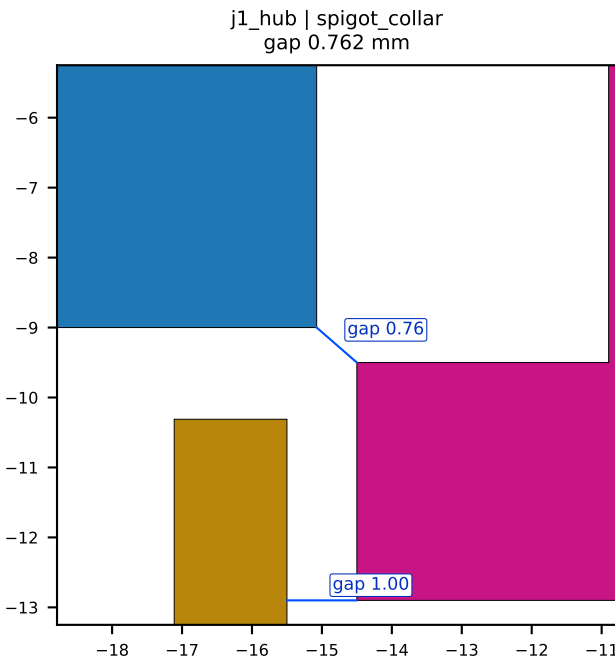
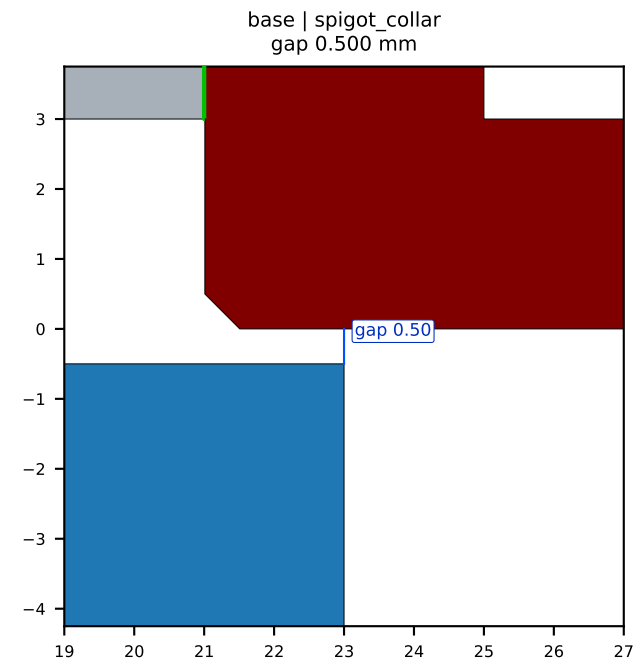
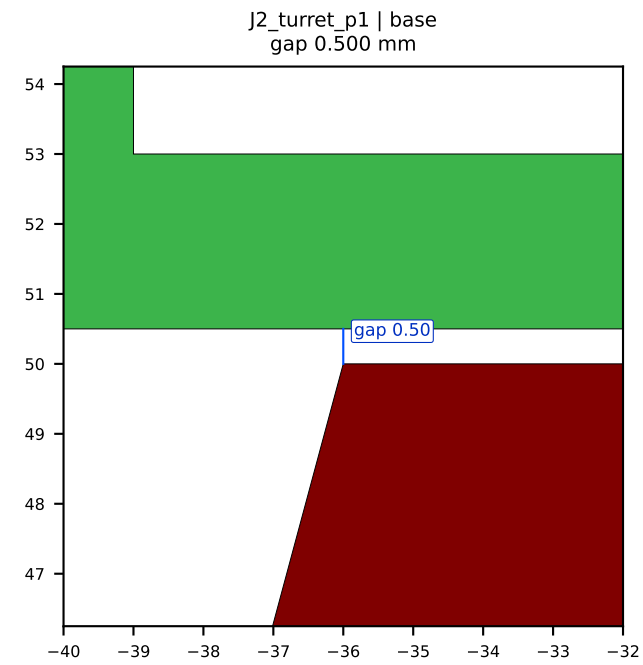
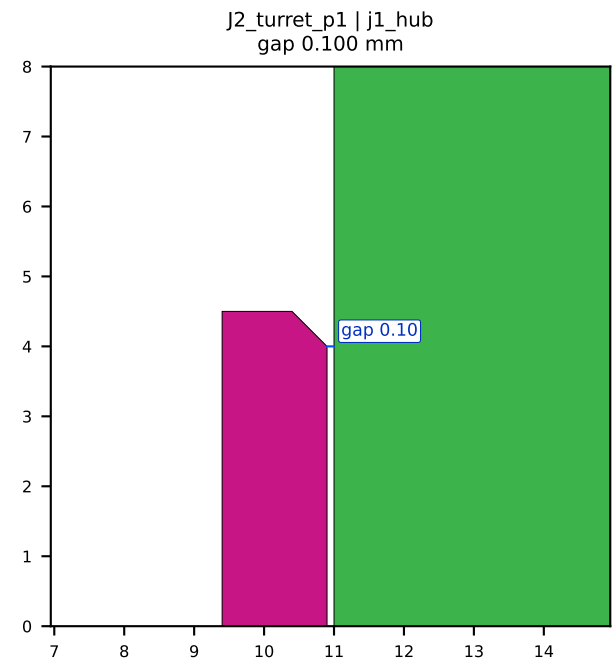
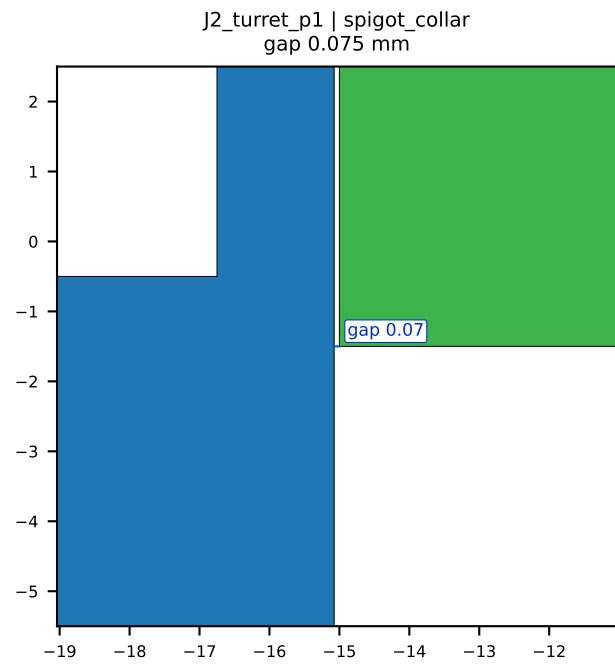


Servo slides into the foot's -0.22 mm pinch channel against a locating wall; cap face under the lips.
Hub bolted to the horn (4 x M2.5, BCD 14, from above), D-plug keys the turret spigot. No grubs.
Base: lower 6806 presses up from underneath; collar (1 x M3 pinch) preloads its INNER race only.

J1 base yaw -- EXACT SECTION (OpenCascade): red = solids intersecting, orange = designed (pinch / thread), green = faces touching, blue = gap < 1 mm

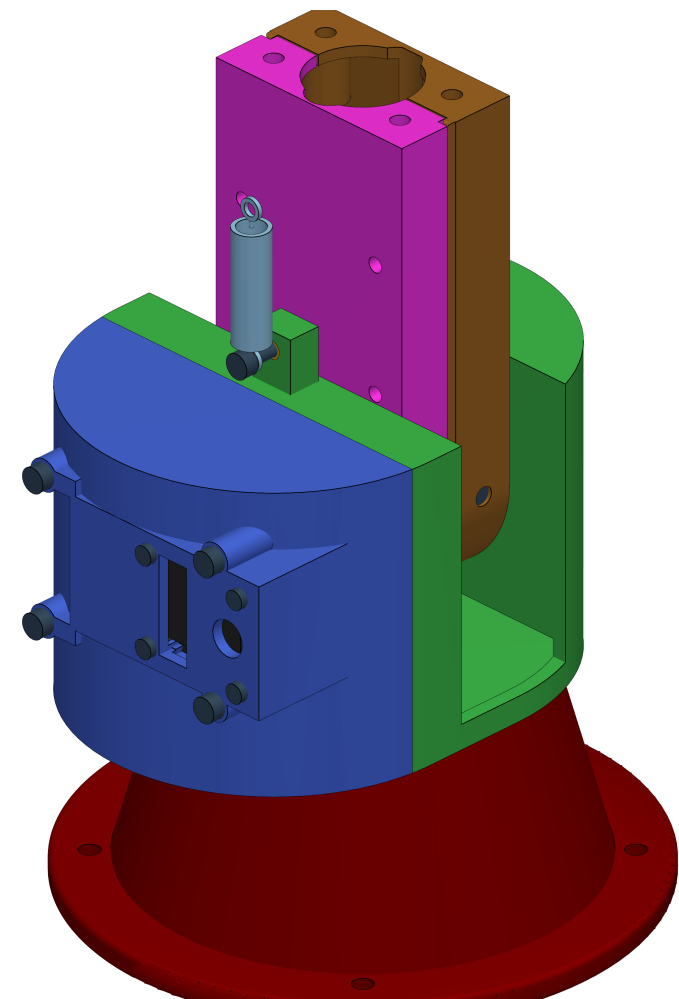


J1 base yaw -- interfaces zoomed (+/-4 mm): measured on the exact section

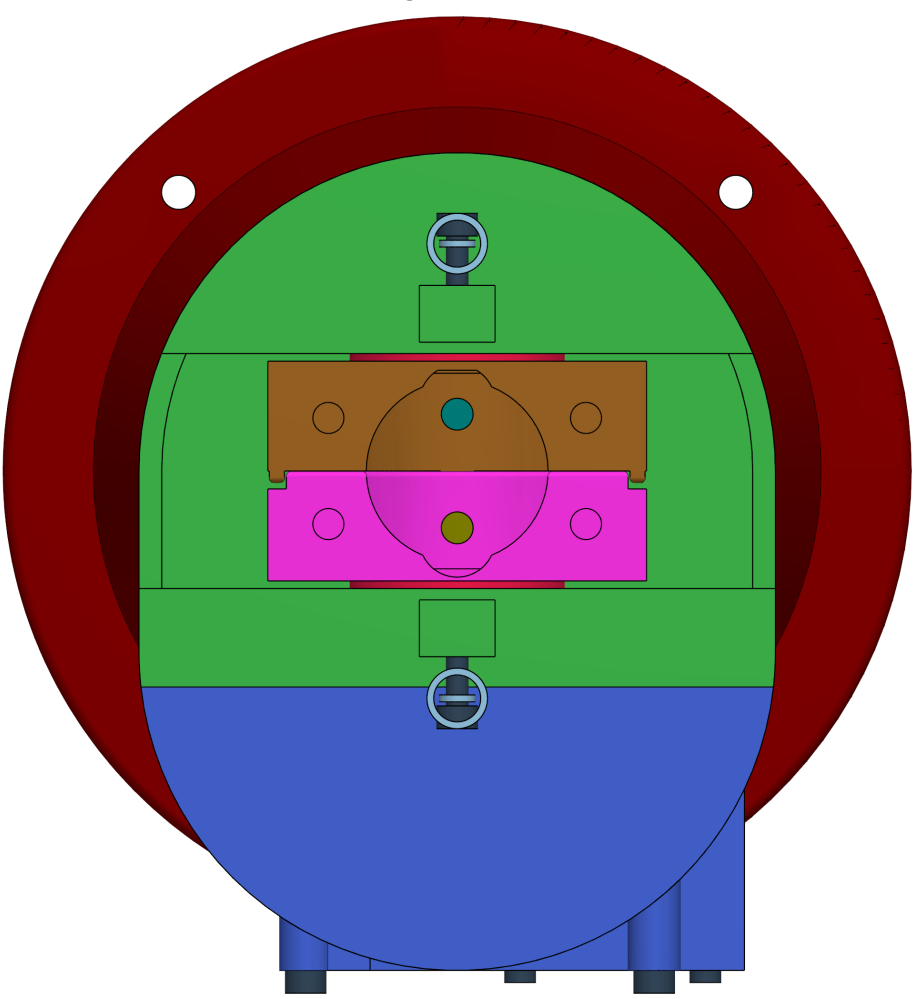


J2 shoulder -- every part opaque in its own colour, black edges (3D, exact solids)

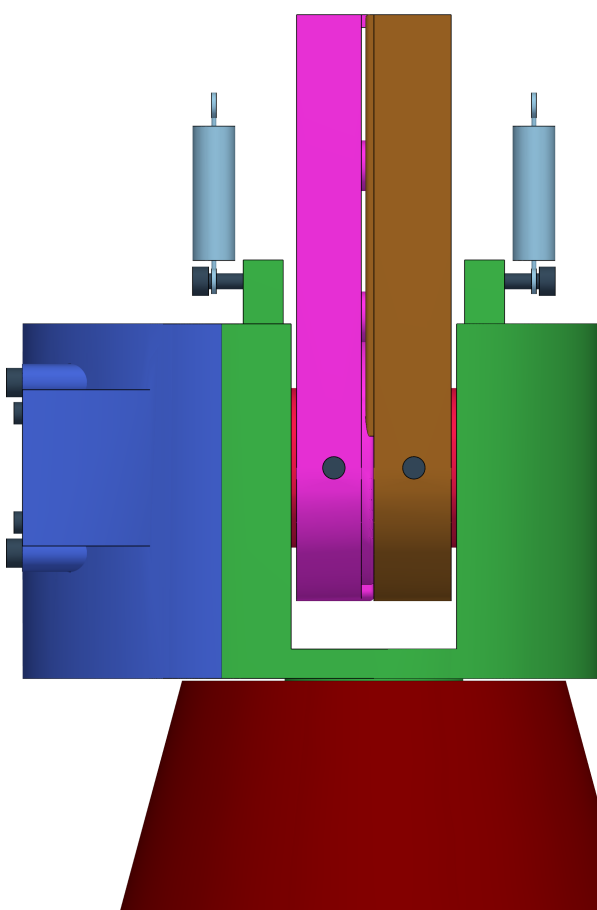
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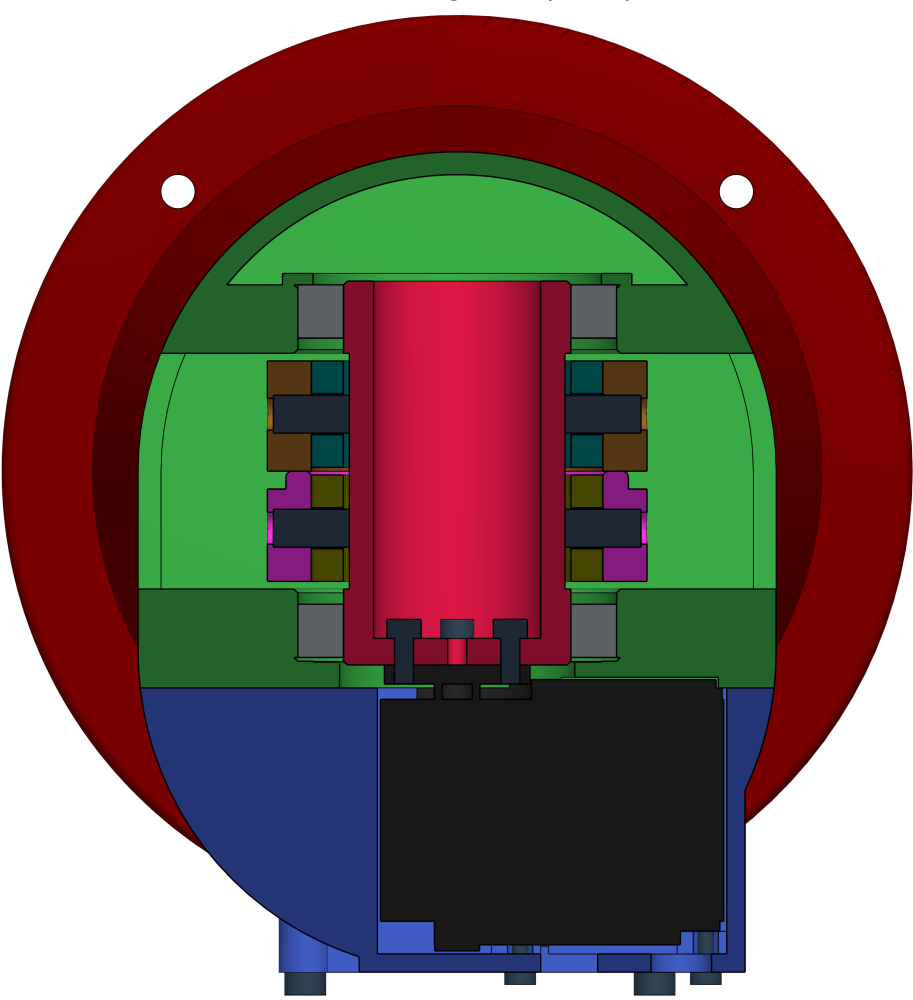
side (along the cut normal)



front

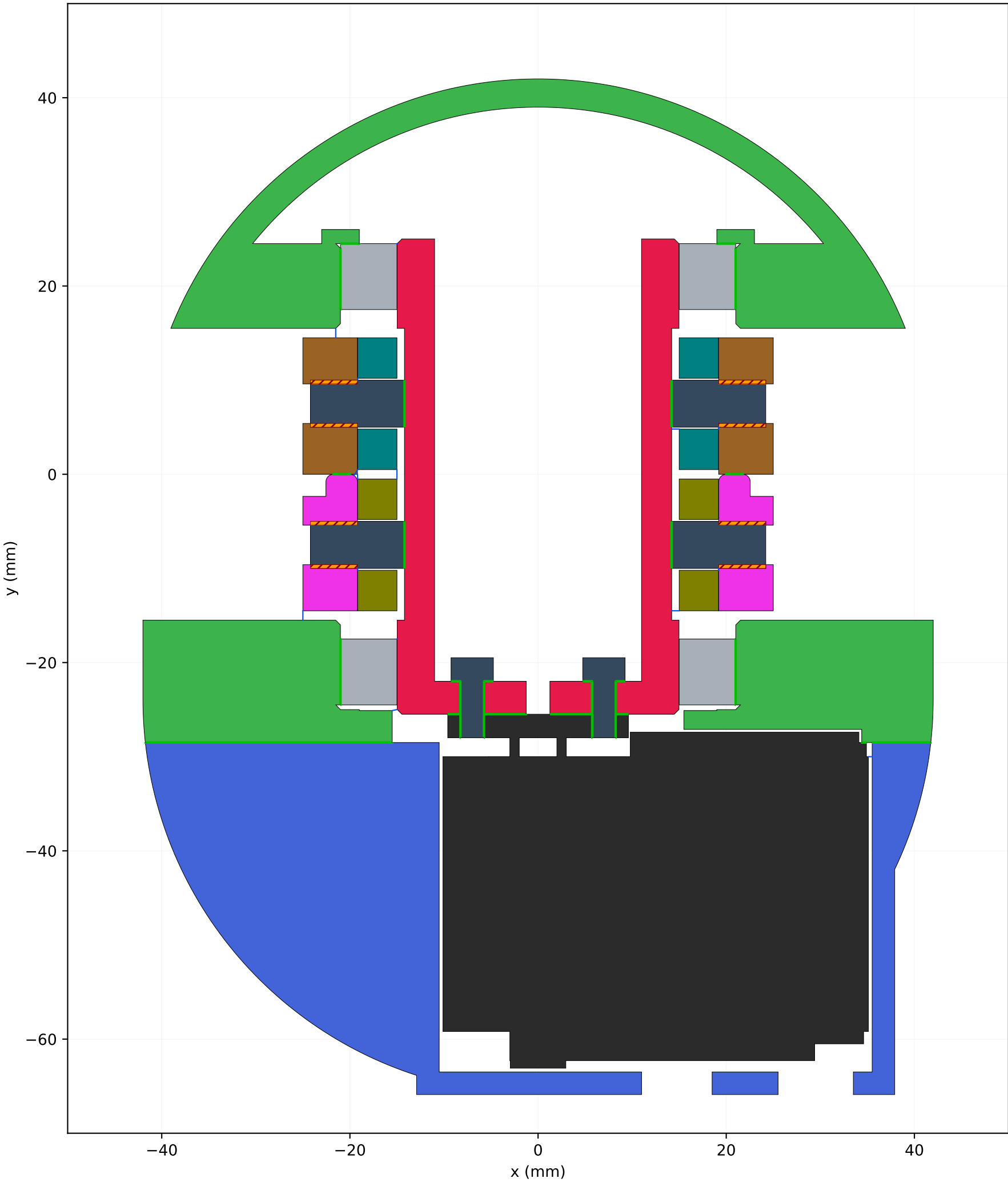


CUT-AWAY through the joint plane



Servo in the turret's servo POD (2.4 mm shell all round), cap face on the parting plane; the pod bolts to the turret with 4 x M3 x 45 into inserts. Printed shaft bolted to the horn (screws through the shaft bore). Both 6806 press in from the fork gap. Each link half: C-clamp keyed by the seam ear, M3 grub onto the shaft's D-flat. Springs: turret lugs -> upper-link collar (7.4 N / 12.4 N, x2).

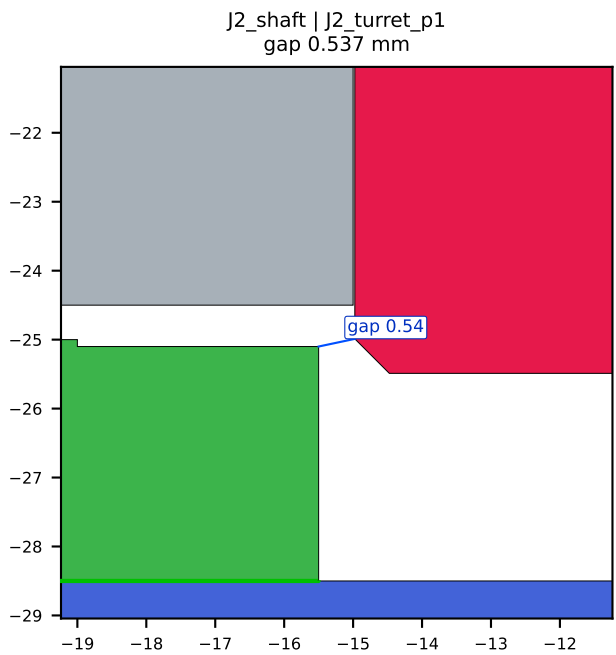
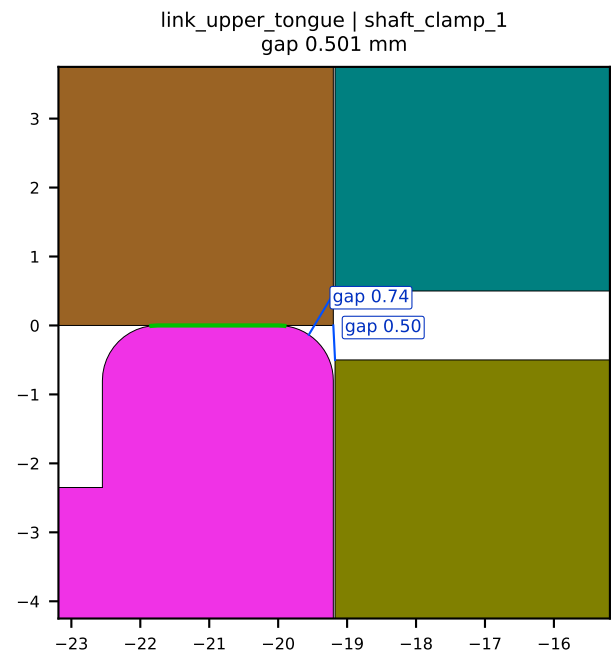
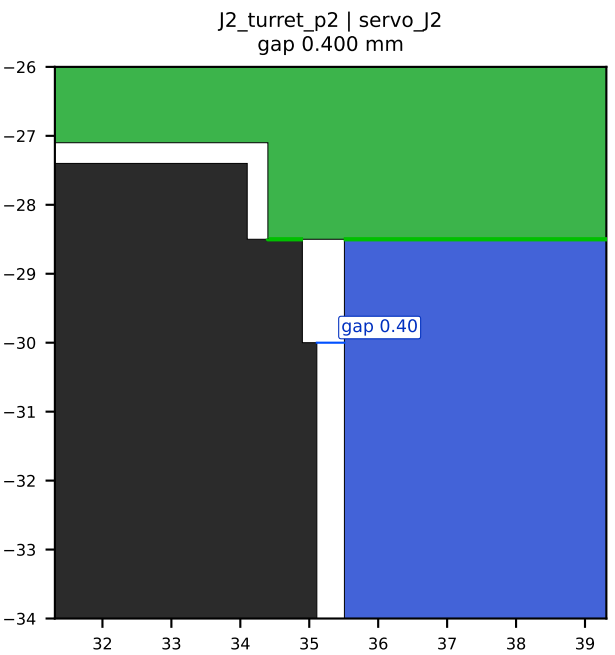
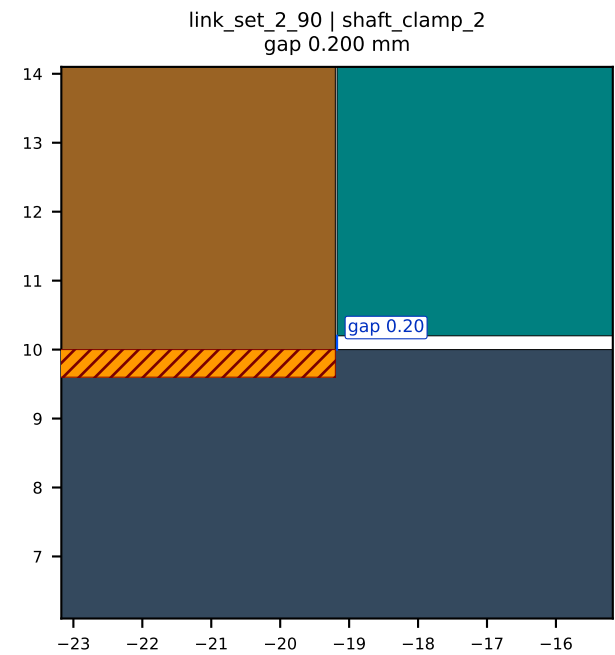
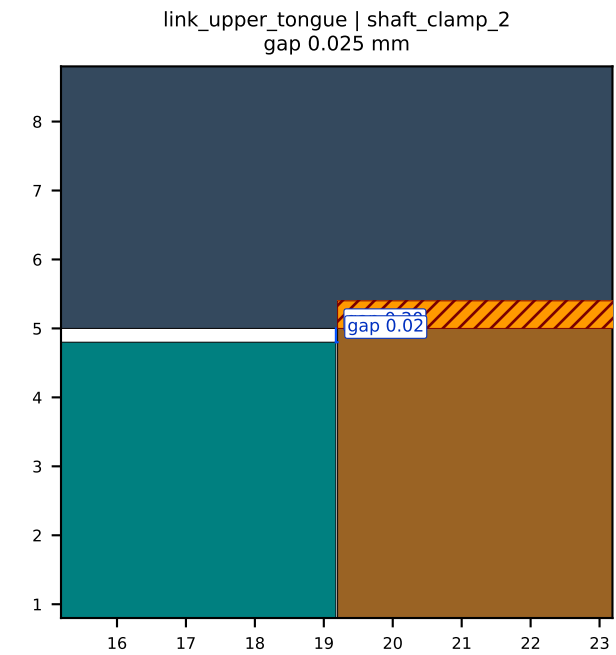
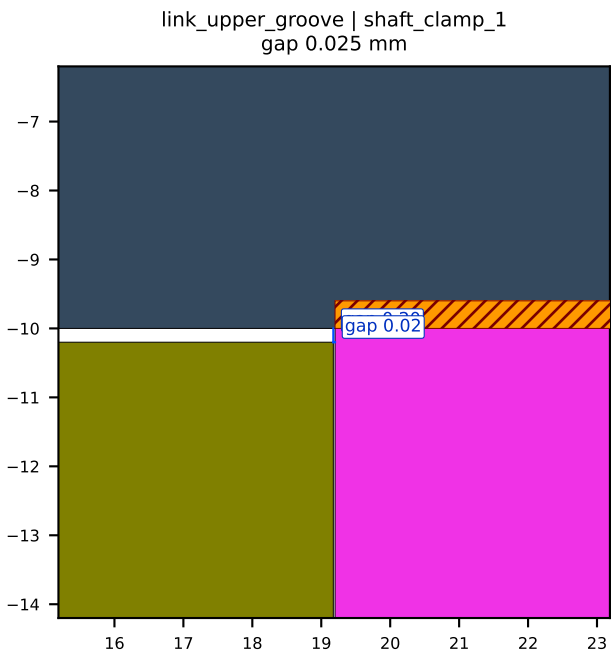
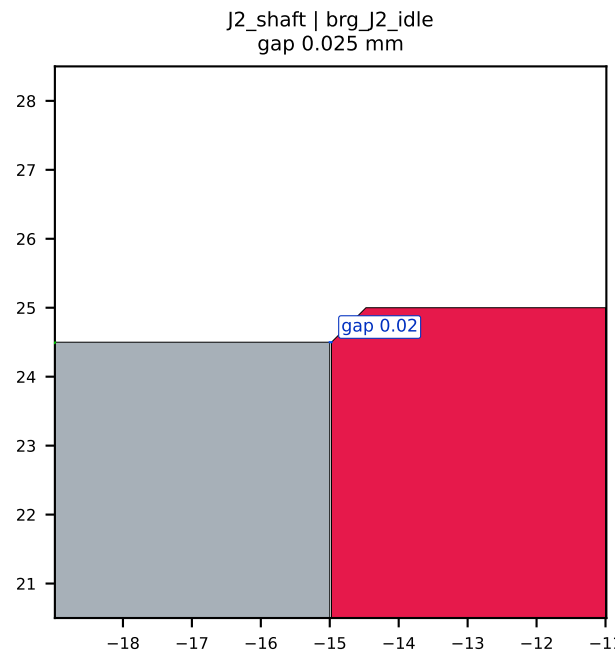
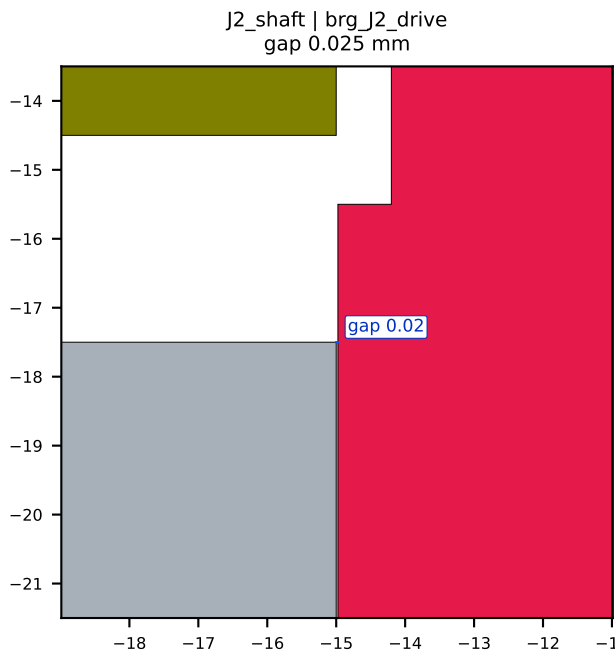
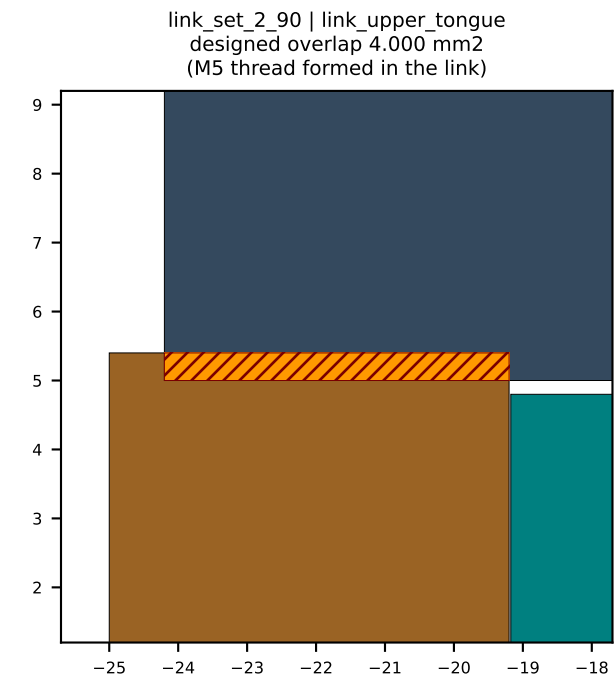
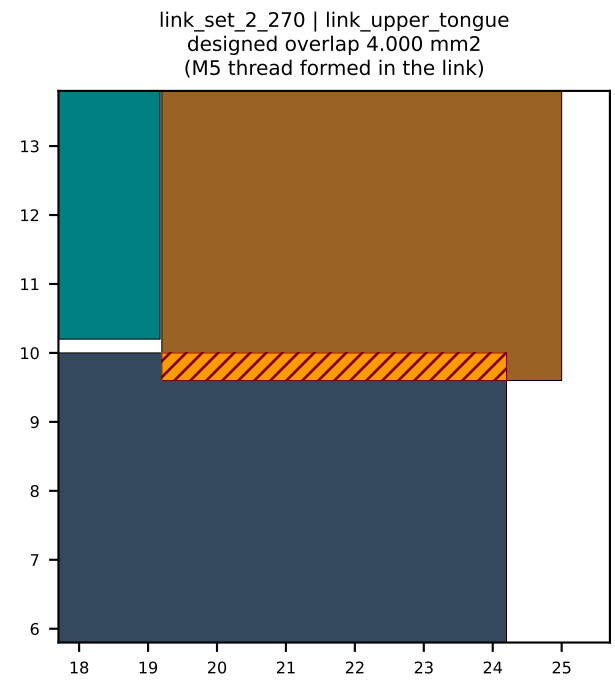
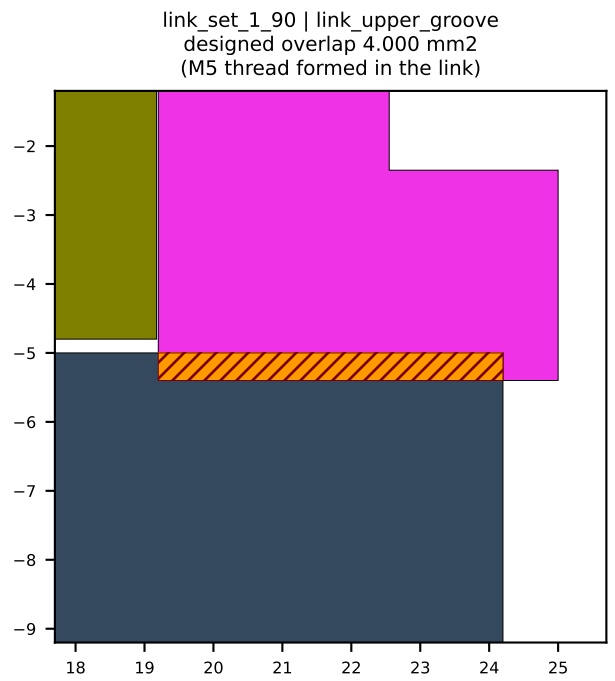
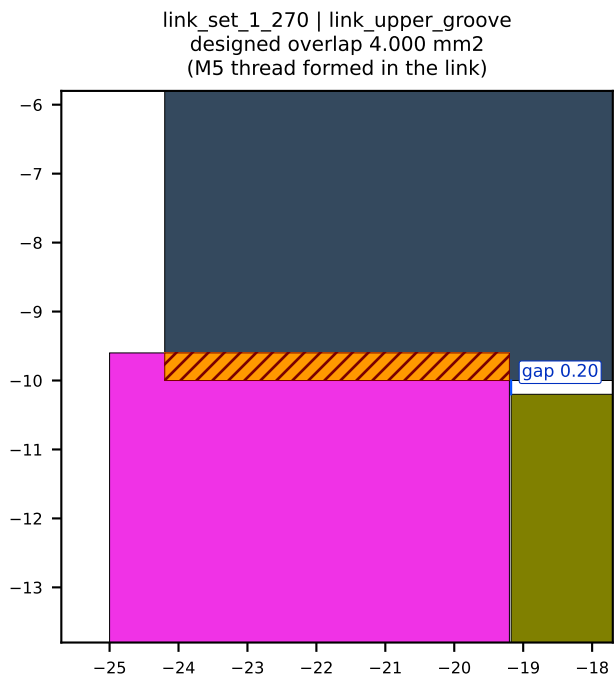
J2 shoulder -- EXACT SECTION (OpenCascade): red = solids intersecting, orange = designed (pinch / thread), green = faces touching, blue = gap < 1 mm



- J2_shaft
- J2_turret_p1
- J2_turret_p2
- brg_J2_drive
- brg_J2_idle
- horn_screw_J2_0
- horn_screw_J2_180
- link_set_1_270
- link_set_1_90
- link_set_2_270
- link_set_2_90
- link_upper_groove
- link_upper_tongue
- servo_J2
- shaft_clamp_1
- shaft_clamp_2

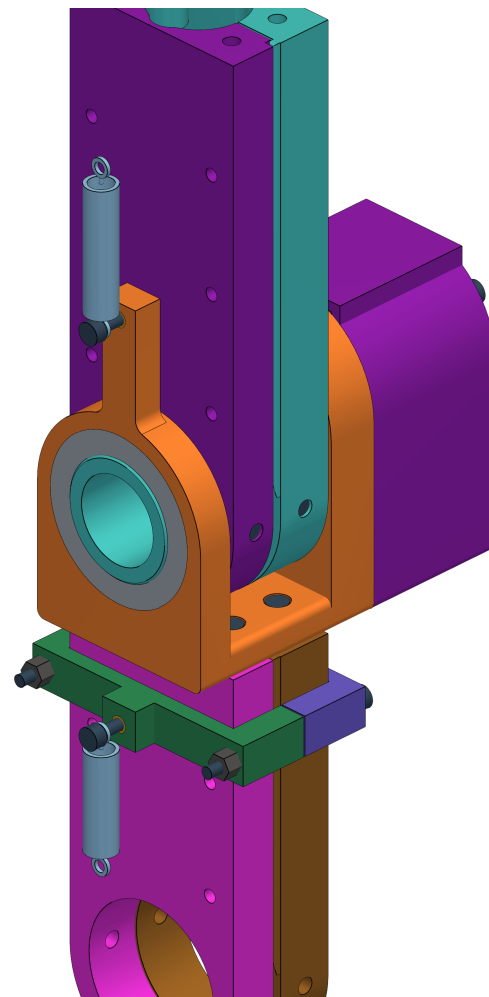
in this cut: 16 parts, 14 touching pairs, 17 gaps < 1 mm, 4 designed overlaps, 0 UNDESIGNED OVERLAPS

J2 shoulder -- interfaces zoomed (+/-4 mm): measured on the exact section

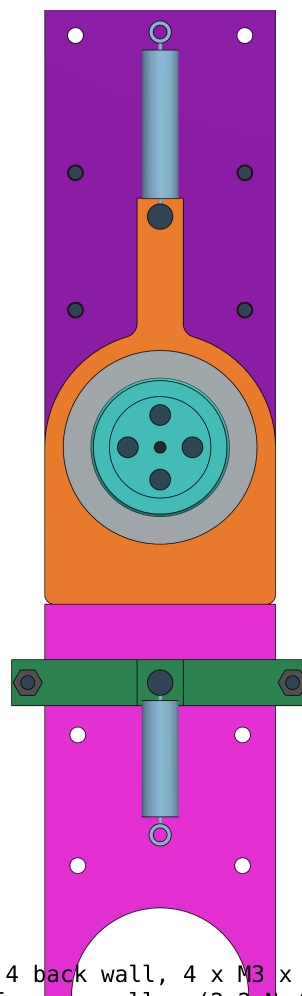


J3 elbow -- every part opaque in its own colour, black edges (3D, exact solids)

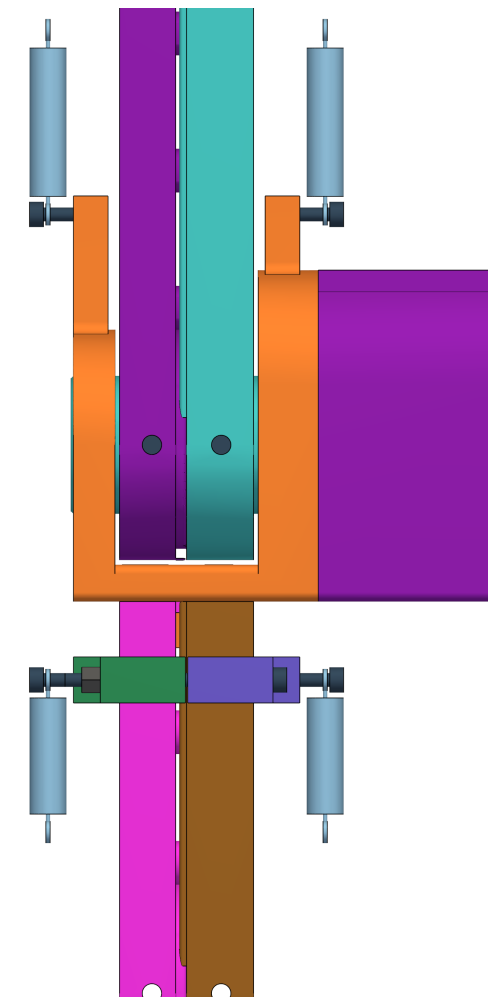
isometric



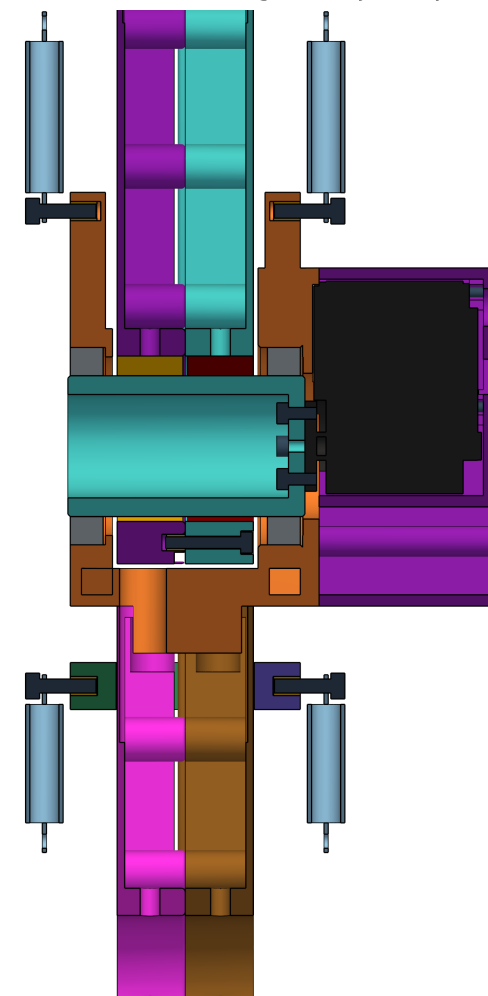
front



side (along the cut normal)

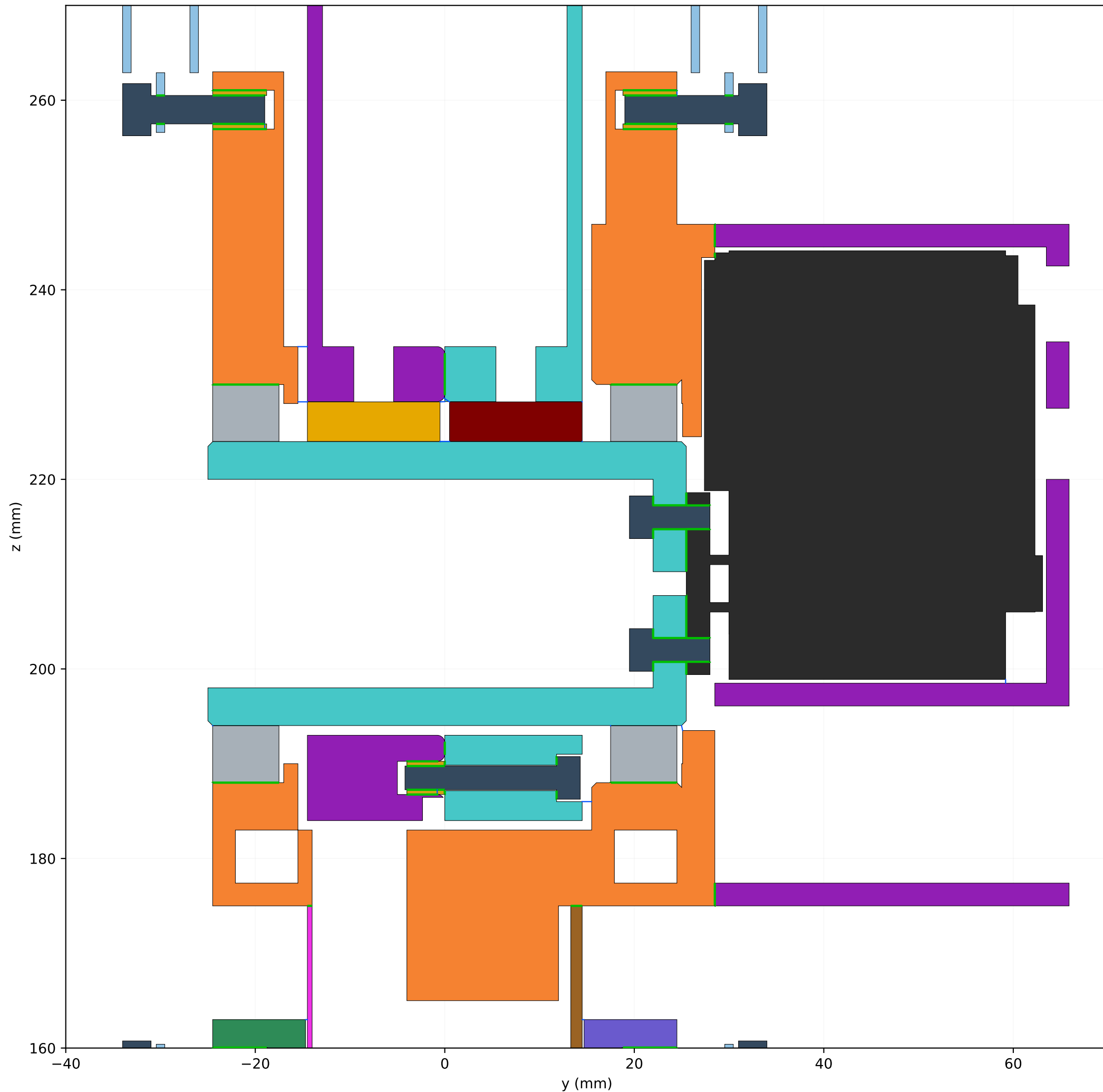


CUT-AWAY through the joint plane



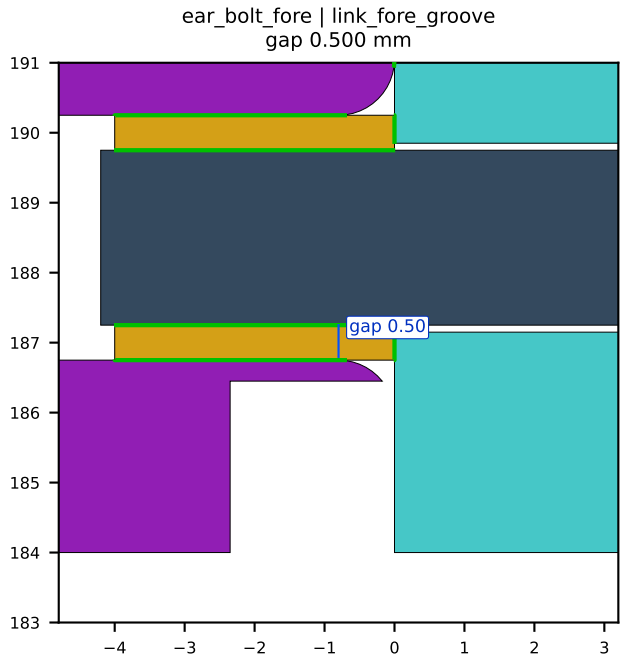
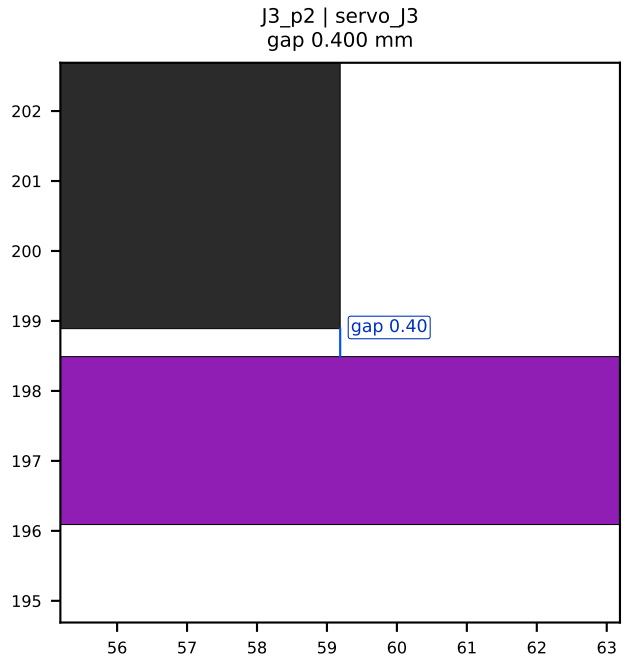
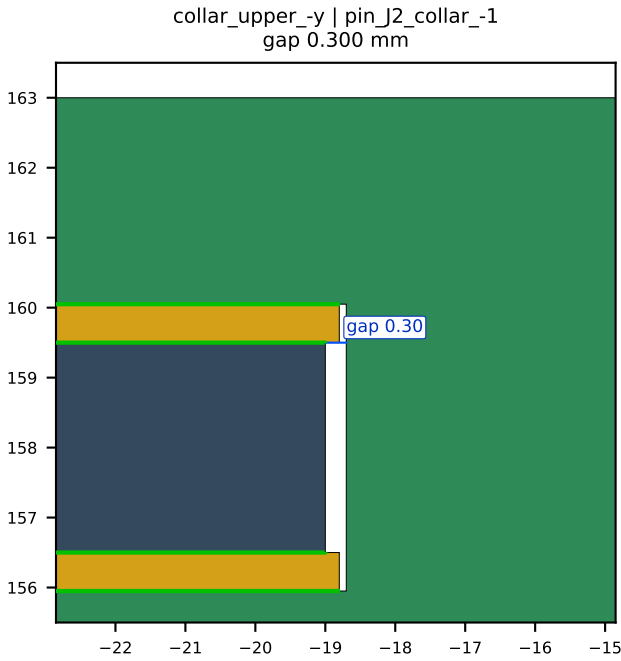
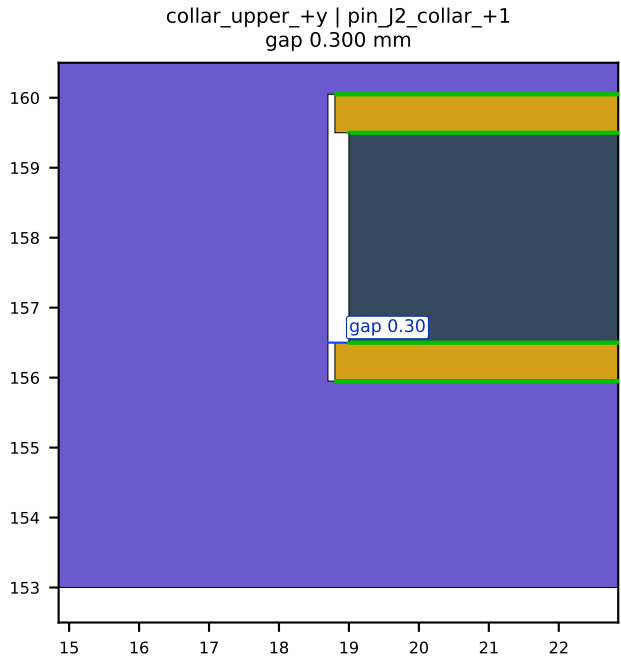
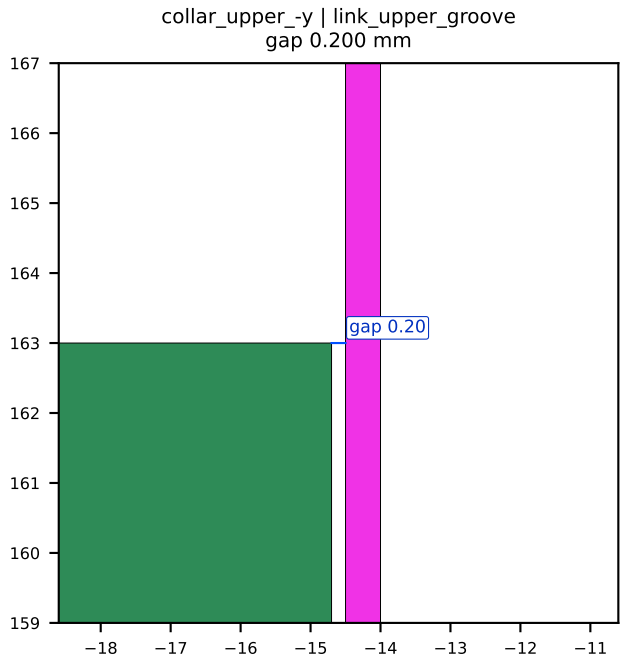
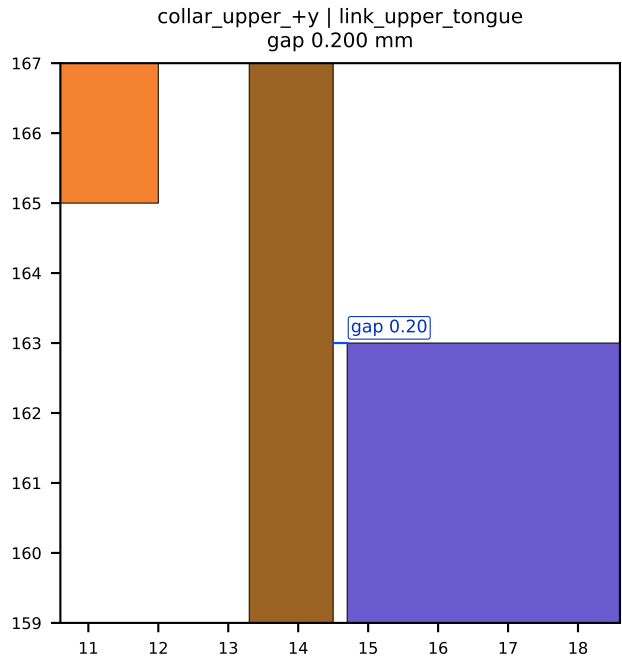
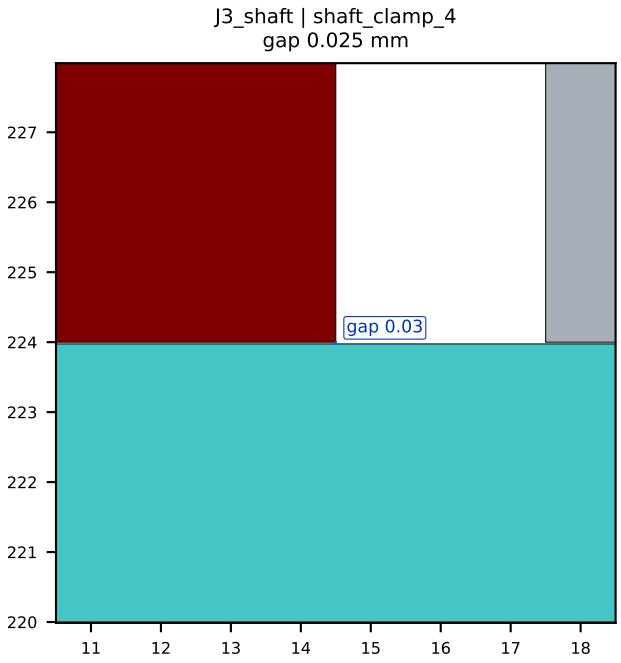
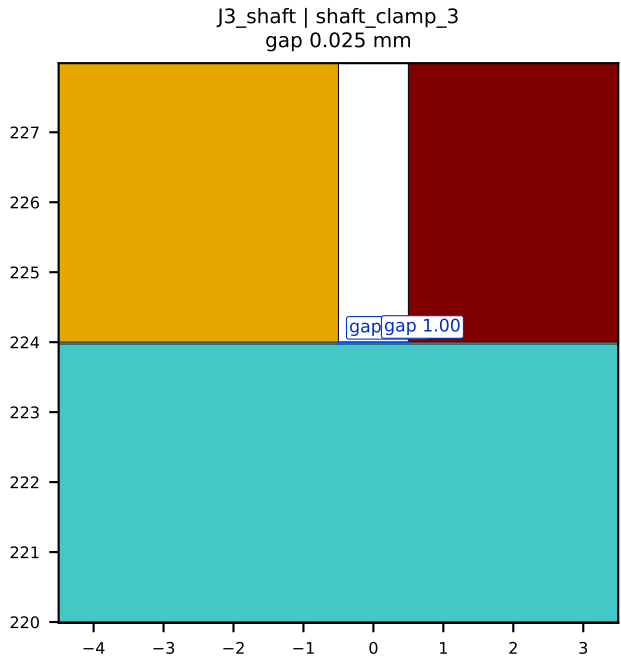
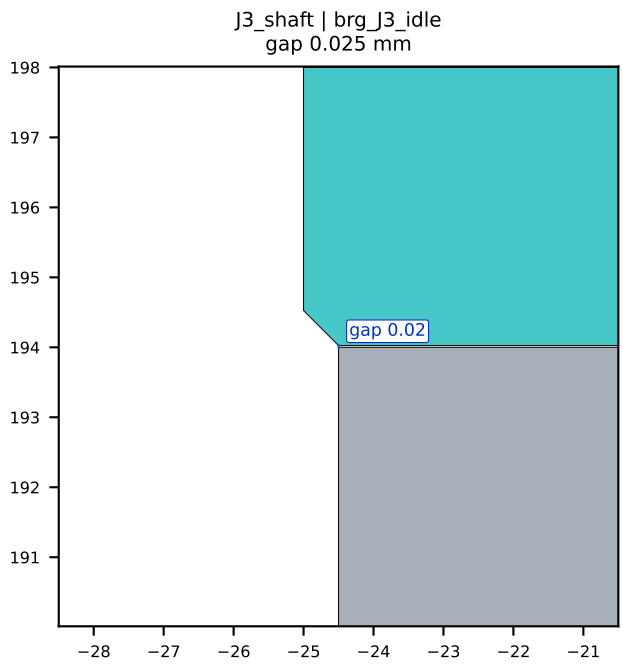
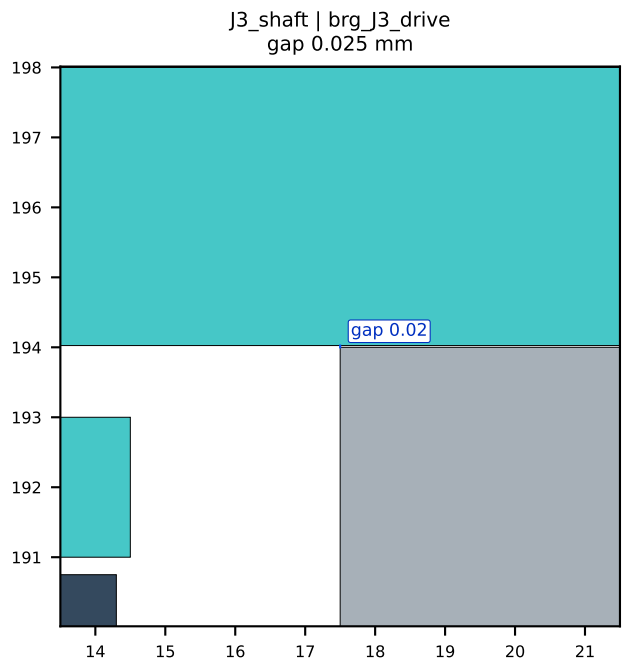
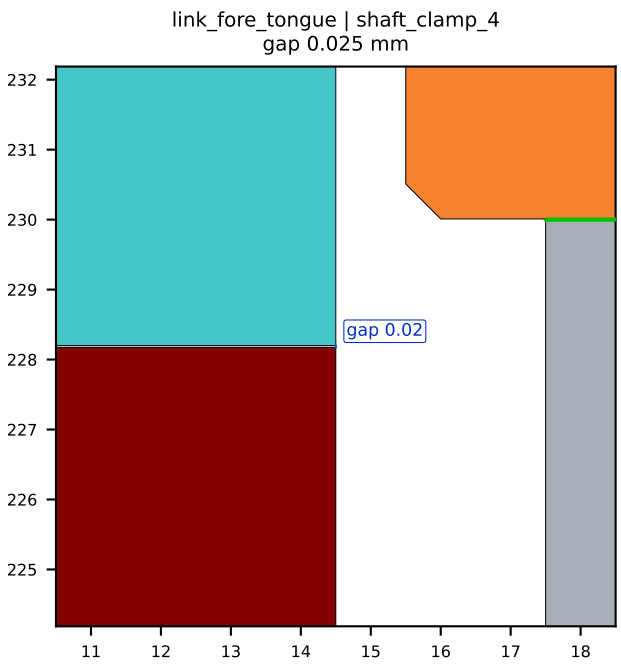
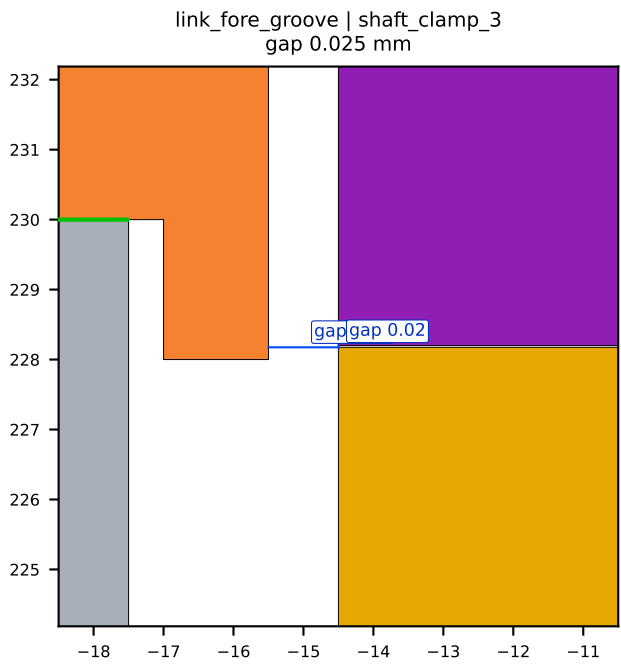
Same drive train as J2 (bay 34.99 deep, 2.4 back wall, 4 x M3 x 45); fork bolts to the upper link's socket end. Springs: fork lugs -> forearm collar (2.2 N / 5.0 N, x2).

J3 elbow -- EXACT SECTION (OpenCascade): red = solids intersecting, orange = designed (pinch / thread), green = faces touching, blue = gap < 1 mm



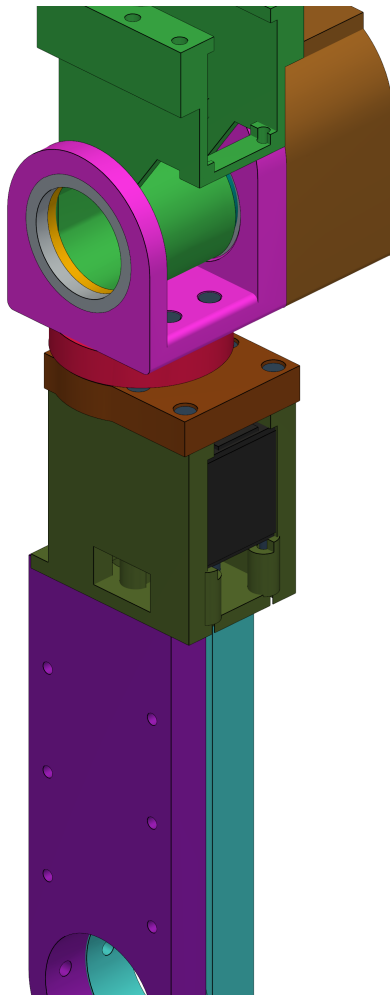
**in this cut: 30 parts, 29 touching pairs, 21 gaps < 1 mm,
0 designed overlaps, 0 UNDESIGNED OVERLAPS**

J3 elbow -- interfaces zoomed (+/-4 mm): measured on the exact section



J4 forearm roll -- every part opaque in its own colour, black edges (3D, exact solids)

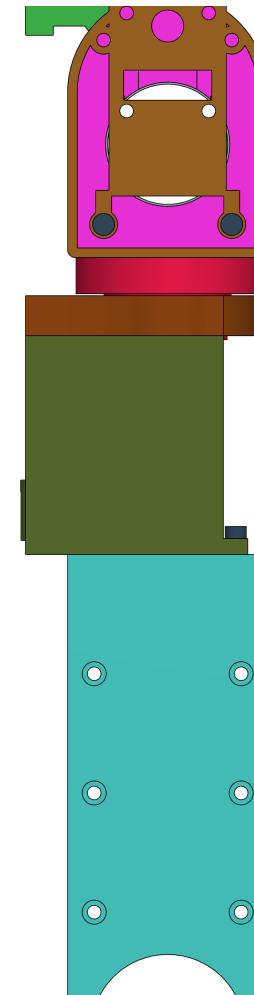
isometric



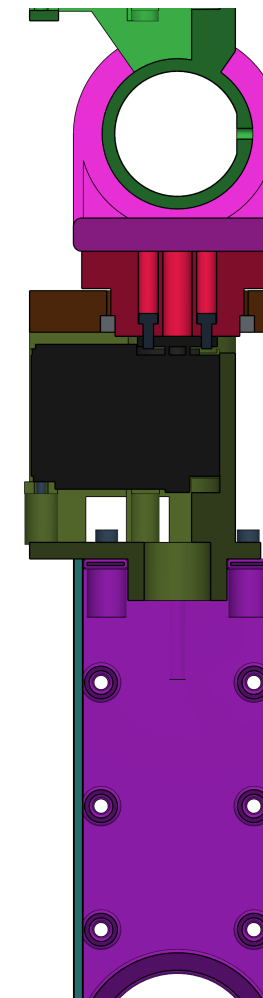
front



side (along the cut normal)



CUT-AWAY through the joint plane



Base bolts to the forearm's socket end (4 x M3 + spigot, cables down the forearm).
 Servo slides in; cap (counterbored bolts) holds the 6706; hub bolted to the horn from above;
 the J5 fork bolts to the hub's inserts. Hub turns a full revolution clear.

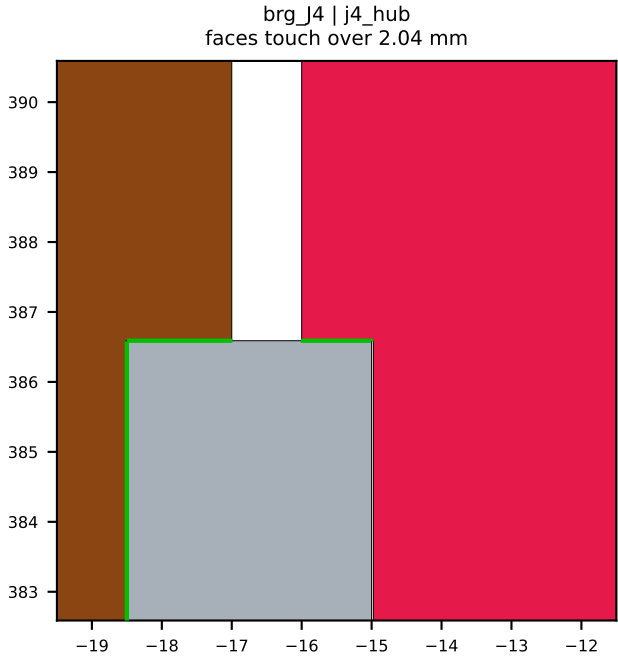
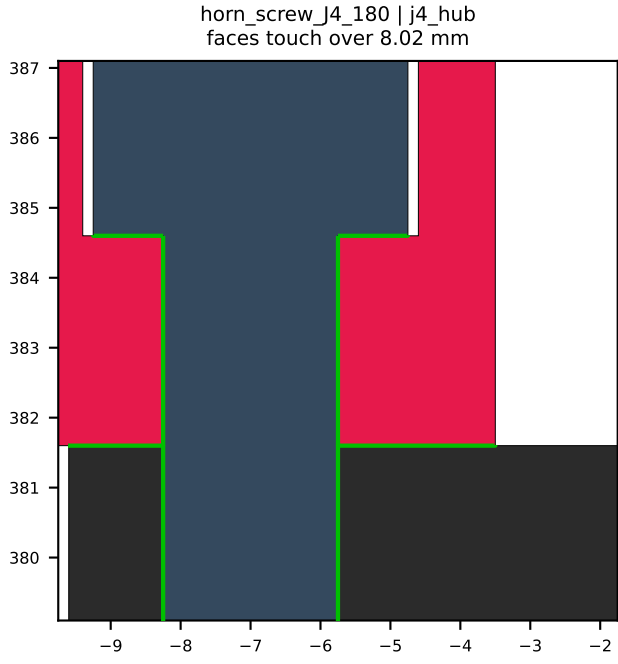
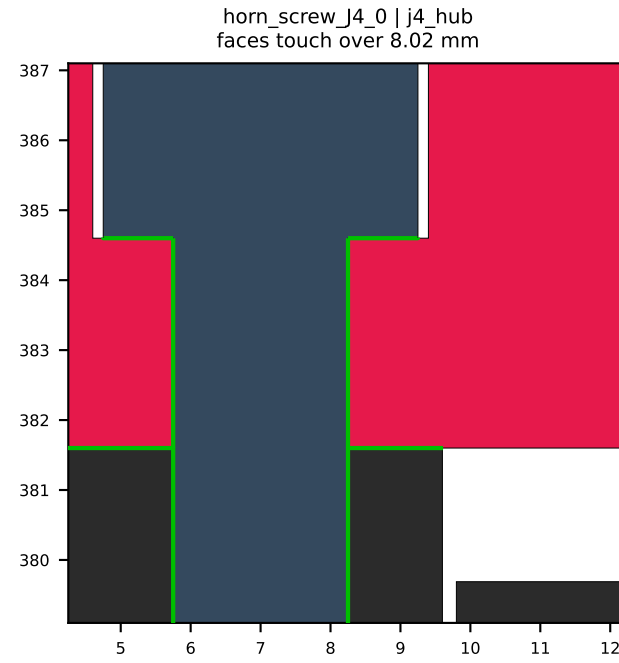
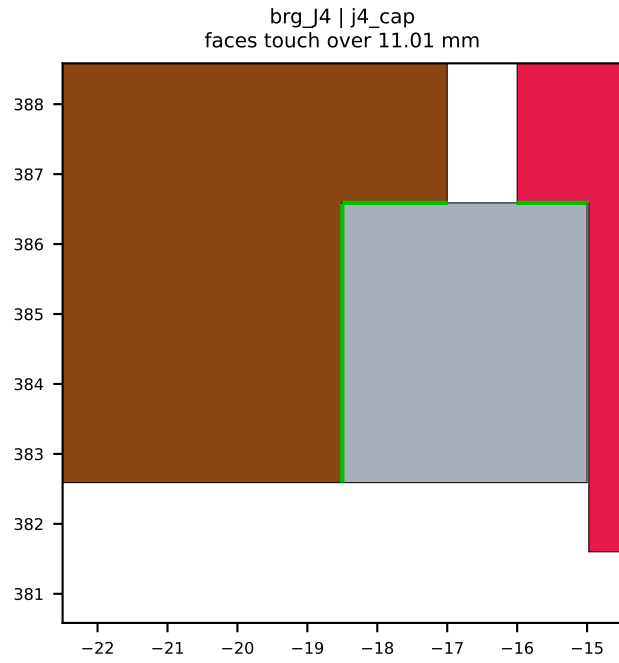
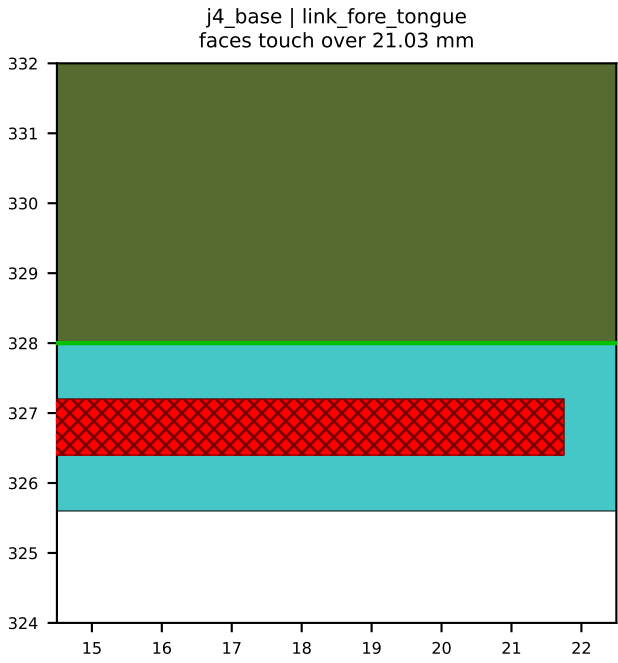
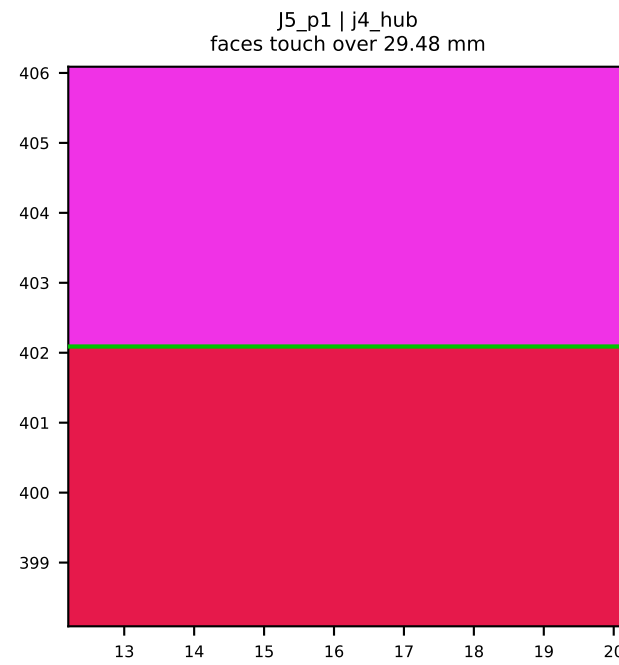
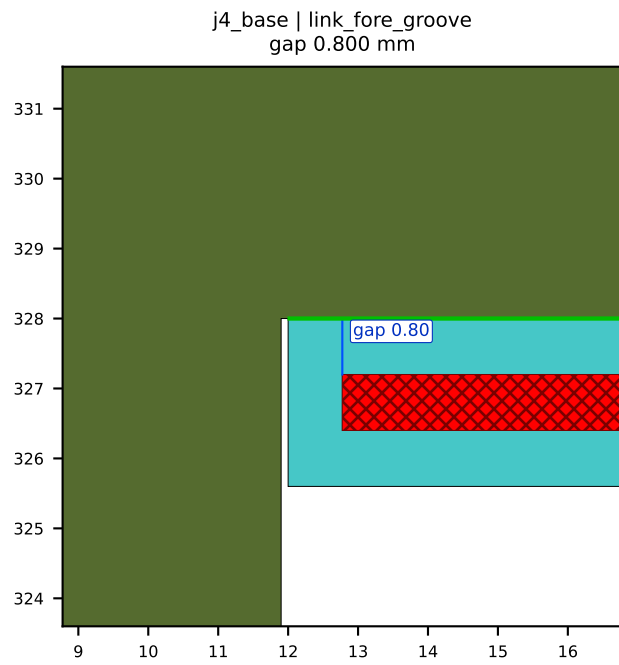
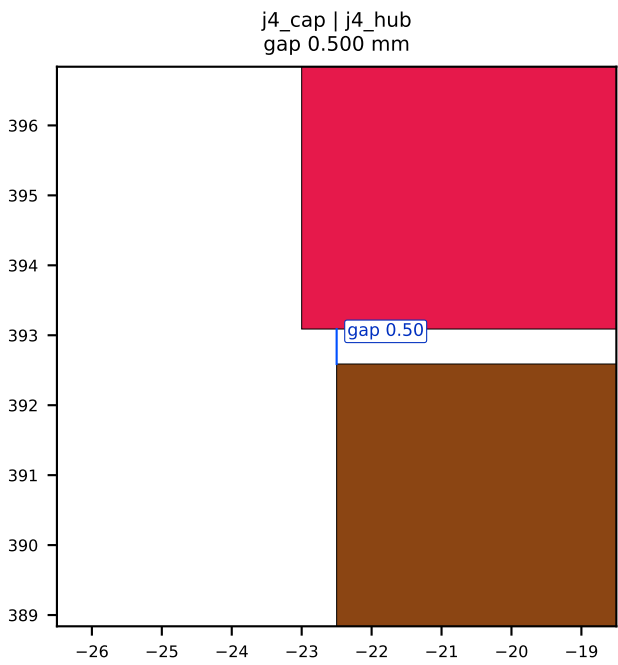
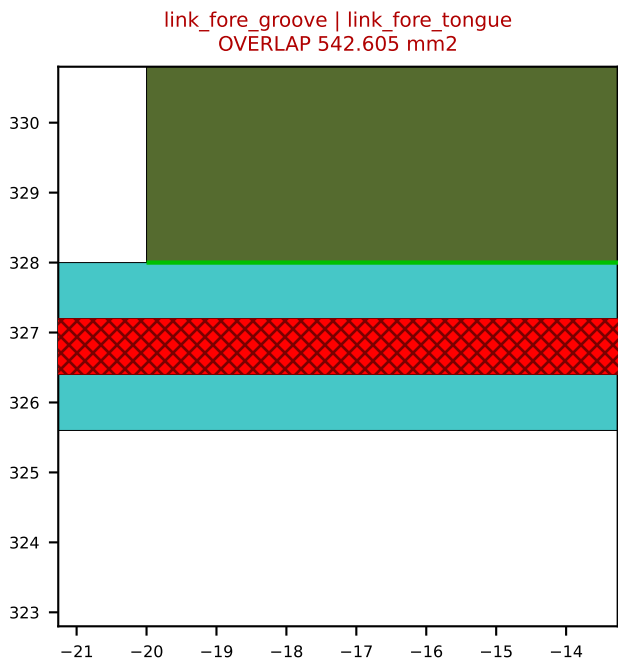
J4 forearm roll -- EXACT SECTION (OpenCascade): red = solids intersecting, orange = designed (pinch / thread), green = faces touching, blue = gap < 1 mm



- J5_p1
- brg_j4
- horn_screw_J4_0
- horn_screw_J4_180
- j4_base
- j4_cap
- j4_hub
- j6_body
- link_fore_groove
- link_fore_tongue
- servo_J4

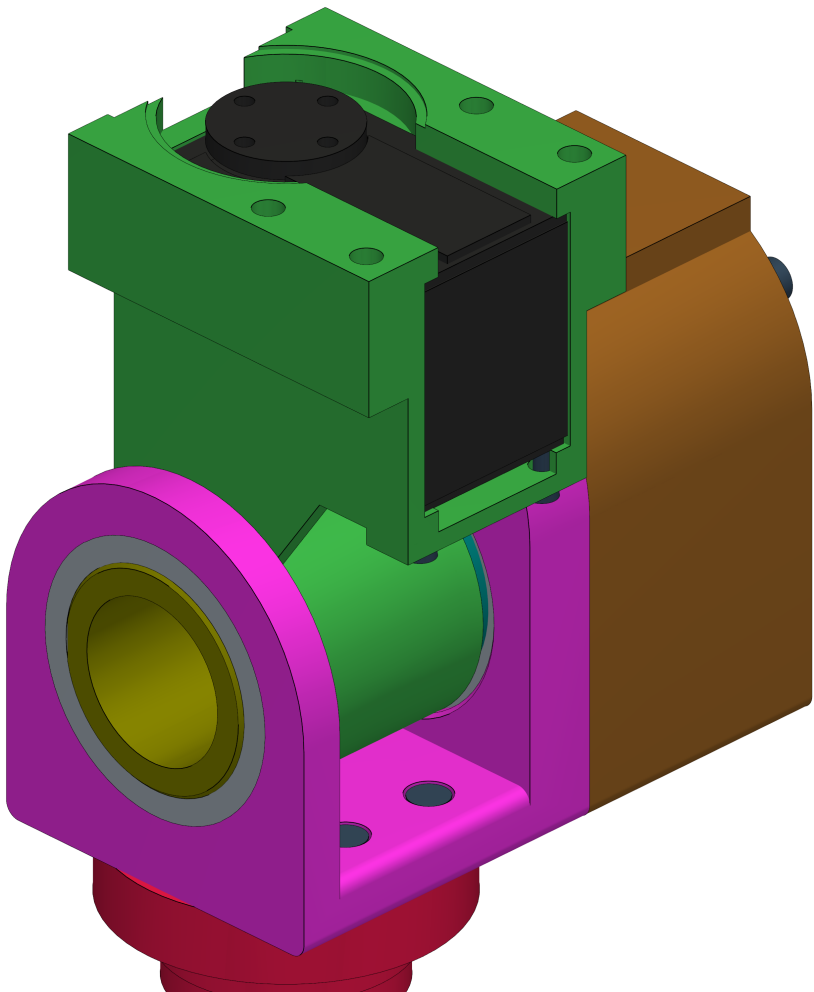
**in this cut: 11 parts, 10 touching pairs, 2 gaps < 1 mm,
0 designed overlaps, 1 UNDESIGNED OVERLAPS**

J4 forearm roll -- interfaces zoomed (+/-4 mm): measured on the exact section

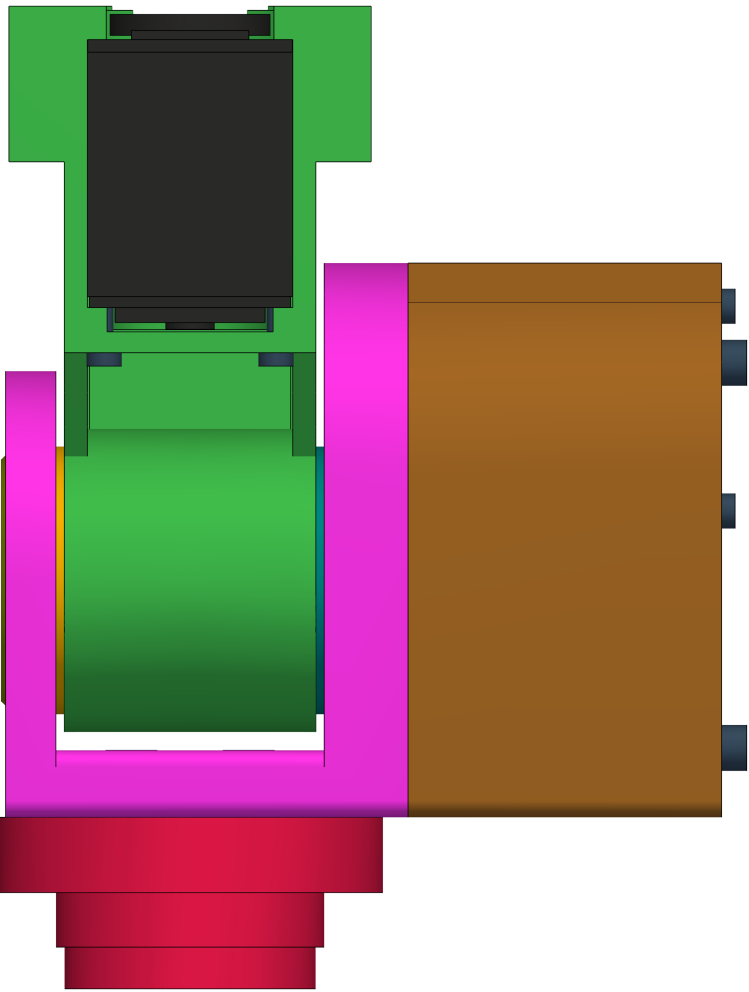


J5 wrist pitch -- every part opaque in its own colour, black edges (3D, exact solids)

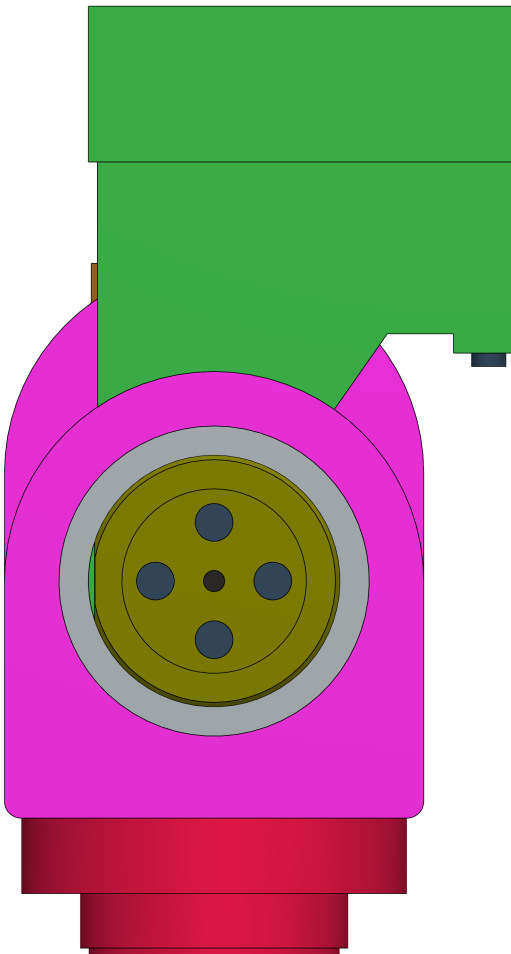
isometric



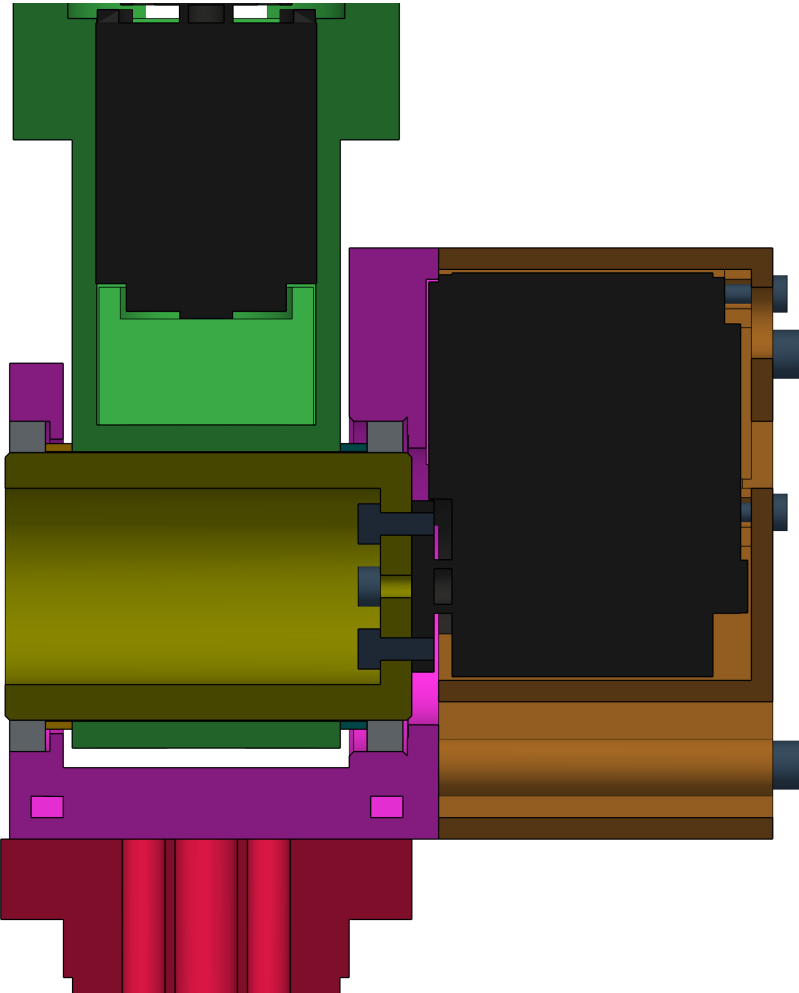
side (along the cut normal)



front

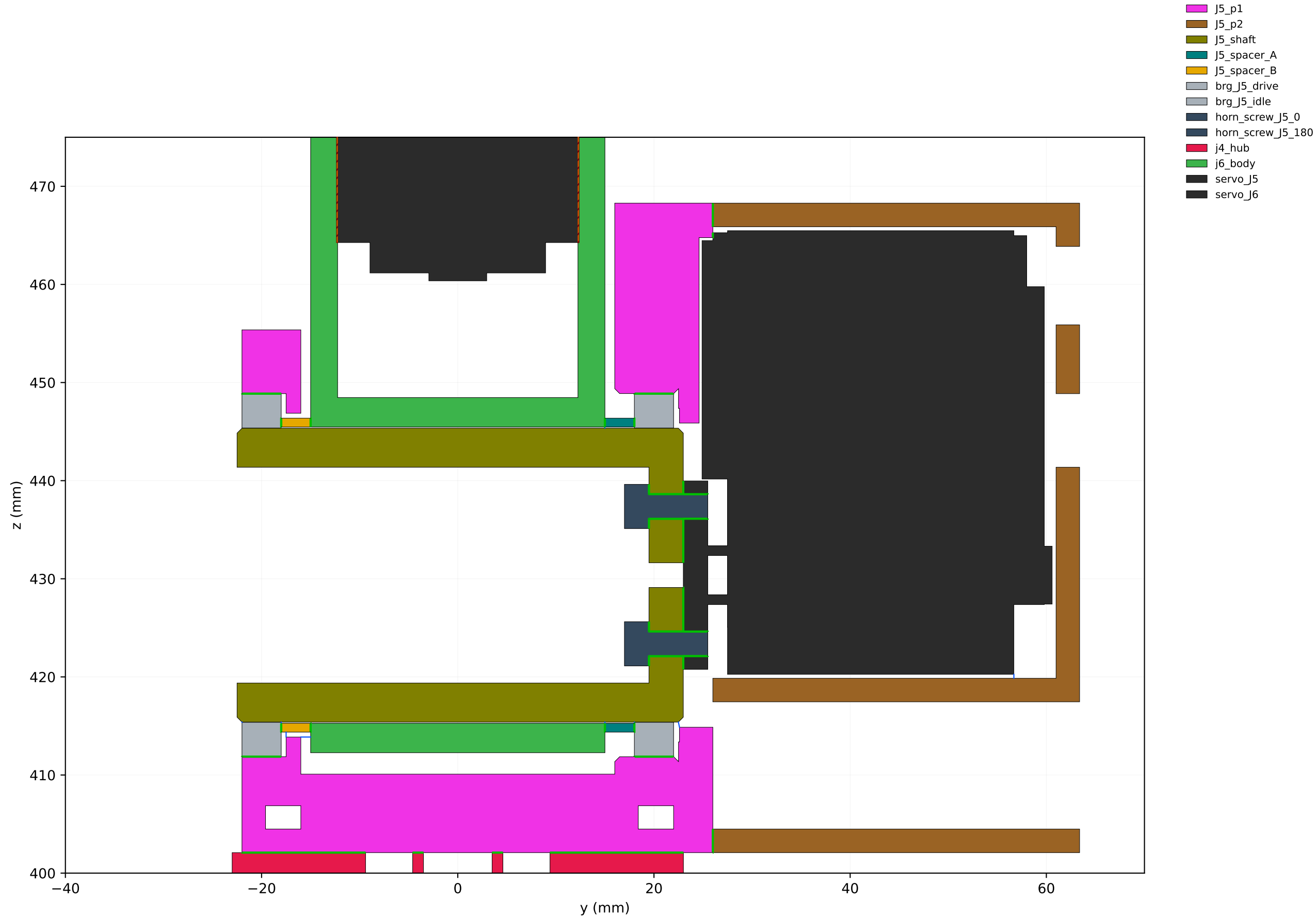


CUT-AWAY through the joint plane

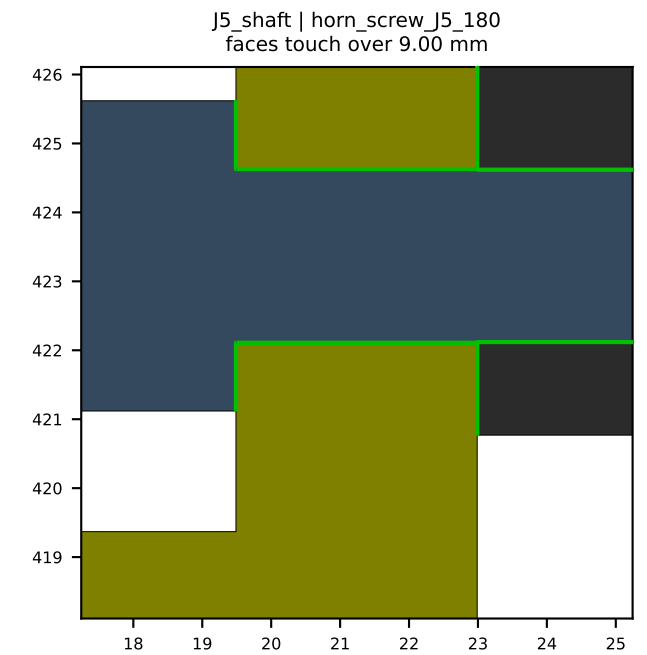
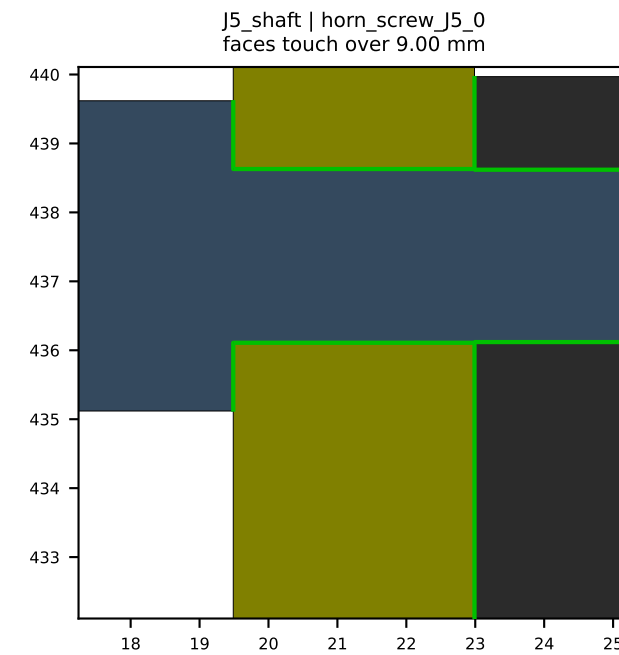
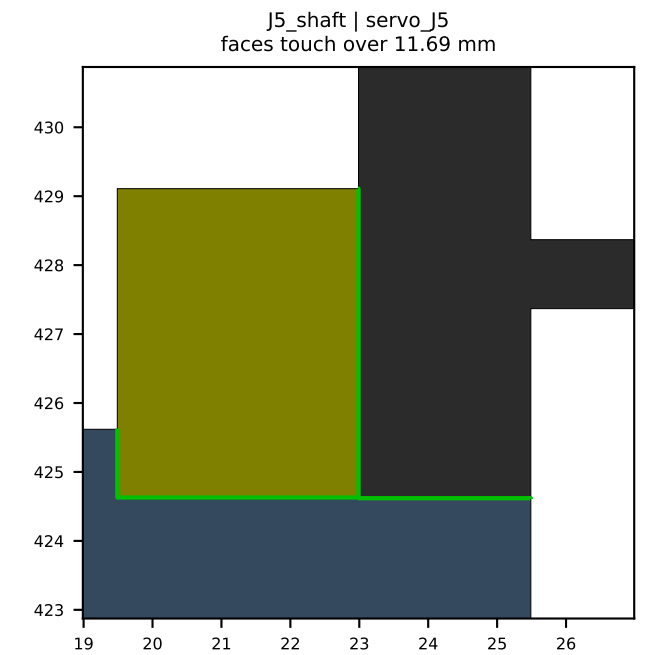
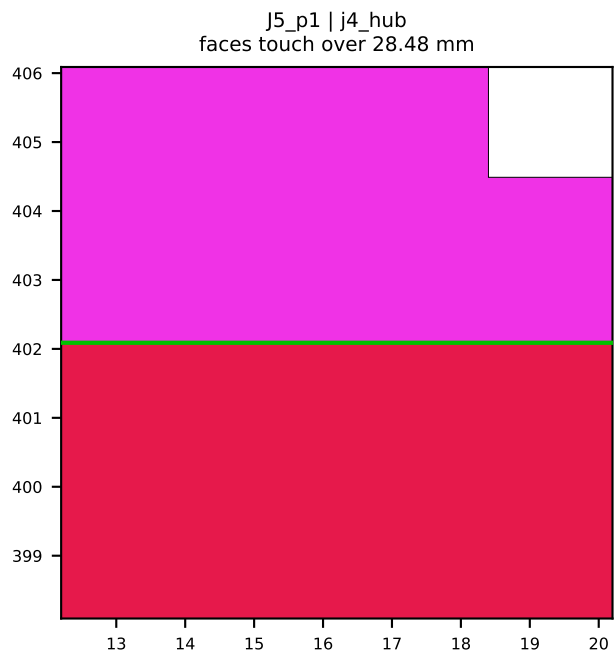
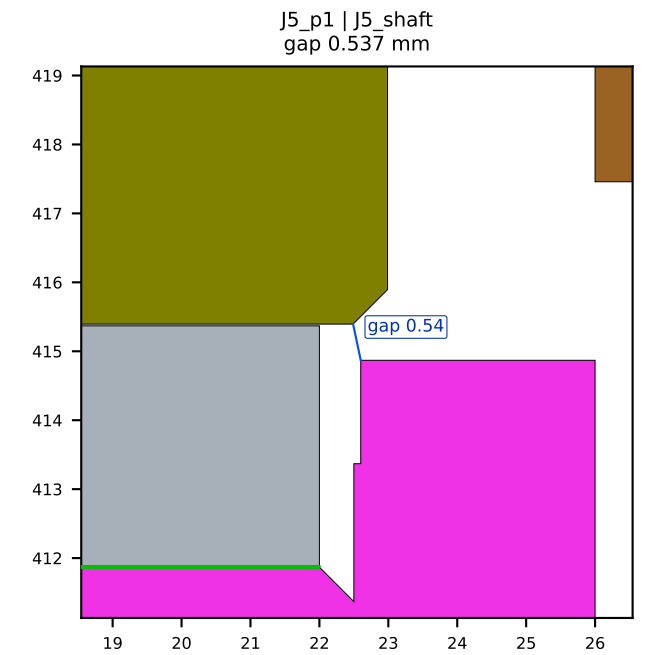
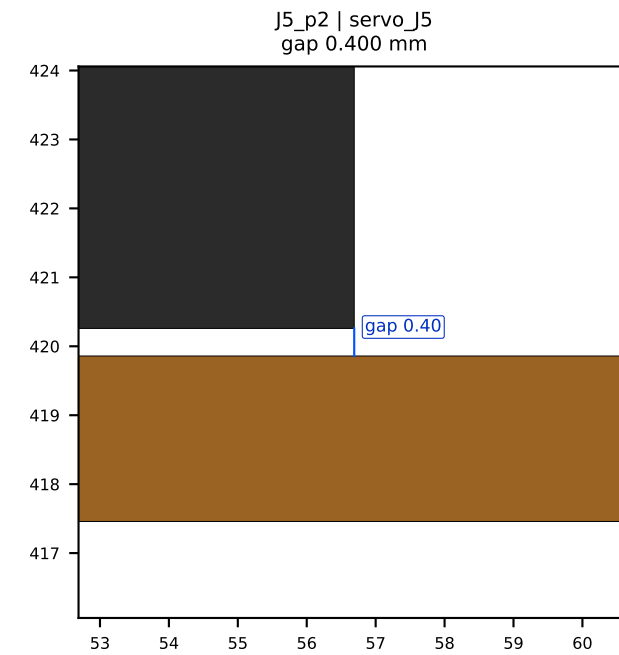
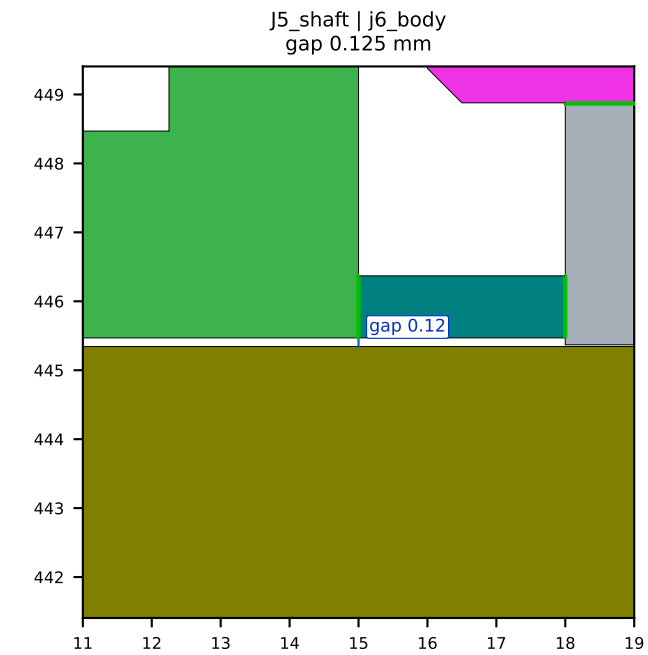
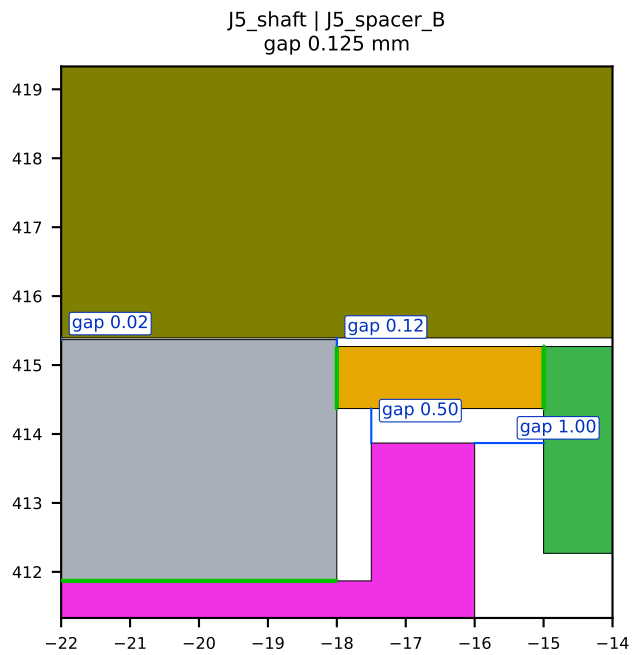
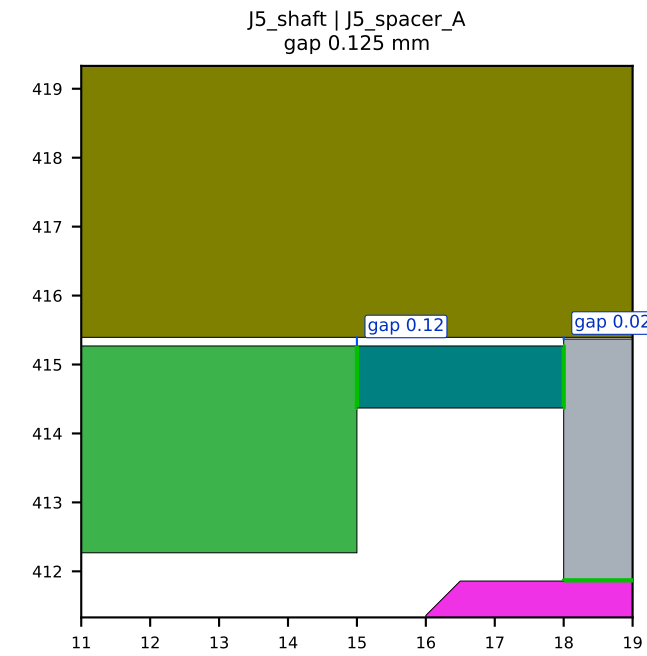
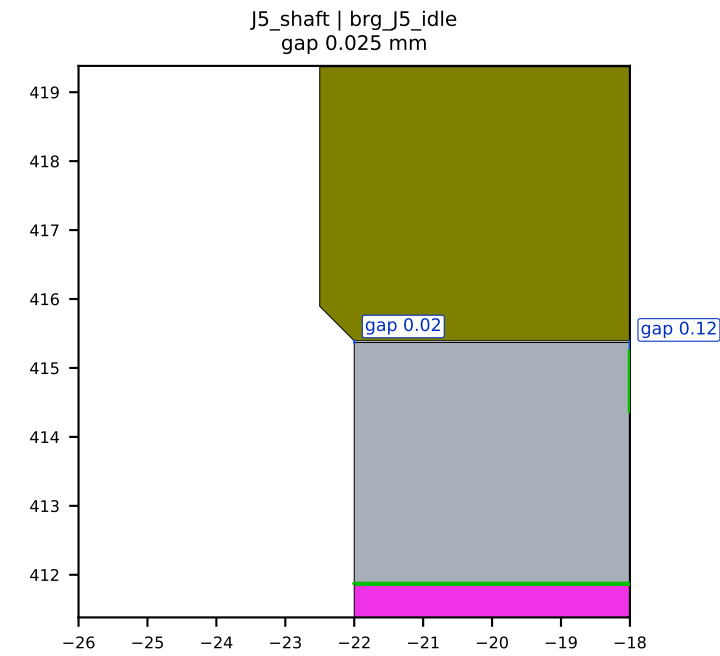
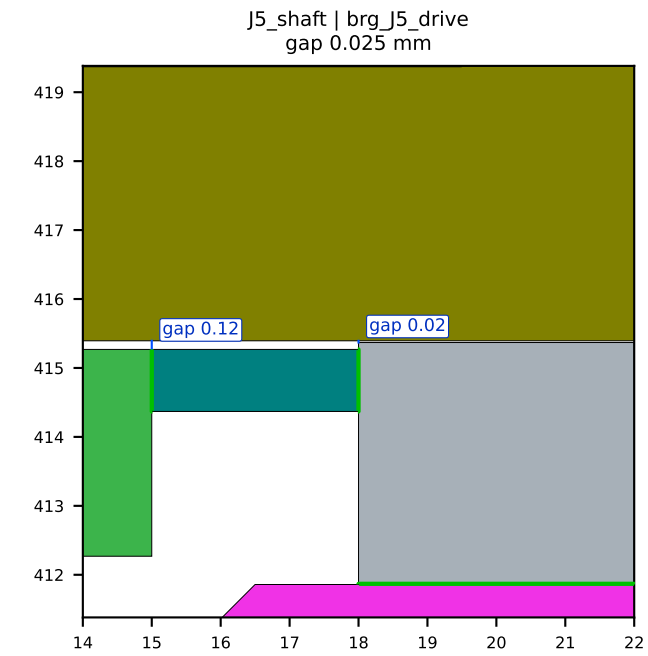
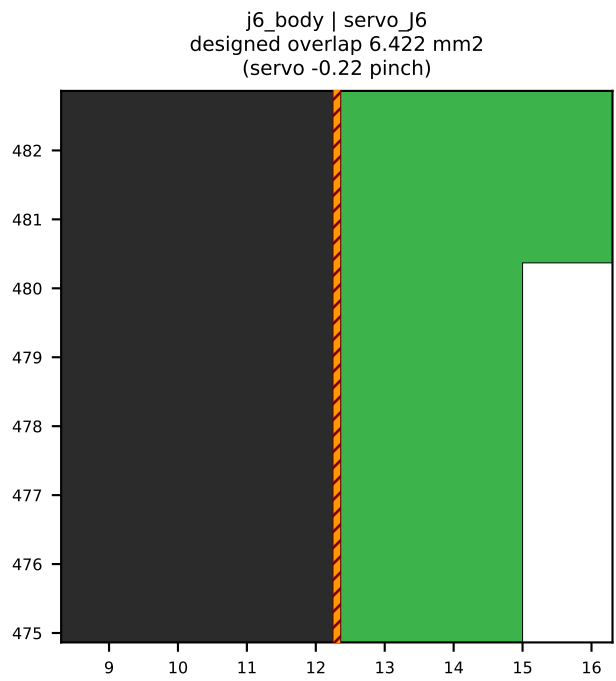


Axle bolted to the horn, D-keys the blade + M3 grub (zero backlash). Spacers clamp the blade between the inner races. Both 6706 press in from the gap. Blade swings +/-93 deg clear.

J5 wrist pitch -- EXACT SECTION (OpenCascade): red = solids intersecting, orange = designed (pinch / thread), green = faces touching, blue = gap < 1 mm

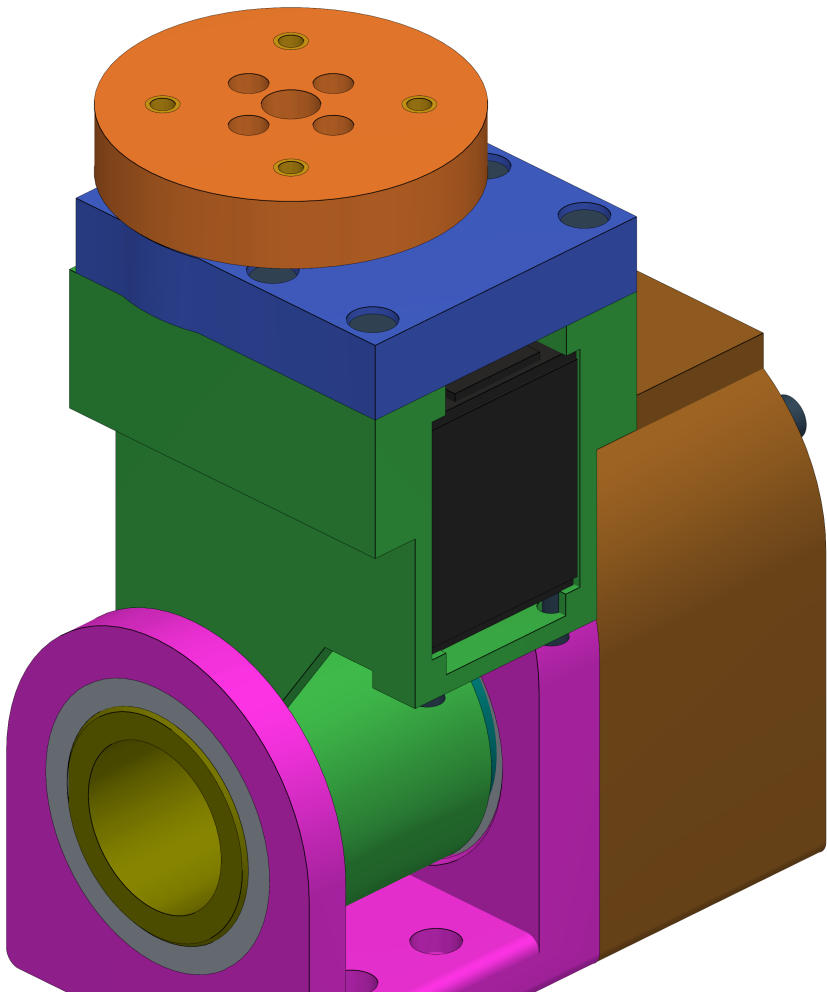


J5 wrist pitch -- interfaces zoomed (+/-4 mm): measured on the exact section

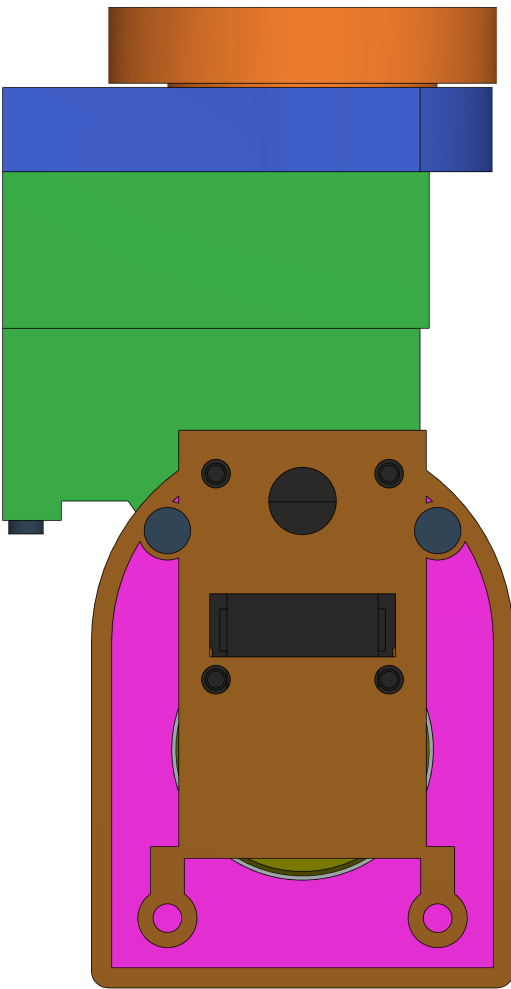


J6 tool roll -- every part opaque in its own colour, black edges (3D, exact solids)

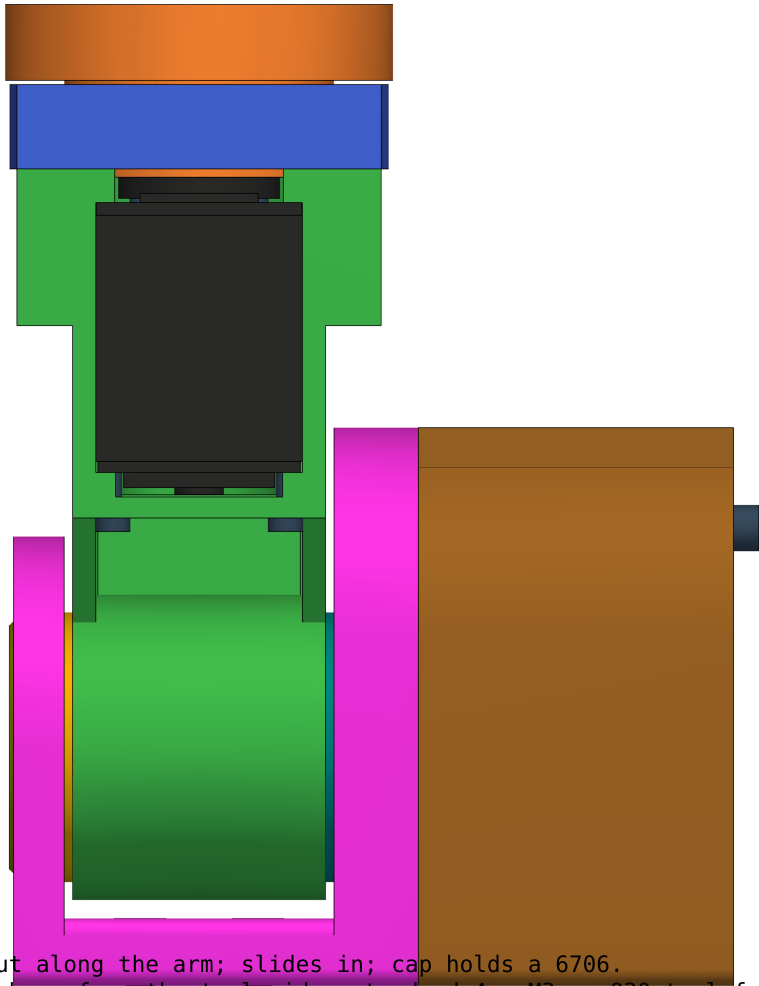
isometric



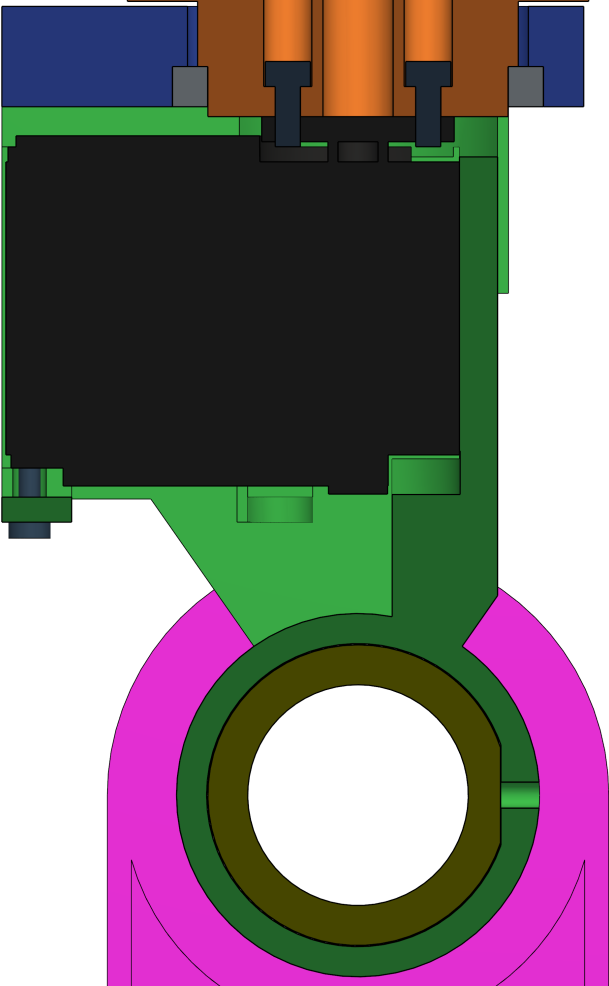
side (along the cut normal)



front

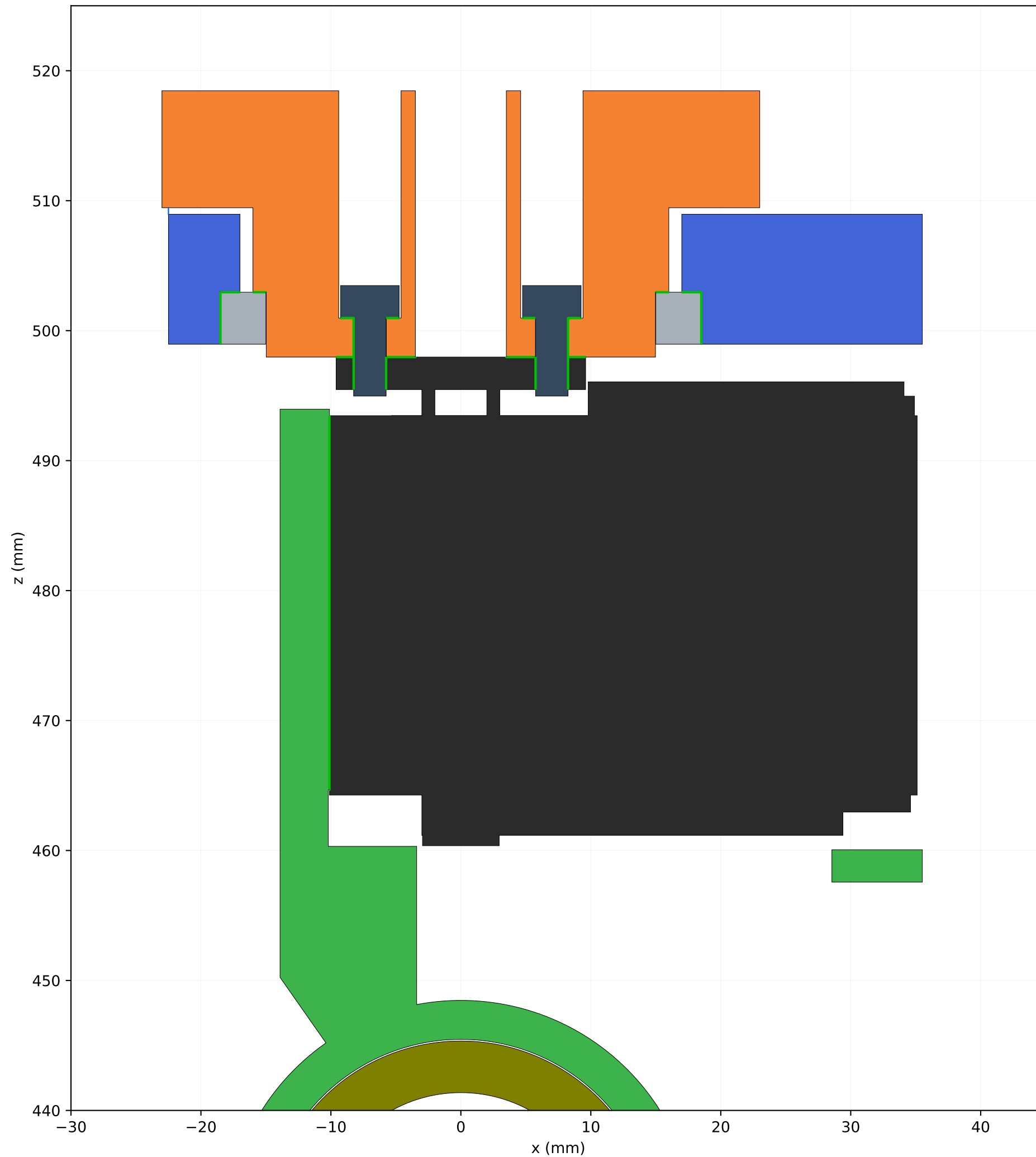


CUT-AWAY through the joint plane



ST3215 in the blade, output along the arm; slides in; cap holds a 6706.
Tool flange bolted to the horn from the tool side; standard 4 x M3 on 030 tool face.

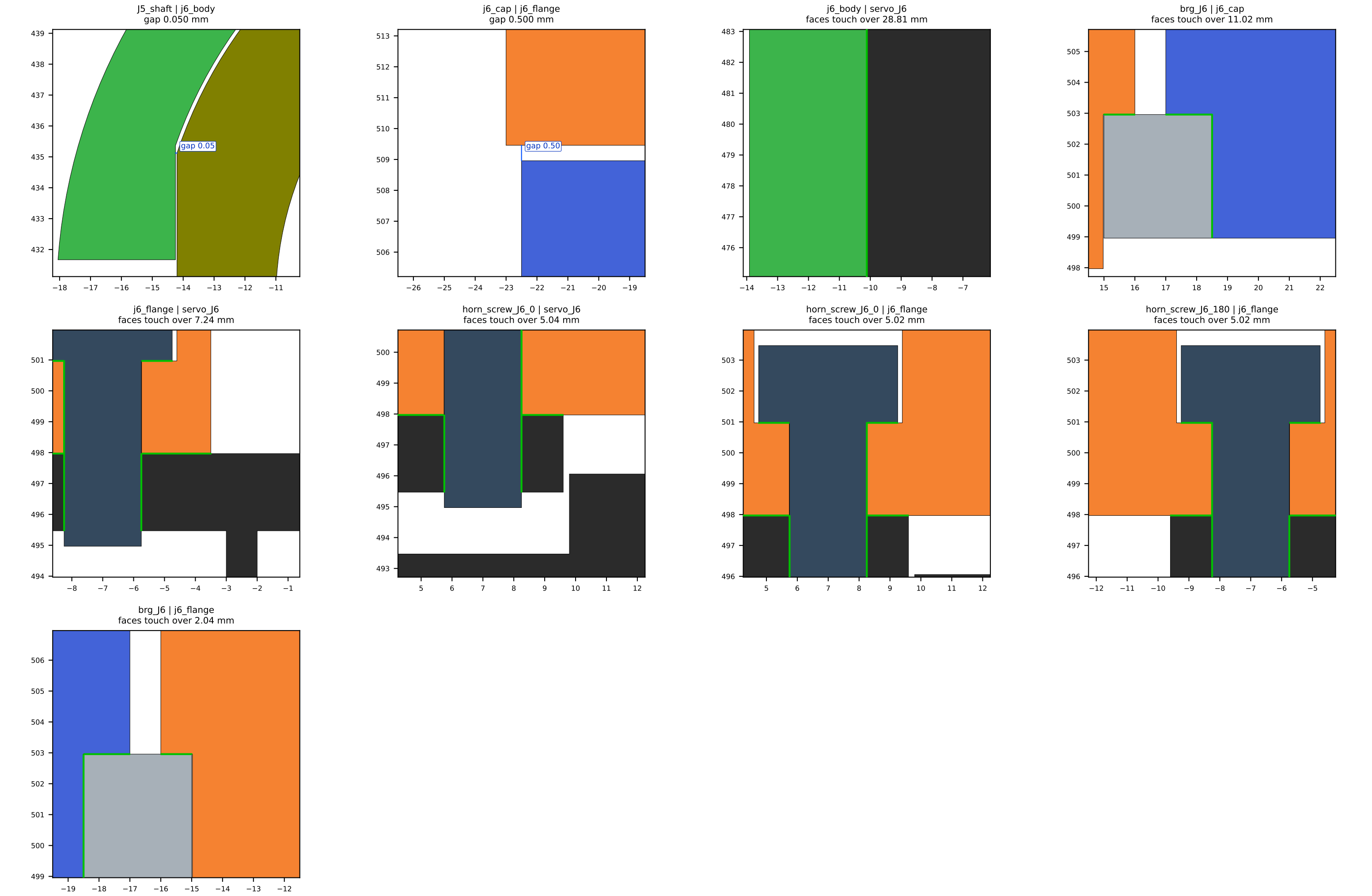
J6 tool roll -- EXACT SECTION (OpenCascade): red = solids intersecting, orange = designed (pinch / thread), green = faces touching, blue = gap < 1 mm



- J5_p1
- J5_shaft
- brg_J6
- horn_screw_J6_0
- horn_screw_J6_180
- j6_body
- j6_cap
- j6_flange
- servo_J6

**in this cut: 9 parts, 8 touching pairs, 2 gaps < 1 mm,
0 designed overlaps, 0 UNDESIGNED OVERLAPS**

J6 tool roll -- interfaces zoomed (+-4 mm): measured on the exact section



RULE 4 -- mate, never overlap: contacts, pathways, attachment (exact solids)

RULE 4 -- mating, overlap and pathways on 232 components (0 unassigned to a body: none)

J2_shaft	brg_J2_drive	b2 bNone cyl. fit clr 0.025
J2_shaft	brg_J2_idle	b2 bNone cyl. fit clr 0.025
J2_shaft	horn_screw_J2_0	b2 b2 cyl. fit clr 0.010; planar mate 10.9 mm2
J2_shaft	horn_screw_J2_180	b2 b2 cyl. fit clr 0.010; planar mate 10.9 mm2
J2_shaft	horn_screw_J2_270	b2 b2 cyl. fit clr 0.010; planar mate 10.9 mm2
J2_shaft	horn_screw_J2_90	b2 b2 cyl. fit clr 0.010; planar mate 10.9 mm2
J2_shaft	link_set_1_270	b2 b2 planar mate 19.6 mm2
J2_shaft	link_set_1_90	b2 b2 planar mate 19.6 mm2
J2_shaft	link_set_2_270	b2 b2 planar mate 19.6 mm2
J2_shaft	link_set_2_90	b2 b2 planar mate 19.6 mm2
J2_shaft	servo_J2	b2 b1 planar mate 264.6 mm2
J2_shaft	shaft_clamp_1	b2 b2 cyl. fit clr 0.025
J2_shaft	shaft_clamp_2	b2 b2 cyl. fit clr 0.025
J2_turret_p1	J2_turret_p2	b1 b1 flush; cyl. fit clr 0.000; planar mate 3666.9 mm2; cyl. fit clr 0.000; flush; cyl. fit clr 0.000; flush; flush; cyl. fit clr 0.000
J2_turret_p1	brg_J1_z3	b1 bNone cyl. fit clr 0.000
J2_turret_p1	brg_J1_z43	b1 bNone planar mate 174.6 mm2; cyl. fit clr 0.000
J2_turret_p1	brg_J2_drive	b1 bNone flush; cyl. fit clr 0.000; flush
J2_turret_p1	brg_J2_idle	b1 bNone flush; cyl. fit clr 0.000; planar mate 251.3 mm2
J2_turret_p1	pin_J2_turret_+1_ins	b1 b1 flush; cyl. fit clr 0.000
J2_turret_p1	pin_J2_turret_-1_ins	b1 b1 flush; cyl. fit clr 0.000
J2_turret_p1	servo_J2	b1 b1 planar mate 232.6 mm2
J2_turret_p1	split_ins_J2_0	b1 b1 flush; cyl. fit clr 0.000
J2_turret_p1	split_ins_J2_1	b1 b1 flush; cyl. fit clr 0.000
J2_turret_p1	split_ins_J2_2	b1 b1 flush; cyl. fit clr 0.000
J2_turret_p1	split_ins_J2_3	b1 b1 flush; cyl. fit clr 0.000
J2_turret_p2	servo_J2	b1 b1 designed overlap 290.384 mm3 (ST3215 pinch (-0.22 channel), y 12.25..12.36); planar mate 5.9 mm2; planar mate 5.9 mm2; p
J2_turret_p2	servo_scr_J2_0	b1 b1 planar mate 4.8 mm2
J2_turret_p2	servo_scr_J2_1	b1 b1 planar mate 4.8 mm2
J2_turret_p2	servo_scr_J2_2	b1 b1 planar mate 4.8 mm2
J2_turret_p2	servo_scr_J2_3	b1 b1 planar mate 4.8 mm2
J2_turret_p2	split_J2_0	b1 b1 planar mate 14.7 mm2
J2_turret_p2	split_J2_1	b1 b1 planar mate 14.7 mm2
J2_turret_p2	split_J2_2	b1 b1 planar mate 14.7 mm2
J2_turret_p2	split_J2_3	b1 b1 planar mate 14.7 mm2
J2_turret_p2	split_ins_J2_0	b1 b1 planar mate 4.1 mm2
J2_turret_p2	split_ins_J2_1	b1 b1 planar mate 4.1 mm2
J2_turret_p2	split_ins_J2_2	b1 b1 planar mate 4.1 mm2
J2_turret_p2	split_ins_J2_3	b1 b1 planar mate 4.1 mm2
J3_p1	J3_p2	b2 b2 flush; flush; planar mate 739.9 mm2; flush
J3_p1	brg_J3_drive	b2 bNone flush; cyl. fit clr 0.000
J3_p1	brg_J3_idle	b2 bNone flush; cyl. fit clr 0.000
J3_p1	j3fork_link_0	b2 b2 planar mate 14.7 mm2
J3_p1	j3fork_link_1	b2 b2 planar mate 14.7 mm2
J3_p1	j3fork_link_2	b2 b2 planar mate 14.7 mm2
J3_p1	j3fork_link_3	b2 b2 planar mate 14.7 mm2
J3_p1	link_end_ins_0	b2 b2 planar mate 4.1 mm2
J3_p1	link_end_ins_1	b2 b2 planar mate 4.1 mm2
J3_p1	link_end_ins_2	b2 b2 planar mate 4.1 mm2
J3_p1	link_end_ins_3	b2 b2 planar mate 4.1 mm2
J3_p1	link_upper_groove	b2 b2 cyl. fit clr 0.000; cyl. fit clr 0.000; cyl. fit clr 0.000; planar mate 382.0 mm2
J3_p1	link_upper_tongue	b2 b2 cyl. fit clr 0.000; cyl. fit clr 0.000; planar mate 403.7 mm2
J3_p1	pin_J3_fork_+1_ins	b2 b2 flush; cyl. fit clr 0.000
J3_p1	pin_J3_fork_-1_ins	b2 b2 flush; cyl. fit clr 0.000
J3_p1	servo_J3	b2 b2 planar mate 232.6 mm2
J3_p1	split_ins_J3_0	b2 b2 flush; cyl. fit clr 0.000
J3_p1	split_ins_J3_1	b2 b2 flush; cyl. fit clr 0.000
J3_p1	split_ins_J3_2	b2 b2 flush; cyl. fit clr 0.000
J3_p1	split_ins_J3_3	b2 b2 flush; cyl. fit clr 0.000
J3_p2	servo_J3	b2 b2 designed overlap 290.384 mm3 (ST3215 pinch (-0.22 channel), y 12.25..12.36); planar mate 5.9 mm2; planar mate 4.9 mm2;

RULE 4 -- mate, never overlap: contacts, pathways, attachment (exact solids) (cont.)

J3_p2	servo_scr_J3_1	b2 b2	planar	mate	4.8 mm2
J3_p2	servo_scr_J3_2	b2 b2	planar	mate	4.8 mm2
J3_p2	servo_scr_J3_3	b2 b2	planar	mate	4.8 mm2
J3_p2	split_J3_0	b2 b2	planar	mate	14.7 mm2
J3_p2	split_J3_1	b2 b2	planar	mate	14.7 mm2
J3_p2	split_J3_2	b2 b2	planar	mate	14.7 mm2
J3_p2	split_J3_3	b2 b2	planar	mate	14.7 mm2
J3_p2	split_ins_J3_0	b2 b2	planar	mate	4.1 mm2
J3_p2	split_ins_J3_1	b2 b2	planar	mate	4.1 mm2
J3_p2	split_ins_J3_2	b2 b2	planar	mate	4.1 mm2
J3_p2	split_ins_J3_3	b2 b2	planar	mate	4.1 mm2
J3_shaft	brg_J3_drive	b3 bNone	cyl. fit	clr 0.025	
J3_shaft	brg_J3_idle	b3 bNone	cyl. fit	clr 0.025	
J3_shaft	horn_screw_J3_0	b3 b3	cyl. fit	clr 0.010; planar	mate 10.9 mm2
J3_shaft	horn_screw_J3_180	b3 b3	cyl. fit	clr 0.010; planar	mate 10.9 mm2
J3_shaft	horn_screw_J3_270	b3 b3	cyl. fit	clr 0.010; planar	mate 10.9 mm2
J3_shaft	horn_screw_J3_90	b3 b3	cyl. fit	clr 0.010; planar	mate 10.9 mm2
J3_shaft	link_set_3_270	b3 b3	planar	mate	19.6 mm2
J3_shaft	link_set_3_90	b3 b3	planar	mate	19.6 mm2
J3_shaft	link_set_4_270	b3 b3	planar	mate	19.6 mm2
J3_shaft	link_set_4_90	b3 b3	planar	mate	19.6 mm2
J3_shaft	servo_J3	b3 b2	planar	mate	264.6 mm2
J3_shaft	shaft_clamp_3	b3 b3	cyl. fit	clr 0.025	
J3_shaft	shaft_clamp_4	b3 b3	cyl. fit	clr 0.025	
J5_p1	J5_p2	b4 b4	flush; flush; planar	mate 712.5 mm2; flush	
J5_p1	brg_J5_drive	b4 bNone	flush; cyl. fit	clr 0.000	
J5_p1	brg_J5_idle	b4 bNone	flush; cyl. fit	clr 0.000	
J5_p1	j4_hub	b4 b4	planar	mate	1489.2 mm2
J5_p1	j4hub_ins_0	b4 b4	planar	mate	4.1 mm2
J5_p1	j4hub_ins_1	b4 b4	planar	mate	4.1 mm2
J5_p1	j4hub_ins_2	b4 b4	planar	mate	4.1 mm2
J5_p1	j4hub_ins_3	b4 b4	planar	mate	4.1 mm2
J5_p1	j5fork_j4hub_0	b4 b4	planar	mate	14.7 mm2
J5_p1	j5fork_j4hub_1	b4 b4	planar	mate	14.7 mm2
J5_p1	j5fork_j4hub_2	b4 b4	planar	mate	14.7 mm2
J5_p1	j5fork_j4hub_3	b4 b4	planar	mate	14.7 mm2
J5_p1	servo_J5	b4 b4	planar	mate	232.6 mm2
J5_p1	split_ins_J5_0	b4 b4	flush; cyl. fit	clr 0.000	
J5_p1	split_ins_J5_1	b4 b4	flush; cyl. fit	clr 0.000	
J5_p1	split_ins_J5_2	b4 b4	flush; cyl. fit	clr 0.000	
J5_p1	split_ins_J5_3	b4 b4	flush; cyl. fit	clr 0.000	
J5_p2	servo_J5	b4 b4	designed overlap	290.384 mm3 (ST3215 pinch (-0.22 channel), y 12.25..12.36); planar	mate 5.9 mm2; planar mate 4.9 mm2; p
J5_p2	servo_scr_J5_0	b4 b4	planar	mate	4.8 mm2
J5_p2	servo_scr_J5_1	b4 b4	planar	mate	4.8 mm2
J5_p2	servo_scr_J5_2	b4 b4	planar	mate	4.8 mm2
J5_p2	servo_scr_J5_3	b4 b4	planar	mate	4.8 mm2
J5_p2	split_J5_0	b4 b4	planar	mate	14.7 mm2
J5_p2	split_J5_1	b4 b4	planar	mate	14.7 mm2
J5_p2	split_J5_2	b4 b4	planar	mate	14.7 mm2
J5_p2	split_J5_3	b4 b4	planar	mate	14.7 mm2
J5_p2	split_ins_J5_0	b4 b4	planar	mate	4.1 mm2
J5_p2	split_ins_J5_1	b4 b4	planar	mate	4.1 mm2
J5_p2	split_ins_J5_2	b4 b4	planar	mate	4.1 mm2
J5_p2	split_ins_J5_3	b4 b4	planar	mate	4.1 mm2
J5_shaft	brg_J5_drive	b5 bNone	cyl. fit	clr 0.025	
J5_shaft	brg_J5_idle	b5 bNone	cyl. fit	clr 0.025	
J5_shaft	horn_screw_J5_0	b5 b5	cyl. fit	clr 0.010; planar	mate 10.9 mm2
J5_shaft	horn_screw_J5_180	b5 b5	cyl. fit	clr 0.010; planar	mate 10.9 mm2
J5_shaft	horn_screw_J5_270	b5 b5	cyl. fit	clr 0.010; planar	mate 10.9 mm2
J5_shaft	horn_screw_J5_90	b5 b5	cyl. fit	clr 0.010; planar	mate 10.9 mm2
J5_shaft	j5_grub	b5 b5	planar	mate	7.1 mm2

RULE 4 -- mate, never overlap: contacts, pathways, attachment (exact solids) (cont.)

J5_spacer_A	brg_J5_drive	b5 bNone planar mate 87.9 mm2
J5_spacer_A	j6_body	b5 b5 planar mate 5.6 mm2; planar mate 50.4 mm2; planar mate 31.9 mm2
J5_spacer_B	brg_J5_idle	b5 bNone planar mate 87.9 mm2
J5_spacer_B	j6_body	b5 b5 planar mate 50.4 mm2; planar mate 31.9 mm2; planar mate 5.6 mm2
base	brg_J1_z3	b0 bNone flush; cyl. fit clr 0.000; planar mate 251.3 mm2
base	brg_J1_z43	b0 bNone flush; cyl. fit clr 0.000; planar mate 251.3 mm2
base	foot_ins_0	b0 b0 planar mate 8.7 mm2
base	foot_ins_1	b0 b0 planar mate 8.7 mm2
base	foot_ins_2	b0 b0 planar mate 8.7 mm2
base	foot_ins_3	b0 b0 planar mate 8.7 mm2
base	j1_mount	b0 b0 planar mate 127.5 mm2; planar mate 127.5 mm2; planar mate 127.5 mm2; planar mate 127.5 mm2
base	m4_base_foot_0	b0 b0 planar mate 22.6 mm2
base	m4_base_foot_1	b0 b0 planar mate 22.6 mm2
base	m4_base_foot_2	b0 b0 planar mate 22.6 mm2
base	m4_base_foot_3	b0 b0 planar mate 22.6 mm2
brg_J1_z3	spigot_collar	bNone b1 planar mate 164.1 mm2
brg_J4	j4_base	bNone b3 planar mate 57.9 mm2; planar mate 57.9 mm2
brg_J4	j4_cap	bNone b3 cyl. fit clr 0.010; flush; planar mate 167.3 mm2
brg_J4	j4_hub	bNone b4 cyl. fit clr 0.025; planar mate 97.4 mm2
brg_J6	j6_body	bNone b5 planar mate 9.1 mm2; planar mate 9.1 mm2; planar mate 7.7 mm2; planar mate 34.0 mm2; planar mate 2.0 mm2; planar mate 5
brg_J6	j6_cap	bNone b5 cyl. fit clr 0.010; flush; planar mate 167.3 mm2
brg_J6	j6_flange	bNone b6 cyl. fit clr 0.025; planar mate 97.4 mm2
cap_J4_0	cap_ins_J4_0	b3 b3 cyl. fit clr 0.000
cap_J4_0	j4_cap	b3 b3 planar mate 14.7 mm2
cap_J4_1	cap_ins_J4_1	b3 b3 cyl. fit clr 0.000
cap_J4_1	j4_cap	b3 b3 planar mate 14.7 mm2
cap_J4_2	cap_ins_J4_2	b3 b3 cyl. fit clr 0.000
cap_J4_2	j4_cap	b3 b3 planar mate 14.7 mm2
cap_J4_3	cap_ins_J4_3	b3 b3 cyl. fit clr 0.000
cap_J4_3	j4_cap	b3 b3 planar mate 14.7 mm2
cap_J6_0	cap_ins_J6_0	b5 b5 cyl. fit clr 0.000
cap_J6_0	j6_cap	b5 b5 planar mate 14.7 mm2
cap_J6_1	cap_ins_J6_1	b5 b5 cyl. fit clr 0.000
cap_J6_1	j6_cap	b5 b5 planar mate 14.7 mm2
cap_J6_2	cap_ins_J6_2	b5 b5 cyl. fit clr 0.000
cap_J6_2	j6_cap	b5 b5 planar mate 14.7 mm2
cap_J6_3	cap_ins_J6_3	b5 b5 cyl. fit clr 0.000
cap_J6_3	j6_cap	b5 b5 planar mate 14.7 mm2
cap_ins_J4_0	j4_base	b3 b3 cyl. fit clr 0.000; flush
cap_ins_J4_0	j4_cap	b3 b3 planar mate 4.1 mm2
cap_ins_J4_1	j4_base	b3 b3 cyl. fit clr 0.000; flush
cap_ins_J4_1	j4_cap	b3 b3 planar mate 4.1 mm2
cap_ins_J4_2	j4_base	b3 b3 cyl. fit clr 0.000; flush
cap_ins_J4_2	j4_cap	b3 b3 planar mate 4.1 mm2
cap_ins_J4_3	j4_base	b3 b3 cyl. fit clr 0.000; flush
cap_ins_J4_3	j4_cap	b3 b3 planar mate 4.1 mm2
cap_ins_J6_0	j6_body	b5 b5 cyl. fit clr 0.000; flush
cap_ins_J6_0	j6_cap	b5 b5 planar mate 4.1 mm2
cap_ins_J6_1	j6_body	b5 b5 cyl. fit clr 0.000; flush
cap_ins_J6_1	j6_cap	b5 b5 planar mate 4.1 mm2
cap_ins_J6_2	j6_body	b5 b5 cyl. fit clr 0.000; flush
cap_ins_J6_2	j6_cap	b5 b5 planar mate 4.1 mm2
cap_ins_J6_3	j6_body	b5 b5 cyl. fit clr 0.000; flush
cap_ins_J6_3	j6_cap	b5 b5 planar mate 4.1 mm2
collar_clamp_158_+1	collar_nut_158_+1	b2 b2 cyl. fit clr 0.000
collar_clamp_158_+1	collar_upper_+y	b2 b2 planar mate 14.7 mm2
collar_clamp_158_-1	collar_nut_158_-1	b2 b2 cyl. fit clr 0.000
collar_clamp_158_-1	collar_upper_+y	b2 b2 planar mate 14.7 mm2
collar_clamp_299_+1	collar_fore_+y	b3 b3 planar mate 14.7 mm2
collar_clamp_299_+1	collar_nut_299_+1	b3 b3 cyl. fit clr 0.000
collar_clamp_299_-1	collar_fore_+y	b3 b3 planar mate 14.7 mm2

RULE 4 -- mate, never overlap: contacts, pathways, attachment (exact solids) (cont.)

collar_fore_+y	pin_J3_collar_+1_ins	b3 b3	flush; cyl. fit clr 0.000
collar_fore_-y	collar_nut_299_+1	b3 b3	planar mate 17.1 mm2
collar_fore_-y	collar_nut_299_-1	b3 b3	planar mate 17.1 mm2
collar_fore_-y	pin_J3_collar_-1_ins	b3 b3	cyl. fit clr 0.000
collar_ins	collar_pinch	b1 b1	cyl. fit clr 0.000
collar_ins	spigot_collar	b1 b1	cyl. fit clr 0.000; flush
collar_nut_158_+1	collar_upper_-y	b2 b2	planar mate 17.1 mm2
collar_nut_158_-1	collar_upper_-y	b2 b2	planar mate 17.1 mm2
collar_pinch	spigot_collar	b1 b1	planar mate 14.7 mm2
collar_upper_+y	pin_J2_collar_+1_ins	b2 b2	flush; cyl. fit clr 0.000
collar_upper_-y	pin_J2_collar_-1_ins	b2 b2	cyl. fit clr 0.000
ear_bolt_fore	ear_ins_fore	b3 b3	cyl. fit clr 0.000
ear_bolt_fore	link_fore_tongue	b3 b3	planar mate 10.2 mm2
ear_bolt_upper	ear_ins_upper	b2 b2	cyl. fit clr 0.000
ear_bolt_upper	link_upper_tongue	b2 b2	planar mate 10.2 mm2
ear_ins_fore	link_fore_groove	b3 b3	cyl. fit clr 0.000
ear_ins_fore	link_fore_tongue	b3 b3	planar mate 3.9 mm2
ear_ins_upper	link_upper_groove	b2 b2	cyl. fit clr 0.000
ear_ins_upper	link_upper_tongue	b2 b2	planar mate 3.9 mm2
foot_ins_0	j1_mount	b0 b0	cyl. fit clr 0.000; flush; planar mate 12.1 mm2
foot_ins_0	m4_base_foot_0	b0 b0	cyl. fit clr 0.000
foot_ins_1	j1_mount	b0 b0	cyl. fit clr 0.000; flush; planar mate 12.1 mm2
foot_ins_1	m4_base_foot_1	b0 b0	cyl. fit clr 0.000
foot_ins_2	j1_mount	b0 b0	cyl. fit clr 0.000; flush; planar mate 12.1 mm2
foot_ins_2	m4_base_foot_2	b0 b0	cyl. fit clr 0.000
foot_ins_3	j1_mount	b0 b0	cyl. fit clr 0.000; flush; planar mate 12.1 mm2
foot_ins_3	m4_base_foot_3	b0 b0	cyl. fit clr 0.000
forearm_ins_0	j4_base	b3 b3	planar mate 4.1 mm2
forearm_ins_0	j4base_forearm_0	b3 b3	cyl. fit clr 0.000
forearm_ins_0	link_fore_groove	b3 b3	cyl. fit clr 0.000; flush
forearm_ins_1	j4_base	b3 b3	planar mate 4.1 mm2
forearm_ins_1	j4base_forearm_1	b3 b3	cyl. fit clr 0.000
forearm_ins_1	link_fore_tongue	b3 b3	cyl. fit clr 0.000; flush
forearm_ins_2	j4_base	b3 b3	planar mate 4.1 mm2
forearm_ins_2	j4base_forearm_2	b3 b3	cyl. fit clr 0.000
forearm_ins_2	link_fore_groove	b3 b3	cyl. fit clr 0.000; flush
forearm_ins_3	j4_base	b3 b3	planar mate 4.1 mm2
forearm_ins_3	j4base_forearm_3	b3 b3	cyl. fit clr 0.000
forearm_ins_3	link_fore_tongue	b3 b3	cyl. fit clr 0.000; flush
horn_screw_J1_0	j1_hub	b1 b1	planar mate 10.9 mm2; cyl. fit clr 0.010
horn_screw_J1_0	servo_J1	b1 b0	cyl. fit clr 0.000
horn_screw_J1_180	j1_hub	b1 b1	planar mate 10.9 mm2; cyl. fit clr 0.010
horn_screw_J1_180	servo_J1	b1 b0	cyl. fit clr 0.000
horn_screw_J1_270	j1_hub	b1 b1	planar mate 10.9 mm2; cyl. fit clr 0.010
horn_screw_J1_270	servo_J1	b1 b0	cyl. fit clr 0.000
horn_screw_J1_90	j1_hub	b1 b1	planar mate 10.9 mm2; cyl. fit clr 0.010
horn_screw_J1_90	servo_J1	b1 b0	cyl. fit clr 0.000
horn_screw_J2_0	servo_J2	b2 b1	cyl. fit clr 0.000; flush
horn_screw_J2_180	servo_J2	b2 b1	cyl. fit clr 0.000; flush
horn_screw_J2_270	servo_J2	b2 b1	cyl. fit clr 0.000; flush
horn_screw_J2_90	servo_J2	b2 b1	cyl. fit clr 0.000; flush
horn_screw_J3_0	servo_J3	b3 b2	cyl. fit clr 0.000; flush
horn_screw_J3_180	servo_J3	b3 b2	cyl. fit clr 0.000; flush
horn_screw_J3_270	servo_J3	b3 b2	cyl. fit clr 0.000; flush
horn_screw_J3_90	servo_J3	b3 b2	cyl. fit clr 0.000; flush
horn_screw_J4_0	j4_hub	b4 b4	planar mate 10.9 mm2; cyl. fit clr 0.010
horn_screw_J4_0	servo_J4	b4 b3	cyl. fit clr 0.000
horn_screw_J4_180	j4_hub	b4 b4	planar mate 10.9 mm2; cyl. fit clr 0.010
horn_screw_J4_180	servo_J4	b4 b3	cyl. fit clr 0.000
horn_screw_J4_270	j4_hub	b4 b4	planar mate 10.9 mm2; cyl. fit clr 0.010
horn_screw_J4_270	servo_J4	b4 b3	cyl. fit clr 0.000

RULE 4 -- mate, never overlap: contacts, pathways, attachment (exact solids) (cont.)

horn_screw_J4_90	servo_J4	b4 b3 cyl. fit clr 0.000
horn_screw_J5_0	servo_J5	b5 b4 cyl. fit clr 0.000; flush
horn_screw_J5_180	servo_J5	b5 b4 cyl. fit clr 0.000; flush
horn_screw_J5_270	servo_J5	b5 b4 cyl. fit clr 0.000; flush
horn_screw_J5_90	servo_J5	b5 b4 cyl. fit clr 0.000; flush
horn_screw_J6_0	j6_flange	b6 b6 planar mate 10.9 mm2; cyl. fit clr 0.010
horn_screw_J6_0	servo_J6	b6 b5 cyl. fit clr 0.000
horn_screw_J6_180	j6_flange	b6 b6 planar mate 10.9 mm2; cyl. fit clr 0.010
horn_screw_J6_180	servo_J6	b6 b5 cyl. fit clr 0.000
horn_screw_J6_270	j6_flange	b6 b6 planar mate 10.9 mm2; cyl. fit clr 0.010
horn_screw_J6_270	servo_J6	b6 b5 cyl. fit clr 0.000
horn_screw_J6_90	j6_flange	b6 b6 planar mate 10.9 mm2; cyl. fit clr 0.010
horn_screw_J6_90	servo_J6	b6 b5 cyl. fit clr 0.000
j1_hub	servo_J1	b1 b0 planar mate 231.1 mm2
j1_mount	servo_J1	b0 b0 designed overlap 290.384 mm3 (ST3215 pinch (-0.22 channel), y 12.25..12.36); planar mate 6.3 mm2; planar mate 53.1 mm2;
j1_mount	servo_scr_J1_0	b0 b0 planar mate 4.8 mm2
j1_mount	servo_scr_J1_1	b0 b0 planar mate 4.8 mm2
j1_mount	servo_scr_J1_2	b0 b0 planar mate 4.8 mm2
j1_mount	servo_scr_J1_3	b0 b0 planar mate 4.8 mm2
j3fork_link_0	link_end_ins_0	b2 b2 cyl. fit clr 0.000
j3fork_link_1	link_end_ins_1	b2 b2 cyl. fit clr 0.000
j3fork_link_2	link_end_ins_2	b2 b2 cyl. fit clr 0.000
j3fork_link_3	link_end_ins_3	b2 b2 cyl. fit clr 0.000
j4_base	j4_cap	b3 b3 flush; flush; flush; planar mate 221.1 mm2; flush; planar mate 221.1 mm2
j4_base	j4base_forearm_0	b3 b3 planar mate 14.7 mm2
j4_base	j4base_forearm_1	b3 b3 planar mate 14.7 mm2
j4_base	j4base_forearm_2	b3 b3 planar mate 14.7 mm2
j4_base	j4base_forearm_3	b3 b3 planar mate 14.7 mm2
j4_base	link_fore_groove	b3 b3 planar mate 371.9 mm2
j4_base	link_fore_tongue	b3 b3 planar mate 394.1 mm2
j4_base	servo_J4	b3 b3 designed overlap 290.384 mm3 (ST3215 pinch (-0.22 channel), y 12.25..12.36); planar mate 6.3 mm2; planar mate 53.1 mm2;
j4_base	servo_scr_J4_0	b3 b3 planar mate 4.8 mm2
j4_base	servo_scr_J4_1	b3 b3 planar mate 4.8 mm2
j4_hub	j4hub_ins_0	b4 b4 flush; cyl. fit clr 0.000
j4_hub	j4hub_ins_1	b4 b4 flush; cyl. fit clr 0.000
j4_hub	j4hub_ins_2	b4 b4 flush; cyl. fit clr 0.000
j4_hub	j4hub_ins_3	b4 b4 flush; cyl. fit clr 0.000
j4_hub	servo_J4	b4 b3 planar mate 231.1 mm2
j4hub_ins_0	j5fork_j4hub_0	b4 b4 cyl. fit clr 0.000
j4hub_ins_1	j5fork_j4hub_1	b4 b4 cyl. fit clr 0.000
j4hub_ins_2	j5fork_j4hub_2	b4 b4 cyl. fit clr 0.000
j4hub_ins_3	j5fork_j4hub_3	b4 b4 cyl. fit clr 0.000
j5_grub	j6_body	b5 b5 designed overlap 6.725 mm3 (set screw forms its thread in the 02.6 pilot, r 1.30..1.50); edge/point; edge/point; edge/point
j6_body	j6_cap	b5 b5 flush; planar mate 4.0 mm2; planar mate 4.0 mm2; flush; flush; planar mate 63.4 mm2; flush; planar mate 250.4 mm2; planar mate
j6_body	servo_J6	b5 b5 designed overlap 290.384 mm3 (ST3215 pinch (-0.22 channel), y 12.25..12.36); planar mate 7.0 mm2; planar mate 53.1 mm2;
j6_body	servo_scr_J6_0	b5 b5 planar mate 4.8 mm2
j6_body	servo_scr_J6_1	b5 b5 planar mate 4.8 mm2
j6_flange	servo_J6	b6 b5 planar mate 231.1 mm2
j6_flange	tool_ins_135	b6 b6 flush; cyl. fit clr 0.000
j6_flange	tool_ins_225	b6 b6 flush; cyl. fit clr 0.000
j6_flange	tool_ins_315	b6 b6 flush; cyl. fit clr 0.000
j6_flange	tool_ins_45	b6 b6 flush; cyl. fit clr 0.000
link_end_ins_0	link_upper_groove	b2 b2 cyl. fit clr 0.000; flush
link_end_ins_1	link_upper_tongue	b2 b2 cyl. fit clr 0.000; flush
link_end_ins_2	link_upper_groove	b2 b2 cyl. fit clr 0.000; flush
link_end_ins_3	link_upper_tongue	b2 b2 cyl. fit clr 0.000; flush
link_fore_groove	link_fore_tongue	b3 b3 planar mate 387.3 mm2; planar mate 7.2 mm2; planar mate 23.5 mm2; planar mate 23.5 mm2; planar mate 23.5 mm2; planar mate
link_fore_groove	link_set_3_270	b3 b3 designed overlap 29.302 mm3 (set screw forms its thread in the 04.2 pilot, r 2.10..2.50); edge/point; edge/point
link_fore_groove	link_set_3_90	b3 b3 designed overlap 29.305 mm3 (set screw forms its thread in the 04.2 pilot, r 2.10..2.50); edge/point; edge/point
link_fore_groove	seam_ins_fore_0	b3 b3 cyl. fit clr 0.000
link_fore_groove	seam_ins_fore_1	b3 b3 cyl. fit clr 0.000

RULE 4 -- mate, never overlap: contacts, pathways, attachment (exact solids) (cont.)

link_fore_groove	seam_ins_fore_3	b3 b3 cyl. fit clr 0.000
link_fore_groove	seam_ins_fore_4	b3 b3 cyl. fit clr 0.000
link_fore_groove	seam_ins_fore_5	b3 b3 cyl. fit clr 0.000
link_fore_groove	shaft_clamp_3	b3 b3 cyl. fit clr 0.025; cyl. fit clr 0.025; flush
link_fore_tongue	link_set_4_270	b3 b3 designed overlap 29.302 mm3 (set screw forms its thread in the 04.2 pilot, r 2.10..2.50); edge/point; edge/point
link_fore_tongue	link_set_4_90	b3 b3 designed overlap 29.305 mm3 (set screw forms its thread in the 04.2 pilot, r 2.10..2.50); edge/point; edge/point
link_fore_tongue	seam_fore_0	b3 b3 planar mate 14.7 mm2
link_fore_tongue	seam_fore_1	b3 b3 planar mate 14.7 mm2
link_fore_tongue	seam_fore_2	b3 b3 planar mate 14.7 mm2
link_fore_tongue	seam_fore_3	b3 b3 planar mate 14.7 mm2
link_fore_tongue	seam_fore_4	b3 b3 planar mate 14.7 mm2
link_fore_tongue	seam_fore_5	b3 b3 planar mate 14.7 mm2
link_fore_tongue	seam_ins_fore_0	b3 b3 planar mate 4.1 mm2
link_fore_tongue	seam_ins_fore_1	b3 b3 planar mate 4.1 mm2
link_fore_tongue	seam_ins_fore_2	b3 b3 planar mate 4.1 mm2
link_fore_tongue	seam_ins_fore_3	b3 b3 planar mate 4.1 mm2
link_fore_tongue	seam_ins_fore_4	b3 b3 planar mate 4.1 mm2
link_fore_tongue	seam_ins_fore_5	b3 b3 planar mate 4.1 mm2
link_fore_tongue	shaft_clamp_4	b3 b3 flush; cyl. fit clr 0.025; cyl. fit clr 0.025
link_set_1_270	link_upper_groove	b2 b2 designed overlap 29.302 mm3 (set screw forms its thread in the 04.2 pilot, r 2.10..2.50); edge/point; edge/point
link_set_1_90	link_upper_groove	b2 b2 designed overlap 29.305 mm3 (set screw forms its thread in the 04.2 pilot, r 2.10..2.50); edge/point; edge/point
link_set_2_270	link_upper_tongue	b2 b2 designed overlap 29.302 mm3 (set screw forms its thread in the 04.2 pilot, r 2.10..2.50); edge/point; edge/point
link_set_2_90	link_upper_tongue	b2 b2 designed overlap 29.305 mm3 (set screw forms its thread in the 04.2 pilot, r 2.10..2.50); edge/point; edge/point
link_upper_groove	link_upper_tongue	b2 b2 planar mate 23.5 mm2; planar mate 23.5 mm2; planar mate 402.6 mm2; planar mate 23.5 mm2; planar mate 23.5 mm2; planar mate
link_upper_groove	seam_ins_upper_0	b2 b2 cyl. fit clr 0.000
link_upper_groove	seam_ins_upper_1	b2 b2 cyl. fit clr 0.000
link_upper_groove	seam_ins_upper_2	b2 b2 cyl. fit clr 0.000
link_upper_groove	seam_ins_upper_3	b2 b2 cyl. fit clr 0.000
link_upper_groove	shaft_clamp_1	b2 b2 cyl. fit clr 0.025; cyl. fit clr 0.025; flush
link_upper_tongue	seam_ins_upper_0	b2 b2 planar mate 4.1 mm2
link_upper_tongue	seam_ins_upper_1	b2 b2 planar mate 4.1 mm2
link_upper_tongue	seam_ins_upper_2	b2 b2 planar mate 4.1 mm2
link_upper_tongue	seam_ins_upper_3	b2 b2 planar mate 4.1 mm2
link_upper_tongue	seam_upper_0	b2 b2 planar mate 14.7 mm2
link_upper_tongue	seam_upper_1	b2 b2 planar mate 14.7 mm2
link_upper_tongue	seam_upper_2	b2 b2 planar mate 14.7 mm2
link_upper_tongue	seam_upper_3	b2 b2 planar mate 14.7 mm2
link_upper_tongue	shaft_clamp_2	b2 b2 flush; cyl. fit clr 0.025; cyl. fit clr 0.025
pin_J2_collar_+1	pin_J2_collar_+1_ins	b2 b2 cyl. fit clr 0.000
pin_J2_collar_+1	spring_J2_+1	b2 bNone cyl. fit clr 0.000; cyl. fit clr 0.000
pin_J2_collar_-1	pin_J2_collar_-1_ins	b2 b2 cyl. fit clr 0.000
pin_J2_collar_-1	spring_J2_-1	b2 bNone cyl. fit clr 0.000; cyl. fit clr 0.000
pin_J2_turret_+1	pin_J2_turret_+1_ins	b1 b1 cyl. fit clr 0.000
pin_J2_turret_+1	spring_J2_+1	b1 bNone cyl. fit clr 0.000; cyl. fit clr 0.000
pin_J2_turret_-1	pin_J2_turret_-1_ins	b1 b1 cyl. fit clr 0.000
pin_J2_turret_-1	spring_J2_-1	b1 bNone cyl. fit clr 0.000; cyl. fit clr 0.000
pin_J3_collar_+1	pin_J3_collar_+1_ins	b3 b3 cyl. fit clr 0.000
pin_J3_collar_+1	spring_J3_+1	b3 bNone cyl. fit clr 0.000; cyl. fit clr 0.000
pin_J3_collar_-1	pin_J3_collar_-1_ins	b3 b3 cyl. fit clr 0.000
pin_J3_collar_-1	spring_J3_-1	b3 bNone cyl. fit clr 0.000; cyl. fit clr 0.000
pin_J3_fork_+1	pin_J3_fork_+1_ins	b2 b2 cyl. fit clr 0.000
pin_J3_fork_+1	spring_J3_+1	b2 bNone cyl. fit clr 0.000; cyl. fit clr 0.000
pin_J3_fork_-1	pin_J3_fork_-1_ins	b2 b2 cyl. fit clr 0.000
pin_J3_fork_-1	spring_J3_-1	b2 bNone cyl. fit clr 0.000; cyl. fit clr 0.000
seam_fore_0	seam_ins_fore_1	b3 b3 cyl. fit clr 0.000
seam_fore_1	seam_ins_fore_0	b3 b3 cyl. fit clr 0.000
seam_fore_2	seam_ins_fore_3	b3 b3 cyl. fit clr 0.000
seam_fore_3	seam_ins_fore_2	b3 b3 cyl. fit clr 0.000
seam_fore_4	seam_ins_fore_5	b3 b3 cyl. fit clr 0.000
seam_fore_5	seam_ins_fore_4	b3 b3 cyl. fit clr 0.000
seam_ins_upper_0	seam_upper_1	b2 b2 cyl. fit clr 0.000

RULE 4 -- mate, never overlap: contacts, pathways, attachment (exact solids) (cont.)

seam_ins_upper_2	seam_upper_3	b2 b2	cyl. fit clr 0.000
seam_ins_upper_3	seam_upper_2	b2 b2	cyl. fit clr 0.000
servo_J1	servo_scr_J1_0	b0 b0	designed overlap 1.525 mm3 (self-tapping screw forms its thread in the servo's 02.0 back hole, r 1.00..1.10); edge/point; c
servo_J1	servo_scr_J1_1	b0 b0	designed overlap 1.525 mm3 (self-tapping screw forms its thread in the servo's 02.0 back hole, r 1.00..1.10); edge/point; c
servo_J1	servo_scr_J1_2	b0 b0	designed overlap 1.525 mm3 (self-tapping screw forms its thread in the servo's 02.0 back hole, r 1.00..1.10); edge/point; c
servo_J1	servo_scr_J1_3	b0 b0	designed overlap 1.525 mm3 (self-tapping screw forms its thread in the servo's 02.0 back hole, r 1.00..1.10); edge/point; c
servo_J2	servo_scr_J2_0	b1 b1	designed overlap 1.525 mm3 (self-tapping screw forms its thread in the servo's 02.0 back hole, r 1.00..1.10); edge/point; c
servo_J2	servo_scr_J2_1	b1 b1	designed overlap 1.525 mm3 (self-tapping screw forms its thread in the servo's 02.0 back hole, r 1.00..1.10); edge/point; c
servo_J2	servo_scr_J2_2	b1 b1	designed overlap 1.525 mm3 (self-tapping screw forms its thread in the servo's 02.0 back hole, r 1.00..1.10); edge/point; c
servo_J2	servo_scr_J2_3	b1 b1	designed overlap 1.525 mm3 (self-tapping screw forms its thread in the servo's 02.0 back hole, r 1.00..1.10); edge/point; c
servo_J3	servo_scr_J3_0	b2 b2	designed overlap 1.525 mm3 (self-tapping screw forms its thread in the servo's 02.0 back hole, r 1.00..1.10); edge/point; c
servo_J3	servo_scr_J3_1	b2 b2	designed overlap 1.525 mm3 (self-tapping screw forms its thread in the servo's 02.0 back hole, r 1.00..1.10); edge/point; c
servo_J3	servo_scr_J3_2	b2 b2	designed overlap 1.525 mm3 (self-tapping screw forms its thread in the servo's 02.0 back hole, r 1.00..1.10); edge/point; c
servo_J3	servo_scr_J3_3	b2 b2	designed overlap 1.525 mm3 (self-tapping screw forms its thread in the servo's 02.0 back hole, r 1.00..1.10); edge/point; c
servo_J4	servo_scr_J4_0	b3 b3	designed overlap 1.525 mm3 (self-tapping screw forms its thread in the servo's 02.0 back hole, r 1.00..1.10); edge/point; c
servo_J4	servo_scr_J4_1	b3 b3	designed overlap 1.525 mm3 (self-tapping screw forms its thread in the servo's 02.0 back hole, r 1.00..1.10); edge/point; c
servo_J5	servo_scr_J5_0	b4 b4	designed overlap 1.525 mm3 (self-tapping screw forms its thread in the servo's 02.0 back hole, r 1.00..1.10); edge/point; c
servo_J5	servo_scr_J5_1	b4 b4	designed overlap 1.525 mm3 (self-tapping screw forms its thread in the servo's 02.0 back hole, r 1.00..1.10); edge/point; c
servo_J5	servo_scr_J5_2	b4 b4	designed overlap 1.525 mm3 (self-tapping screw forms its thread in the servo's 02.0 back hole, r 1.00..1.10); edge/point; c
servo_J5	servo_scr_J5_3	b4 b4	designed overlap 1.525 mm3 (self-tapping screw forms its thread in the servo's 02.0 back hole, r 1.00..1.10); edge/point; c
servo_J6	servo_scr_J6_0	b5 b5	designed overlap 1.525 mm3 (self-tapping screw forms its thread in the servo's 02.0 back hole, r 1.00..1.10); edge/point; c
servo_J6	servo_scr_J6_1	b5 b5	designed overlap 1.525 mm3 (self-tapping screw forms its thread in the servo's 02.0 back hole, r 1.00..1.10); edge/point; c
split_J2_0	split_ins_J2_0	b1 b1	cyl. fit clr 0.000
split_J2_1	split_ins_J2_1	b1 b1	cyl. fit clr 0.000
split_J2_2	split_ins_J2_2	b1 b1	cyl. fit clr 0.000
split_J2_3	split_ins_J2_3	b1 b1	cyl. fit clr 0.000
split_J3_0	split_ins_J3_0	b2 b2	cyl. fit clr 0.000
split_J3_1	split_ins_J3_1	b2 b2	cyl. fit clr 0.000
split_J3_2	split_ins_J3_2	b2 b2	cyl. fit clr 0.000
split_J3_3	split_ins_J3_3	b2 b2	cyl. fit clr 0.000
split_J5_0	split_ins_J5_0	b4 b4	cyl. fit clr 0.000
split_J5_1	split_ins_J5_1	b4 b4	cyl. fit clr 0.000
split_J5_2	split_ins_J5_2	b4 b4	cyl. fit clr 0.000
split_J5_3	split_ins_J5_3	b4 b4	cyl. fit clr 0.000

SPLIT CLAMPS -- printed open, closed by their bolt: travel needed vs available

spigot collar + slit spigot (collet) on the J1 hub plug: clearances 0.075 + 0.100 mm -> slot travel 1.10 of 2.0, spigot slits 0.63 of 2.0 ok

collar_upper halves on the link_upper: back plates 0.200 + 0.200 mm off -> arm gap travel 0.40 of 0.50 ok

collar_fore halves on the link_fore: back plates 0.200 + 0.200 mm off -> arm gap travel 0.40 of 0.50 ok

BOLTED JOINTS -- the two parts a screw joins share a planar mate normal to the screw

cap_J4_0	head on j4_cap	holds in j4_base	clamped
cap_J4_1	head on j4_cap	holds in j4_base	clamped
cap_J4_2	head on j4_cap	holds in j4_base	clamped
cap_J4_3	head on j4_cap	holds in j4_base	clamped
cap_J6_0	head on j6_cap	holds in j6_body	clamped
cap_J6_1	head on j6_cap	holds in j6_body	clamped
cap_J6_2	head on j6_cap	holds in j6_body	clamped
cap_J6_3	head on j6_cap	holds in j6_body	clamped
collar_clamp_158_+1	head on collar_upper_+y	holds in collar_upper_-y	clamped
collar_clamp_158_-1	head on collar_upper_+y	holds in collar_upper_-y	clamped
collar_clamp_299_+1	head on collar_fore_+y	holds in collar_fore_-y	clamped
collar_clamp_299_-1	head on collar_fore_+y	holds in collar_fore_-y	clamped
collar_pinch	head on spigot_collar	holds in spigot_collar	same part
ear_bolt_fore	head on link_fore_tongue	holds in link_fore_groove	clamped
ear_bolt_upper	head on link_upper_tongue	holds in link_upper_groove	clamped
horn_screw_J1_0	head on j1_hub	holds in j1_hub,servo_J1	clamped
horn_screw_J1_180	head on j1_hub	holds in j1_hub,servo_J1	clamped
horn_screw_J1_270	head on j1_hub	holds in j1_hub,servo_J1	clamped
horn_screw_J1_90	head on j1_hub	holds in j1_hub,servo_J1	clamped
horn_screw_J2_0	head on J2 shaft	holds in J2 shaft,servo_J2	clamped

RULE 4 -- mate, never overlap: contacts, pathways, attachment (exact solids) (cont.)

horn_screw_J2_270	head on J2_shaft	holds in J2_shaft,servo_J2	clamped
horn_screw_J2_90	head on J2_shaft	holds in J2_shaft,servo_J2	clamped
horn_screw_J3_0	head on J3_shaft	holds in J3_shaft,servo_J3	clamped
horn_screw_J3_180	head on J3_shaft	holds in J3_shaft,servo_J3	clamped
horn_screw_J3_270	head on J3_shaft	holds in J3_shaft,servo_J3	clamped
horn_screw_J3_90	head on J3_shaft	holds in J3_shaft,servo_J3	clamped
horn_screw_J4_0	head on j4_hub	holds in j4_hub,servo_J4	clamped
horn_screw_J4_180	head on j4_hub	holds in j4_hub,servo_J4	clamped
horn_screw_J4_270	head on j4_hub	holds in j4_hub,servo_J4	clamped
horn_screw_J4_90	head on j4_hub	holds in j4_hub,servo_J4	clamped
horn_screw_J5_0	head on J5_shaft	holds in J5_shaft,servo_J5	clamped
horn_screw_J5_180	head on J5_shaft	holds in J5_shaft,servo_J5	clamped
horn_screw_J5_270	head on J5_shaft	holds in J5_shaft,servo_J5	clamped
horn_screw_J5_90	head on J5_shaft	holds in J5_shaft,servo_J5	clamped
horn_screw_J6_0	head on j6_flange	holds in j6_flange,servo_J6	clamped
horn_screw_J6_180	head on j6_flange	holds in j6_flange,servo_J6	clamped
horn_screw_J6_270	head on j6_flange	holds in j6_flange,servo_J6	clamped
horn_screw_J6_90	head on j6_flange	holds in j6_flange,servo_J6	clamped
j3fork_link_0	head on J3_p1	holds in link_upper_groove	clamped
j3fork_link_1	head on J3_p1	holds in link_upper_tongue	clamped
j3fork_link_2	head on J3_p1	holds in link_upper_groove	clamped
j3fork_link_3	head on J3_p1	holds in link_upper_tongue	clamped
j4base_forearm_0	head on j4_base	holds in link_fore_groove	clamped
j4base_forearm_1	head on j4_base	holds in link_fore_tongue	clamped
j4base_forearm_2	head on j4_base	holds in link_fore_groove	clamped
j4base_forearm_3	head on j4_base	holds in link_fore_tongue	clamped
j5fork_j4hub_0	head on J5_p1	holds in j4_hub	clamped
j5fork_j4hub_1	head on J5_p1	holds in j4_hub	clamped
j5fork_j4hub_2	head on J5_p1	holds in j4_hub	clamped
j5fork_j4hub_3	head on J5_p1	holds in j4_hub	clamped
m4_base_foot_0	head on base	holds in j1_mount	clamped
m4_base_foot_1	head on base	holds in j1_mount	clamped
m4_base_foot_2	head on base	holds in j1_mount	clamped
m4_base_foot_3	head on base	holds in j1_mount	clamped
seam_fore_0	head on link_fore_tongue	holds in link_fore_groove	clamped
seam_fore_1	head on link_fore_tongue	holds in link_fore_groove	clamped
seam_fore_2	head on link_fore_tongue	holds in link_fore_groove	clamped
seam_fore_3	head on link_fore_tongue	holds in link_fore_groove	clamped
seam_fore_4	head on link_fore_tongue	holds in link_fore_groove	clamped
seam_fore_5	head on link_fore_tongue	holds in link_fore_groove	clamped
seam_upper_0	head on link_upper_tongue	holds in link_upper_groove	clamped
seam_upper_1	head on link_upper_tongue	holds in link_upper_groove	clamped
seam_upper_2	head on link_upper_tongue	holds in link_upper_groove	clamped
seam_upper_3	head on link_upper_tongue	holds in link_upper_groove	clamped
servo_scr_J1_0	head on j1_mount	holds in servo_J1	clamped
servo_scr_J1_1	head on j1_mount	holds in servo_J1	clamped
servo_scr_J1_2	head on j1_mount	holds in servo_J1	clamped
servo_scr_J1_3	head on j1_mount	holds in servo_J1	clamped
servo_scr_J2_0	head on J2_turret_p2	holds in servo_J2	clamped
servo_scr_J2_1	head on J2_turret_p2	holds in servo_J2	clamped
servo_scr_J2_2	head on J2_turret_p2	holds in servo_J2	clamped
servo_scr_J2_3	head on J2_turret_p2	holds in servo_J2	clamped
servo_scr_J3_0	head on J3_p2	holds in servo_J3	clamped
servo_scr_J3_1	head on J3_p2	holds in servo_J3	clamped
servo_scr_J3_2	head on J3_p2	holds in servo_J3	clamped
servo_scr_J3_3	head on J3_p2	holds in servo_J3	clamped
servo_scr_J4_0	head on j4_base	holds in servo_J4	clamped
servo_scr_J4_1	head on j4_base	holds in servo_J4	clamped
servo_scr_J5_0	head on J5_p2	holds in servo_J5	clamped
servo_scr_J5_1	head on J5_p2	holds in servo_J5	clamped
servo_scr_J5_2	head on J5_p2	holds in servo_J5	clamped

RULE 4 -- mate, never overlap: contacts, pathways, attachment (exact solids) (cont.)

servo_scr_J6_0	head on j6_body	holds in servo_J6	clamped
servo_scr_J6_1	head on j6_body	holds in servo_J6	clamped
split_J2_0	head on J2_turret_p2	holds in J2_turret_p1	clamped
split_J2_1	head on J2_turret_p2	holds in J2_turret_p1	clamped
split_J2_2	head on J2_turret_p2	holds in J2_turret_p1	clamped
split_J2_3	head on J2_turret_p2	holds in J2_turret_p1	clamped
split_J3_0	head on J3_p2	holds in J3_p1	clamped
split_J3_1	head on J3_p2	holds in J3_p1	clamped
split_J3_2	head on J3_p2	holds in J3_p1	clamped
split_J3_3	head on J3_p2	holds in J3_p1	clamped
split_J5_0	head on J5_p2	holds in J5_p1	clamped
split_J5_1	head on J5_p2	holds in J5_p1	clamped
split_J5_2	head on J5_p2	holds in J5_p1	clamped
split_J5_3	head on J5_p2	holds in J5_p1	clamped

ATTACHED -- each body reached from what drives it, through touching parts only

body 0 (from j1_mount): 15 parts, 15 reached
body 1 (from servo_J1): 27 parts, 27 reached
body 2 (from servo_J2): 60 parts, 60 reached
body 3 (from servo_J3): 58 parts, 58 reached
body 4 (from servo_J4): 28 parts, 28 reached
body 5 (from servo_J5): 21 parts, 21 reached
body 6 (from servo_J6): 9 parts, 9 reached

RULE 4: 0 failure(s) (63 s)

CONTROLS

J3_p2 pushed 0.3 mm into J3_p1 (along the joint axis)	-> DETECTED
j4_cap lifted 0.2 mm off j4_base	-> DETECTED
upper link tongue moved 1.1 mm onto the J2 cheek	-> DETECTED

Manual check by slicing -- every pair measured in each joint's cut

Manual verification by slicing -- measured in each joint's section

Total clashes: 0

J1 base yaw (cut: point (np.float64(0.0), np.float64(0.0), np.float64(0.0)), normal (np.float64(0.0), np.float64(1.0), np.float64(0.0)); 10 parts in the cut)

part A	part B	overlap mm2	overlap thickness	gap	verdict
J2_turret_p1	base	0.00	0.000	0.50	gap 0.50
J2_turret_p1	brg_J1_z3	0.00	0.000	0.00	contact
J2_turret_p1	brg_J1_z43	0.00	0.000	0.00	contact
J2_turret_p1	j1_hub	0.00	0.000	0.12	gap 0.12
J2_turret_p1	spigot_collar	0.00	0.000	0.07	gap 0.07
base	brg_J1_z3	0.00	0.000	0.00	contact
base	brg_J1_z43	0.00	0.000	0.00	contact
base	spigot_collar	0.00	0.000	0.50	gap 0.50
brg_J1_z3	spigot_collar	0.00	0.000	0.00	contact
horn_screw_J1_0	j1_hub	0.00	0.000	0.00	contact
horn_screw_J1_0	servo_J1	0.00	0.000	0.00	contact
horn_screw_J1_180	j1_hub	0.00	0.000	0.00	contact
horn_screw_J1_180	servo_J1	0.00	0.000	0.00	contact
j1_hub	j1_mount	0.00	0.000	0.96	gap 0.96
j1_hub	servo_J1	0.00	0.000	0.00	contact
j1_hub	spigot_collar	0.00	0.000	0.76	gap 0.76
j1_mount	servo_J1	0.00	0.000	0.00	contact
j1_mount	spigot_collar	0.00	0.000	1.31	gap 1.31

J2 shoulder (cut: point (np.float64(0.0), np.float64(0.0), np.float64(90.0)), normal (np.float64(0.0), np.float64(0.0), np.float64(1.0)); 16 parts in the cut)

part A	part B	overlap mm2	overlap thickness	gap	verdict
J2_shaft	J2_turret_p1	0.00	0.000	0.54	gap 0.54
J2_shaft	brg_J2_drive	0.00	0.000	0.02	gap 0.02
J2_shaft	brg_J2_idle	0.00	0.000	0.02	gap 0.02
J2_shaft	horn_screw_J2_0	0.00	0.000	0.00	contact
J2_shaft	horn_screw_J2_180	0.00	0.000	0.00	contact
J2_shaft	link_set_1_270	0.00	0.000	0.00	contact
J2_shaft	link_set_1_90	0.00	0.000	0.00	contact
J2_shaft	link_set_2_270	0.00	0.000	0.00	contact
J2_shaft	link_set_2_90	0.00	0.000	0.00	contact
J2_shaft	servo_J2	0.00	0.000	0.00	contact
J2_shaft	shaft_clamp_1	0.00	0.000	0.79	gap 0.79
J2_shaft	shaft_clamp_2	0.00	0.000	0.79	gap 0.79
J2_turret_p1	J2_turret_p2	0.00	0.000	0.00	contact
J2_turret_p1	brg_J2_drive	0.00	0.000	0.00	contact
J2_turret_p1	brg_J2_idle	0.00	0.000	0.00	contact
J2_turret_p1	link_upper_groove	0.00	0.000	1.00	gap 1.00
J2_turret_p1	link_upper_tongue	0.00	0.000	1.00	gap 1.00
J2_turret_p1	servo_J2	0.00	0.000	0.00	contact
J2_turret_p2	servo_J2	0.00	0.000	0.40	gap 0.40
horn_screw_J2_0	servo_J2	0.00	0.000	0.00	contact
horn_screw_J2_180	servo_J2	0.00	0.000	0.00	contact
link_set_1_270	link_upper_groove	3.96	0.397	0.00	thread 0.40 (M5 set screw forms its thread in the link's 04.2 hole)
link_set_1_270	shaft_clamp_1	0.00	0.000	0.20	gap 0.20
link_set_1_90	link_upper_groove	3.96	0.397	0.00	thread 0.40 (M5 set screw forms its thread in the link's 04.2 hole)
link_set_1_90	shaft_clamp_1	0.00	0.000	0.20	gap 0.20
link_set_2_270	link_upper_tongue	3.96	0.397	0.00	thread 0.40 (M5 set screw forms its thread in the link's 04.2 hole)
link_set_2_270	shaft_clamp_2	0.00	0.000	0.20	gap 0.20
link_set_2_90	link_upper_tongue	3.96	0.397	0.00	thread 0.40 (M5 set screw forms its thread in the link's 04.2 hole)
link_set_2_90	shaft_clamp_2	0.00	0.000	0.20	gap 0.20
link_upper_groove	link_upper_tongue	0.00	0.000	0.00	contact

Manual check by slicing -- every pair measured in each joint's cut (cont.)

link_upper_groove shaft_clamp_2 0.00 0.000 0.74 gap 0.74
link_upper_tongue shaft_clamp_1 0.00 0.000 0.50 gap 0.50
link_upper_tongue shaft_clamp_2 0.00 0.000 0.02 gap 0.02
shaft_clamp_1 shaft_clamp_2 0.00 0.000 1.00 gap 1.00

J3 elbow (cut: point (np.float64(0.0), np.float64(0.0), np.float64(0.0)), normal (np.float64(1.0), np.float64(0.0), np.float64(0.0))); 30 parts in the cut)

part A	part B	overlap mm2	overlap thickness	gap	verdict
--- ---	--- ---	--- ---	--- ---		
J3_p1	J3_p2	0.00 0.000 0.00	contact		
J3_p1	J3_shaft	0.00 0.000 0.54	gap 0.54		
J3_p1	brg_J3_drive	0.00 0.000 0.00	contact		
J3_p1	brg_J3_idle	0.00 0.000 0.00	contact		
J3_p1	ear_bolt_fore	0.00 0.000 1.20	gap 1.20		
J3_p1	link_fore_groove	0.00 0.000 1.00	gap 1.00		
J3_p1	link_fore_tongue	0.00 0.000 1.00	gap 1.00		
J3_p1	link_upper_tongue	0.00 0.000 0.00	contact		
J3_p1	pin_J3_fork_+1	0.00 0.000 0.55	gap 0.55		
J3_p1	pin_J3_fork_+1_ins	0.00 0.000 0.00	contact		
J3_p1	pin_J3_fork_-1	0.00 0.000 0.55	gap 0.55		
J3_p1	pin_J3_fork_-1_ins	0.00 0.000 0.00	contact		
J3_p1	servo_J3	0.00 0.000 0.00	contact		
J3_p1	shaft_clamp_3	0.00 0.000 1.00	gap 1.00		
J3_p1	spring_J3_+1	0.00 0.000 1.51	gap 1.51		
J3_p1	spring_J3_-1	0.00 0.000 1.51	gap 1.51		
J3_p2	servo_J3	0.00 0.000 0.40	gap 0.40		
J3_shaft	brg_J3_drive	0.00 0.000 0.02	gap 0.02		
J3_shaft	brg_J3_idle	0.00 0.000 0.02	gap 0.02		
J3_shaft	horn_screw_J3_0	0.00 0.000 0.00	contact		
J3_shaft	horn_screw_J3_180	0.00 0.000 0.00	contact		
J3_shaft	link_fore_groove	0.00 0.000 1.02	gap 1.02		
J3_shaft	link_fore_tongue	0.00 0.000 1.02	gap 1.02		
J3_shaft	servo_J3	0.00 0.000 0.00	contact		
J3_shaft	shaft_clamp_3	0.23 0.016 0.00	fit/seat 0.016		
J3_shaft	shaft_clamp_4	0.23 0.016 0.00	fit/seat 0.016		
collar_upper_+y	link_upper_tongue	0.00 0.000 0.20	gap 0.20		
collar_upper_+y	pin_J2_collar_+1	0.00 0.000 0.30	gap 0.30		
collar_upper_+y	pin_J2_collar_+1_ins	0.00 0.000 0.00	contact		
collar_upper_+y	spring_J2_+1	0.00 0.000 1.51	gap 1.51		
collar_upper_-y	link_upper_groove	0.00 0.000 0.20	gap 0.20		
collar_upper_-y	pin_J2_collar_-1	0.00 0.000 0.30	gap 0.30		
collar_upper_-y	pin_J2_collar_-1_ins	0.00 0.000 0.00	contact		
collar_upper_-y	spring_J2_-1	0.00 0.000 1.51	gap 1.51		
ear_bolt_fore	ear_ins_fore	0.00 0.000 0.00	contact		
ear_bolt_fore	link_fore_groove	0.00 0.000 0.50	gap 0.50		
ear_bolt_fore	link_fore_tongue	0.00 0.000 0.00	contact		
ear_ins_fore	link_fore_groove	0.00 0.000 0.00	contact		
ear_ins_fore	link_fore_tongue	0.00 0.000 0.00	contact		
horn_screw_J3_0	servo_J3	0.00 0.000 0.00	contact		
horn_screw_J3_180	servo_J3	0.00 0.000 0.00	contact		
link_fore_groove	link_fore_tongue	0.00 0.000 0.00	contact		
link_fore_groove	shaft_clamp_3	0.00 0.000 0.08	gap 0.08		
link_fore_groove	shaft_clamp_4	0.00 0.000 0.77	gap 0.77		
link_fore_tongue	shaft_clamp_3	0.00 0.000 0.51	gap 0.51		
link_fore_tongue	shaft_clamp_4	0.00 0.000 0.08	gap 0.08		
link_upper_groove	link_upper_tongue	0.00 0.000 0.00	contact		
pin_J2_collar_+1	pin_J2_collar_+1_ins	0.00 0.000 0.00	contact		
pin_J2_collar_+1	spring_J2_+1	0.00 0.000 0.00	contact		
pin_J2_collar_-1	pin_J2_collar_-1_ins	0.00 0.000 0.00	contact		
pin_J2_collar_-1	spring_J2_-1	0.00 0.000 0.00	contact		
pin_J3_fork_+1	pin_J3_fork_+1_ins	0.00 0.000 0.00	contact		

Manual check by slicing -- every pair measured in each joint's cut (cont.)

```
| pin_J3_fork_-1 | pin_J3_fork_-1_ins | 0.00 | 0.000 | 0.00 | contact |
| pin_J3_fork_-1 | spring_J3_-1 | 0.00 | 0.000 | 0.00 | contact |
| shaft_clamp_3 | shaft_clamp_4 | 0.00 | 0.000 | 1.00 | gap 1.00 |
```

J4 forearm roll (cut: point (np.float64(0.0), np.float64(0.0), np.float64(0.0)), normal (np.float64(0.0), np.float64(1.0), np.float64(0.0)); 10 parts in the cut)

```
| part A | part B | overlap mm2 | overlap thickness | gap | verdict |
|---|---|---|---|---|---|
| J5_p1 | j4_hub | 0.00 | 0.000 | 0.00 | contact |
| brg_J4 | j4_cap | 0.00 | 0.000 | 0.00 | contact |
| brg_J4 | j4_hub | 0.00 | 0.000 | 0.00 | contact |
| horn_screw_J4_0 | j4_hub | 0.00 | 0.000 | 0.00 | contact |
| horn_screw_J4_0 | servo_J4 | 0.00 | 0.000 | 0.00 | contact |
| horn_screw_J4_180 | j4_hub | 0.00 | 0.000 | 0.00 | contact |
| horn_screw_J4_180 | servo_J4 | 0.00 | 0.000 | 0.00 | contact |
| j4_base | link_fore_tongue | 0.00 | 0.000 | 0.00 | contact |
| j4_base | servo_J4 | 0.00 | 0.000 | 0.00 | contact |
| j4_cap | j4_hub | 0.00 | 0.000 | 0.50 | gap 0.50 |
| j4_hub | servo_J4 | 0.00 | 0.000 | 0.00 | contact |
```

J5 wrist pitch (cut: point (np.float64(0.0), np.float64(0.0), np.float64(0.0)), normal (np.float64(1.0), np.float64(0.0), np.float64(0.0)); 13 parts in the cut)

```
| part A | part B | overlap mm2 | overlap thickness | gap | verdict |
|---|---|---|---|---|---|
| J5_p1 | J5_p2 | 0.00 | 0.000 | 0.00 | contact |
| J5_p1 | J5_shaft | 0.00 | 0.000 | 0.54 | gap 0.54 |
| J5_p1 | J5_spacer_B | 0.00 | 0.000 | 0.50 | gap 0.50 |
| J5_p1 | brg_J5_drive | 0.00 | 0.000 | 0.00 | contact |
| J5_p1 | brg_J5_idle | 0.00 | 0.000 | 0.00 | contact |
| J5_p1 | j4_hub | 0.00 | 0.000 | 0.00 | contact |
| J5_p1 | j6_body | 0.00 | 0.000 | 1.00 | gap 1.00 |
| J5_p1 | servo_J5 | 0.00 | 0.000 | 0.00 | contact |
| J5_p2 | servo_J5 | 0.00 | 0.000 | 0.40 | gap 0.40 |
| J5_shaft | J5_spacer_A | 0.00 | 0.000 | 0.12 | gap 0.12 |
| J5_shaft | J5_spacer_B | 0.00 | 0.000 | 0.12 | gap 0.12 |
| J5_shaft | brg_J5_drive | 0.00 | 0.000 | 0.02 | gap 0.02 |
| J5_shaft | brg_J5_idle | 0.00 | 0.000 | 0.02 | gap 0.02 |
| J5_shaft | horn_screw_J5_0 | 0.00 | 0.000 | 0.00 | contact |
| J5_shaft | horn_screw_J5_180 | 0.00 | 0.000 | 0.00 | contact |
| J5_shaft | j6_body | 0.00 | 0.000 | 0.09 | gap 0.09 |
| J5_shaft | servo_J5 | 0.00 | 0.000 | 0.00 | contact |
| J5_spacer_A | brg_J5_drive | 0.00 | 0.000 | 0.00 | contact |
| J5_spacer_A | j6_body | 0.00 | 0.000 | 0.00 | contact |
| J5_spacer_B | brg_J5_idle | 0.00 | 0.000 | 0.00 | contact |
| J5_spacer_B | j6_body | 0.00 | 0.000 | 0.00 | contact |
| horn_screw_J5_0 | servo_J5 | 0.00 | 0.000 | 0.00 | contact |
| horn_screw_J5_180 | servo_J5 | 0.00 | 0.000 | 0.00 | contact |
| j6_body | servo_J6 | 6.42 | 0.110 | 0.00 | fit/seat 0.110 |
```

J6 tool roll (cut: point (np.float64(0.0), np.float64(0.0), np.float64(0.0)), normal (np.float64(0.0), np.float64(1.0), np.float64(0.0)); 9 parts in the cut)

```
| part A | part B | overlap mm2 | overlap thickness | gap | verdict |
|---|---|---|---|---|---|
| J5_shaft | j6_body | 0.00 | 0.000 | 0.05 | gap 0.05 |
| brg_J6 | j6_cap | 0.00 | 0.000 | 0.00 | contact |
| brg_J6 | j6_flange | 0.00 | 0.000 | 0.00 | contact |
| horn_screw_J6_0 | j6_flange | 0.00 | 0.000 | 0.00 | contact |
| horn_screw_J6_0 | servo_J6 | 0.00 | 0.000 | 0.00 | contact |
| horn_screw_J6_180 | j6_flange | 0.00 | 0.000 | 0.00 | contact |
| horn_screw_J6_180 | servo_J6 | 0.00 | 0.000 | 0.00 | contact |
| j6_body | servo_J6 | 0.00 | 0.000 | 0.00 | contact |
```

Manual check by slicing -- every pair measured in each joint's cut (cont.)

| j6_flange | servo_J6 | 0.00 | 0.000 | 0.00 | contact |

Link lock J2/J3 -- M5 set screws on the double-D flats, centring rings, key access

```
ring 1 in link_upper_groove (axis [0. 1. 0.])
  1 ring x link 0.0000 mm3, x shaft 0.0000 mm3; touching link 0.025 / shaft 0.025 mm ok
  2 seam ear inside the ring gap: 420.9 mm3 (ring slides in over it) ok
  3 set screw 90 deg: link hole 04.24 (M5 self-tap), thread in link 5.6 mm, ring hole clears M5 True, shaft surface on the axis r 14.195 -> ON the flat ok
  3 set screw 270 deg: link hole 04.24 (M5 self-tap), thread in link 5.6 mm, ring hole clears M5 True, shaft surface on the axis r 14.195 -> ON the flat ok
ring 2 in link_upper_tongue (axis [ 0. -1. 0.])
  1 ring x link 0.0000 mm3, x shaft 0.0000 mm3; touching link 0.025 / shaft 0.025 mm ok
  2 seam ear inside the ring gap: 414.7 mm3 (ring slides in over it) ok
  3 set screw 90 deg: link hole 04.24 (M5 self-tap), thread in link 5.6 mm, ring hole clears M5 True, shaft surface on the axis r 14.195 -> ON the flat ok
  3 set screw 270 deg: link hole 04.24 (M5 self-tap), thread in link 5.6 mm, ring hole clears M5 True, shaft surface on the axis r 14.195 -> ON the flat ok
ring 3 in link_fore_groove (axis [0. 1. 0.])
  1 ring x link 0.0000 mm3, x shaft 0.0000 mm3; touching link 0.025 / shaft 0.025 mm ok
  2 seam ear inside the ring gap: 420.9 mm3 (ring slides in over it) ok
  3 set screw 90 deg: link hole 04.24 (M5 self-tap), thread in link 5.6 mm, ring hole clears M5 True, shaft surface on the axis r 14.195 -> ON the flat ok
  3 set screw 270 deg: link hole 04.24 (M5 self-tap), thread in link 5.6 mm, ring hole clears M5 True, shaft surface on the axis r 14.195 -> ON the flat ok
ring 4 in link_fore_tongue (axis [ 0. -1. 0.])
  1 ring x link 0.0000 mm3, x shaft 0.0000 mm3; touching link 0.025 / shaft 0.025 mm ok
  2 seam ear inside the ring gap: 414.7 mm3 (ring slides in over it) ok
  3 set screw 90 deg: link hole 04.24 (M5 self-tap), thread in link 5.6 mm, ring hole clears M5 True, shaft surface on the axis r 14.195 -> ON the flat ok
  3 set screw 270 deg: link hole 04.24 (M5 self-tap), thread in link 5.6 mm, ring hole clears M5 True, shaft surface on the axis r 14.195 -> ON the flat ok

4 hex key (03.2 path from the screw's outer end, 80 mm) in the whole assembled arm
  ring 1 screw 90: CLEAR
  ring 1 screw 270: CLEAR
  ring 2 screw 90: CLEAR
  ring 2 screw 270: CLEAR
  ring 3 screw 90: CLEAR
  ring 3 screw 270: CLEAR
  ring 4 screw 90: CLEAR
  ring 4 screw 270: CLEAR
```

LINK LOCK VERIFICATION: 0 failure(s)

6 DOF -- axes measured twice, Jacobian rank

1 JOINT AXES -- A: servo horn B: bearing pockets on the printed part

J1 base yaw axis [0. -0. -1.] through [0. 0. -14.31] | 3 pocket faces (base.step), coaxial to 0.0000 | servo vs bearings: 0.0000 deg, 0.0000 mm ok
J2 shoulder axis [0. -1. 0.] through [0. -28.5 90.] | 4 pocket faces (J2_turret_p1.step), coaxial to 0.0000 | servo vs bearings: 0.0000 deg, 0.0000 mm ok
J3 elbow axis [0. 1. 0.] through [0. 28.5 209.] | 3 pocket faces (J3_p1.step), coaxial to 0.0000 | servo vs bearings: 0.0000 deg, 0.0000 mm ok
J4 forearm roll axis [0. -0. -1.] through [0. 0. 378.59] | 1 pocket faces (j4_cap.step), coaxial to 0.0000 | servo vs bearings: 0.0000 deg, 0.0000 mm ok
J5 wrist pitch axis [0. 1. 0.] through [0. 26. 430.37] | 3 pocket faces (J5_p1.step), coaxial to 0.0000 | servo vs bearings: 0.0000 deg, 0.0000 mm ok
J6 tool roll axis [-0. -0. -1.] through [0. 0. 494.96] | 1 pocket faces (j6_cap.step), coaxial to 0.0000 | servo vs bearings: 0.0000 deg, 0.0000 mm ok

2 six distinct ST3215 (one per joint): True; swept ranges without a clash: J1 yaw +-90.0, J2 shoulder +-54.0, J3 elbow +-72.0, J4 roll +-90.0, J5 pitch +-46.5, J6 roll +-90.0

3 Jacobian rank over 3000 random poses inside the swept ranges (linear rows / 430 mm): rank 6 at 100.0 %; sigma_min/sigma_max min 3.96e-06, median 0.024
107 near-singular poses (sigma ratio < 1e-3): elbow straight (|J3| < 8 deg) 70; wrist aligned (|J5| < 8 deg) 28; wrist centre within 25 mm of the J1 axis 9; other 0
home (all 0: arm straight up) rank 3 sigma_min/sigma_max 0.00e+00
working pose (30,40,-70,20,35,10) rank 6 sigma_min/sigma_max 8.09e-02
J5 = 0 only (wrist aligned) rank 5 sigma_min/sigma_max 3.73e-17
J3 = 0 only (elbow straight) rank 5 sigma_min/sigma_max 3.75e-17
wrist: J4-J5 0.000 mm, J5-J6 0.000 mm, J4-J6 0.000 mm apart -> spherical wrist (axes meet in one point)

4 CONTROL: J5 axis moved onto J4 -> max rank 4 over 300 poses -> DETECTED

6-DOF CHECK: 0 failure(s)

Tool access -- a hex key reaches every screw at its build step

ok	m4_base_foot_2	step	3 (base + J1 bearings onto the foot): clear then; in the finished arm J2_turret_p2 cover(s) it
ok	m4_base_foot_3	step	3 (base + J1 bearings onto the foot): clear then; in the finished arm J2_turret_p2 cover(s) it
ok	ear_bolt_upper	step	5 (BENCH upper link): clear then; in the finished arm J2_turret_p1, brg_J2_idle cover(s) it
ok	seam_upper_0	step	5 (BENCH upper link): clear then; in the finished arm J2_turret_p1 cover(s) it
ok	seam_upper_1	step	5 (BENCH upper link): clear then; in the finished arm J2_turret_p1 cover(s) it
ok	horn_screw_J2_0	step	6 (BENCH J2 servo + shaft): clear then; in the finished arm J2_turret_p1 cover(s) it
ok	horn_screw_J2_180	step	6 (BENCH J2 servo + shaft): clear then; in the finished arm J2_turret_p1 cover(s) it
ok	horn_screw_J2_270	step	6 (BENCH J2 servo + shaft): clear then; in the finished arm J2_turret_p1 cover(s) it
ok	horn_screw_J2_90	step	6 (BENCH J2 servo + shaft): clear then; in the finished arm J2_turret_p1 cover(s) it
ok	j3fork_link_0	step	9 (J3 fork p1 onto the upper link): clear then; in the finished arm shaft_clamp_3, link_fore_groove, seam_ins_fore_0, link_set_3_270 cover(s) it
ok	j3fork_link_1	step	9 (J3 fork p1 onto the upper link): clear then; in the finished arm shaft_clamp_4, link_fore_tongue, seam_fore_1, link_set_4_90 cover(s) it
ok	j3fork_link_2	step	9 (J3 fork p1 onto the upper link): clear then; in the finished arm shaft_clamp_3, link_fore_groove, seam_ins_fore_1, link_set_3_90 cover(s) it
ok	j3fork_link_3	step	9 (J3 fork p1 onto the upper link): clear then; in the finished arm shaft_clamp_4, link_fore_tongue, seam_fore_0, link_set_4_270 cover(s) it
ok	ear_bolt_fore	step	10 (BENCH forearm): clear then; in the finished arm J3_p1, brg_J3_drive cover(s) it
ok	seam_fore_0	step	10 (BENCH forearm): clear then; in the finished arm J3_p1, J3_p2 cover(s) it
ok	seam_fore_1	step	10 (BENCH forearm): clear then; in the finished arm J3_p1, J3_p2 cover(s) it
ok	seam_fore_4	step	10 (BENCH forearm): clear then; in the finished arm collar_fore_+y cover(s) it
ok	seam_fore_5	step	10 (BENCH forearm): clear then; in the finished arm collar_fore_+y cover(s) it
ok	j4base_forearm_0	step	11 (BENCH J4 base on the forearm): clear then; in the finished arm j4_cap, j4_hub, brg_J4 cover(s) it
ok	j4base_forearm_1	step	11 (BENCH J4 base on the forearm): clear then; in the finished arm j4_cap, j4_hub, brg_J4 cover(s) it
ok	j4base_forearm_2	step	11 (BENCH J4 base on the forearm): clear then; in the finished arm j4_cap, j4_hub, servo_J4, brg_J4 cover(s) it
ok	j4base_forearm_3	step	11 (BENCH J4 base on the forearm): clear then; in the finished arm j4_cap, j4_hub, servo_J4, brg_J4 cover(s) it
ok	cap_J4_0	step	12 (BENCH J4 servo (2 screws from under the plate), bearing, cap): clear then; in the finished arm j4_hub, J5_p1, J5_shaft, j6_body, J5_spacer
ok	cap_J4_1	step	12 (BENCH J4 servo (2 screws from under the plate), bearing, cap): clear then; in the finished arm j4_hub, J5_p1, J5_shaft, j6_body, J5_spacer
ok	servo_scr_J4_0	step	12 (BENCH J4 servo (2 screws from under the plate), bearing, cap): clear then; in the finished arm collar_fore_-y cover(s) it
ok	servo_scr_J4_1	step	12 (BENCH J4 servo (2 screws from under the plate), bearing, cap): clear then; in the finished arm collar_fore_+y cover(s) it
ok	horn_screw_J4_0	step	13 (BENCH J4 hub onto the horn): clear then; in the finished arm J5_p1, J5_shaft, j6_body cover(s) it
ok	horn_screw_J4_180	step	13 (BENCH J4 hub onto the horn): clear then; in the finished arm J5_p1, J5_shaft, j6_body cover(s) it
ok	horn_screw_J4_270	step	13 (BENCH J4 hub onto the horn): clear then; in the finished arm J5_p1, J5_shaft, j6_body cover(s) it
ok	horn_screw_J4_90	step	13 (BENCH J4 hub onto the horn): clear then; in the finished arm J5_p1, J5_shaft, j6_body cover(s) it
ok	j5fork_j4hub_0	step	14 (BENCH J5 fork p1 onto the hub): clear then; in the finished arm j6_body cover(s) it
ok	j5fork_j4hub_1	step	14 (BENCH J5 fork p1 onto the hub): clear then; in the finished arm j6_body cover(s) it
ok	j5fork_j4hub_2	step	14 (BENCH J5 fork p1 onto the hub): clear then; in the finished arm j6_body, servo_J6 cover(s) it
ok	j5fork_j4hub_3	step	14 (BENCH J5 fork p1 onto the hub): clear then; in the finished arm j6_body, servo_J6 cover(s) it
ok	cap_J6_0	step	15 (BENCH blade: J6 servo, bearing, cap): clear then; in the finished arm j6_flange cover(s) it
ok	cap_J6_1	step	15 (BENCH blade: J6 servo, bearing, cap): clear then; in the finished arm j6_flange cover(s) it

TOOL ACCESS: 114 screws, 78 clear even in the finished arm, 36 clear at their build step (covered later), 0 BLOCKED at their step; 0 components outside the sequence

Every fastener as a solid: free, engaged, seated, tip clear

FASTENERS: 114 screws/grubs/pins, 61 inserts, 4 nuts, against 49 parts

1 cap_J4_0	free
1 cap_J4_1	free
1 cap_J4_2	free
1 cap_J4_3	free
1 cap_J6_0	free
1 cap_J6_1	free
1 cap_J6_2	free
1 cap_J6_3	free
1 collar_clamp_158_+1	free
1 collar_clamp_158_-1	free
1 collar_clamp_299_+1	free
1 collar_clamp_299_-1	free
1 collar_pinch	free
1 ear_bolt_fore	free
1 ear_bolt_upper	free
1 horn_screw_J1_0	free
1 horn_screw_J1_180	free
1 horn_screw_J1_270	free
1 horn_screw_J1_90	free
1 horn_screw_J2_0	free
1 horn_screw_J2_180	free
1 horn_screw_J2_270	free
1 horn_screw_J2_90	free
1 horn_screw_J3_0	free
1 horn_screw_J3_180	free
1 horn_screw_J3_270	free
1 horn_screw_J3_90	free
1 horn_screw_J4_0	free
1 horn_screw_J4_180	free
1 horn_screw_J4_270	free
1 horn_screw_J4_90	free
1 horn_screw_J5_0	free
1 horn_screw_J5_180	free
1 horn_screw_J5_270	free
1 horn_screw_J5_90	free
1 horn_screw_J6_0	free
1 horn_screw_J6_180	free
1 horn_screw_J6_270	free
1 horn_screw_J6_90	free
1 j3fork_link_0	free
1 j3fork_link_1	free
1 j3fork_link_2	free
1 j3fork_link_3	free
1 j4base_forearm_0	free
1 j4base_forearm_1	free
1 j4base_forearm_2	free
1 j4base_forearm_3	free
1 j5_grub	j6_body 6.725 mm3 (self-tapping, designed)
1 j5fork_j4hub_0	free
1 j5fork_j4hub_1	free
1 j5fork_j4hub_2	free
1 j5fork_j4hub_3	free
1 link_set_1_270	link_upper_groove 29.302 mm3 (self-tapping, designed)
1 link_set_1_90	link_upper_groove 29.305 mm3 (self-tapping, designed)
1 link_set_2_270	link_upper_tongue 29.302 mm3 (self-tapping, designed)
1 link_set_2_90	link_upper_tongue 29.305 mm3 (self-tapping, designed)
1 link_set_3_270	link_fore_groove 29.302 mm3 (self-tapping, designed)
1 link_set_3_90	link_fore_groove 29.305 mm3 (self-tapping, designed)
1 link_set_4_270	link_fore_tongue 29.302 mm3 (self-tapping, designed)

Every fastener as a solid: free, engaged, seated, tip clear (cont.)

1 m4_base_foot_0	free
1 m4_base_foot_1	free
1 m4_base_foot_2	free
1 m4_base_foot_3	free
1 pin_J2_collar_+1	free
1 pin_J2_collar_-1	free
1 pin_J2_turret_+1	free
1 pin_J2_turret_-1	free
1 pin_J3_collar_+1	free
1 pin_J3_collar_-1	free
1 pin_J3_fork_+1	free
1 pin_J3_fork_-1	free
1 seam_fore_0	free
1 seam_fore_1	free
1 seam_fore_2	free
1 seam_fore_3	free
1 seam_fore_4	free
1 seam_fore_5	free
1 seam_upper_0	free
1 seam_upper_1	free
1 seam_upper_2	free
1 seam_upper_3	free
1 servo_scr_J1_0	servo_J1 1.525 mm3 (self-tapping, designed)
1 servo_scr_J1_1	servo_J1 1.525 mm3 (self-tapping, designed)
1 servo_scr_J1_2	servo_J1 1.525 mm3 (self-tapping, designed)
1 servo_scr_J1_3	servo_J1 1.525 mm3 (self-tapping, designed)
1 servo_scr_J2_0	servo_J2 1.525 mm3 (self-tapping, designed)
1 servo_scr_J2_1	servo_J2 1.525 mm3 (self-tapping, designed)
1 servo_scr_J2_2	servo_J2 1.525 mm3 (self-tapping, designed)
1 servo_scr_J2_3	servo_J2 1.525 mm3 (self-tapping, designed)
1 servo_scr_J3_0	servo_J3 1.525 mm3 (self-tapping, designed)
1 servo_scr_J3_1	servo_J3 1.525 mm3 (self-tapping, designed)
1 servo_scr_J3_2	servo_J3 1.525 mm3 (self-tapping, designed)
1 servo_scr_J3_3	servo_J3 1.525 mm3 (self-tapping, designed)
1 servo_scr_J4_0	servo_J4 1.525 mm3 (self-tapping, designed)
1 servo_scr_J4_1	servo_J4 1.525 mm3 (self-tapping, designed)
1 servo_scr_J5_0	servo_J5 1.525 mm3 (self-tapping, designed)
1 servo_scr_J5_1	servo_J5 1.525 mm3 (self-tapping, designed)
1 servo_scr_J5_2	servo_J5 1.525 mm3 (self-tapping, designed)
1 servo_scr_J5_3	servo_J5 1.525 mm3 (self-tapping, designed)
1 servo_scr_J6_0	servo_J6 1.525 mm3 (self-tapping, designed)
1 servo_scr_J6_1	servo_J6 1.525 mm3 (self-tapping, designed)
1 split_J2_0	free
1 split_J2_1	free
1 split_J2_2	free
1 split_J2_3	free
1 split_J3_0	free
1 split_J3_1	free
1 split_J3_2	free
1 split_J3_3	free
1 split_J5_0	free
1 split_J5_1	free
1 split_J5_2	free
1 split_J5_3	free

CLAMP / THREAD ENGAGEMENT, HEAD SEAT, TIP

cap_J4_0	engaged	5.7 mm in cap_ins_J4_0	head on j4_cap	tip free	ok
cap_J4_1	engaged	5.7 mm in cap_ins_J4_1	head on j4_cap	tip free	ok
cap_J4_2	engaged	5.7 mm in cap_ins_J4_2	head on j4_cap	tip free	ok
cap_J4_3	engaged	5.7 mm in cap_ins_J4_3	head on j4_cap	tip free	ok
cap_J6_0	engaged	5.7 mm in cap_ins_J6_0	head on j6_cap	tip free	ok

Every fastener as a solid: free, engaged, seated, tip clear (cont.)

cap_J6_2	engaged	5.7 mm in cap_ins_J6_2	head on j6_cap	tip free	ok
cap_J6_3	engaged	5.7 mm in cap_ins_J6_3	head on j6_cap	tip free	ok
collar_clamp_158_+1	engaged	4.0 mm in collar_nut_158_+1	head on collar_upper_+y	tip free	ok
collar_clamp_158_-1	engaged	4.0 mm in collar_nut_158_-1	head on collar_upper_+y	tip free	ok
collar_clamp_299_+1	engaged	4.0 mm in collar_nut_299_+1	head on collar_fore_+y	tip free	ok
collar_clamp_299_-1	engaged	4.0 mm in collar_nut_299_-1	head on collar_fore_+y	tip free	ok
collar_pinch	engaged	3.1 mm in collar_ins	head on spigot_collar	tip free	ok
ear_bolt_fore	engaged	4.0 mm in ear_ins_fore	head on link_fore_tongue	tip free	ok
ear_bolt_upper	engaged	4.0 mm in ear_ins_upper	head on link_upper_tongue	tip free	ok
horn_screw_J1_0	engaged	3.0 mm in servo_J1	head on j1_hub	tip free	ok
horn_screw_J1_180	engaged	3.0 mm in servo_J1	head on j1_hub	tip free	ok
horn_screw_J1_270	engaged	3.0 mm in servo_J1	head on j1_hub	tip free	ok
horn_screw_J1_90	engaged	3.0 mm in servo_J1	head on j1_hub	tip free	ok
horn_screw_J2_0	engaged	2.5 mm in servo_J2	head on J2_shaft	tip free	ok
horn_screw_J2_180	engaged	2.5 mm in servo_J2	head on J2_shaft	tip free	ok
horn_screw_J2_270	engaged	2.5 mm in servo_J2	head on J2_shaft	tip free	ok
horn_screw_J2_90	engaged	2.5 mm in servo_J2	head on J2_shaft	tip free	ok
horn_screw_J3_0	engaged	2.5 mm in servo_J3	head on J3_shaft	tip free	ok
horn_screw_J3_180	engaged	2.5 mm in servo_J3	head on J3_shaft	tip free	ok
horn_screw_J3_270	engaged	2.5 mm in servo_J3	head on J3_shaft	tip free	ok
horn_screw_J3_90	engaged	2.5 mm in servo_J3	head on J3_shaft	tip free	ok
horn_screw_J4_0	engaged	3.0 mm in servo_J4	head on j4_hub	tip free	ok
horn_screw_J4_180	engaged	3.0 mm in servo_J4	head on j4_hub	tip free	ok
horn_screw_J4_270	engaged	3.0 mm in servo_J4	head on j4_hub	tip free	ok
horn_screw_J4_90	engaged	3.0 mm in servo_J4	head on j4_hub	tip free	ok
horn_screw_J5_0	engaged	2.5 mm in servo_J5	head on J5_shaft	tip free	ok
horn_screw_J5_180	engaged	2.5 mm in servo_J5	head on J5_shaft	tip free	ok
horn_screw_J5_270	engaged	2.5 mm in servo_J5	head on J5_shaft	tip free	ok
horn_screw_J5_90	engaged	2.5 mm in servo_J5	head on J5_shaft	tip free	ok
horn_screw_J6_0	engaged	3.0 mm in servo_J6	head on j6_flange	tip free	ok
horn_screw_J6_180	engaged	3.0 mm in servo_J6	head on j6_flange	tip free	ok
horn_screw_J6_270	engaged	3.0 mm in servo_J6	head on j6_flange	tip free	ok
horn_screw_J6_90	engaged	3.0 mm in servo_J6	head on j6_flange	tip free	ok
j3fork_link_0	engaged	5.2 mm in link_end_ins_0	head on J3_p1	tip free	ok
j3fork_link_1	engaged	5.2 mm in link_end_ins_1	head on J3_p1	tip free	ok
j3fork_link_2	engaged	5.2 mm in link_end_ins_2	head on J3_p1	tip free	ok
j3fork_link_3	engaged	5.2 mm in link_end_ins_3	head on J3_p1	tip free	ok
j4base_forearm_0	engaged	5.7 mm in forearm_ins_0	head on j4_base	tip free	ok
j4base_forearm_1	engaged	5.7 mm in forearm_ins_1	head on j4_base	tip free	ok
j4base_forearm_2	engaged	5.7 mm in forearm_ins_2	head on j4_base	tip free	ok
j4base_forearm_3	engaged	5.7 mm in forearm_ins_3	head on j4_base	tip free	ok
j5_grub	engaged	3.8 mm in j6_body	head on (grub)	tip on J5_shaft	ok
j5fork_j4hub_0	engaged	5.2 mm in j4hub_ins_0	head on J5_p1	tip free	ok
j5fork_j4hub_1	engaged	5.2 mm in j4hub_ins_1	head on J5_p1	tip free	ok
j5fork_j4hub_2	engaged	5.2 mm in j4hub_ins_2	head on J5_p1	tip free	ok
j5fork_j4hub_3	engaged	5.2 mm in j4hub_ins_3	head on J5_p1	tip free	ok
link_set_1_270	engaged	5.0 mm in link_upper_groove	head on link_upper_groove	tip on J2_shaft	ok
link_set_1_90	engaged	5.0 mm in link_upper_groove	head on link_upper_groove	tip on J2_shaft	ok
link_set_2_270	engaged	5.0 mm in link_upper_tongue	head on link_upper_tongue	tip on J2_shaft	ok
link_set_2_90	engaged	5.0 mm in link_upper_tongue	head on link_upper_tongue	tip on J2_shaft	ok
link_set_3_270	engaged	5.0 mm in link_fore_groove	head on link_fore_groove	tip on J3_shaft	ok
link_set_3_90	engaged	5.0 mm in link_fore_groove	head on link_fore_groove	tip on J3_shaft	ok
link_set_4_270	engaged	5.0 mm in link_fore_tongue	head on link_fore_tongue	tip on J3_shaft	ok
link_set_4_90	engaged	5.0 mm in link_fore_tongue	head on link_fore_tongue	tip on J3_shaft	ok
m4_base_foot_0	engaged	6.5 mm in foot_ins_0	head on base	tip free	ok
m4_base_foot_1	engaged	6.5 mm in foot_ins_1	head on base	tip free	ok
m4_base_foot_2	engaged	6.5 mm in foot_ins_2	head on base	tip free	ok
m4_base_foot_3	engaged	6.5 mm in foot_ins_3	head on base	tip free	ok
pin_J2_collar_+1	engaged	5.5 mm in pin_J2_collar_+1_ins	head on (pin: spring side)	tip free	ok
pin_J2_collar_-1	engaged	5.5 mm in pin_J2_collar_-1_ins	head on (pin: spring side)	tip free	ok
pin_J2_turret_+1	engaged	5.5 mm in pin_J2_turret_+1_ins	head on (pin: spring side)	tip free	ok

Every fastener as a solid: free, engaged, seated, tip clear (cont.)

pin_J3_collar_+1	engaged	5.5 mm	in pin_J3_collar_+1_ins	head on (pin: spring side)	tip free	ok
pin_J3_collar_-1	engaged	5.5 mm	in pin_J3_collar_-1_ins	head on (pin: spring side)	tip free	ok
pin_J3_fork_+1	engaged	5.5 mm	in pin_J3_fork_+1_ins	head on (pin: spring side)	tip free	ok
pin_J3_fork_-1	engaged	5.5 mm	in pin_J3_fork_-1_ins	head on (pin: spring side)	tip free	ok
seam_fore_0	engaged	4.7 mm	in seam_ins_fore_1	head on link_fore_tongue	tip free	ok
seam_fore_1	engaged	4.7 mm	in seam_ins_fore_0	head on link_fore_tongue	tip free	ok
seam_fore_2	engaged	4.7 mm	in seam_ins_fore_3	head on link_fore_tongue	tip free	ok
seam_fore_3	engaged	4.7 mm	in seam_ins_fore_2	head on link_fore_tongue	tip free	ok
seam_fore_4	engaged	4.7 mm	in seam_ins_fore_5	head on link_fore_tongue	tip free	ok
seam_fore_5	engaged	4.7 mm	in seam_ins_fore_4	head on link_fore_tongue	tip free	ok
seam_upper_0	engaged	4.7 mm	in seam_ins_upper_1	head on link_upper_tongue	tip free	ok
seam_upper_1	engaged	4.7 mm	in seam_ins_upper_0	head on link_upper_tongue	tip free	ok
seam_upper_2	engaged	4.7 mm	in seam_ins_upper_3	head on link_upper_tongue	tip free	ok
seam_upper_3	engaged	4.7 mm	in seam_ins_upper_2	head on link_upper_tongue	tip free	ok
servo_scr_J1_0	engaged	4.4 mm	in servo_J1	head on j1_mount	tip free	ok
servo_scr_J1_1	engaged	2.6 mm	in servo_J1	head on j1_mount	tip free	ok
servo_scr_J1_2	engaged	2.6 mm	in servo_J1	head on j1_mount	tip free	ok
servo_scr_J1_3	engaged	2.6 mm	in servo_J1	head on j1_mount	tip free	ok
servo_scr_J2_0	engaged	2.6 mm	in servo_J2	head on J2_turret_p2	tip free	ok
servo_scr_J2_1	engaged	2.6 mm	in servo_J2	head on J2_turret_p2	tip free	ok
servo_scr_J2_2	engaged	2.6 mm	in servo_J2	head on J2_turret_p2	tip free	ok
servo_scr_J2_3	engaged	2.6 mm	in servo_J2	head on J2_turret_p2	tip free	ok
servo_scr_J3_0	engaged	4.4 mm	in servo_J3	head on J3_p2	tip free	ok
servo_scr_J3_1	engaged	2.6 mm	in servo_J3	head on J3_p2	tip free	ok
servo_scr_J3_2	engaged	2.6 mm	in servo_J3	head on J3_p2	tip free	ok
servo_scr_J3_3	engaged	2.6 mm	in servo_J3	head on J3_p2	tip free	ok
servo_scr_J4_0	engaged	2.6 mm	in servo_J4	head on j4_base	tip free	ok
servo_scr_J4_1	engaged	2.6 mm	in servo_J4	head on j4_base	tip free	ok
servo_scr_J5_0	engaged	4.4 mm	in servo_J5	head on J5_p2	tip free	ok
servo_scr_J5_1	engaged	2.6 mm	in servo_J5	head on J5_p2	tip free	ok
servo_scr_J5_2	engaged	2.6 mm	in servo_J5	head on J5_p2	tip free	ok
servo_scr_J5_3	engaged	2.6 mm	in servo_J5	head on J5_p2	tip free	ok
servo_scr_J6_0	engaged	2.6 mm	in servo_J6	head on j6_body	tip free	ok
servo_scr_J6_1	engaged	2.6 mm	in servo_J6	head on j6_body	tip free	ok
split_J2_0	engaged	5.7 mm	in split_ins_J2_0	head on J2_turret_p2	tip free	ok
split_J2_1	engaged	5.7 mm	in split_ins_J2_1	head on J2_turret_p2	tip free	ok
split_J2_2	engaged	5.7 mm	in split_ins_J2_2	head on J2_turret_p2	tip free	ok
split_J2_3	engaged	5.7 mm	in split_ins_J2_3	head on J2_turret_p2	tip free	ok
split_J3_0	engaged	5.7 mm	in split_ins_J3_0	head on J3_p2	tip free	ok
split_J3_1	engaged	5.7 mm	in split_ins_J3_1	head on J3_p2	tip free	ok
split_J3_2	engaged	5.7 mm	in split_ins_J3_2	head on J3_p2	tip free	ok
split_J3_3	engaged	5.7 mm	in split_ins_J3_3	head on J3_p2	tip free	ok
split_J5_0	engaged	5.7 mm	in split_ins_J5_0	head on J5_p2	tip free	ok
split_J5_1	engaged	5.7 mm	in split_ins_J5_1	head on J5_p2	tip free	ok
split_J5_2	engaged	5.7 mm	in split_ins_J5_2	head on J5_p2	tip free	ok
split_J5_3	engaged	5.7 mm	in split_ins_J5_3	head on J5_p2	tip free	ok

INSERTS / NUTS in their pockets

cap_ins_J4_0	host j4_base	material round	it 100 %	ok
cap_ins_J4_1	host j4_base	material round	it 100 %	ok
cap_ins_J4_2	host j4_base	material round	it 100 %	ok
cap_ins_J4_3	host j4_base	material round	it 100 %	ok
cap_ins_J6_0	host j6_body	material round	it 100 %	ok
cap_ins_J6_1	host j6_body	material round	it 100 %	ok
cap_ins_J6_2	host j6_body	material round	it 100 %	ok
cap_ins_J6_3	host j6_body	material round	it 100 %	ok
collar_ins	host spigot_collar	material round	it 100 %	ok
collar_nut_158_+1	host collar_upper_-y	material round	it 100 %	ok
collar_nut_158_-1	host collar_upper_-y	material round	it 100 %	ok
collar_nut_299_+1	host collar_fore_-y	material round	it 100 %	ok
collar_nut_299_-1	host collar_fore_-y	material round	it 100 %	ok

Every fastener as a solid: free, engaged, seated, tip clear (cont.)

ear_ins_upper	host link_upper_groove	material round it 100 % ok
foot_ins_0	host j1_mount	material round it 100 % ok
foot_ins_1	host j1_mount	material round it 100 % ok
foot_ins_2	host j1_mount	material round it 100 % ok
foot_ins_3	host j1_mount	material round it 100 % ok
forearm_ins_0	host link_fore_groove	material round it 100 % ok
forearm_ins_1	host link_fore_tongue	material round it 100 % ok
forearm_ins_2	host link_fore_groove	material round it 100 % ok
forearm_ins_3	host link_fore_tongue	material round it 100 % ok
j4hub_ins_0	host j4_hub	material round it 100 % ok
j4hub_ins_1	host j4_hub	material round it 100 % ok
j4hub_ins_2	host j4_hub	material round it 100 % ok
j4hub_ins_3	host j4_hub	material round it 100 % ok
link_end_ins_0	host link_upper_groove	material round it 100 % ok
link_end_ins_1	host link_upper_tongue	material round it 100 % ok
link_end_ins_2	host link_upper_groove	material round it 100 % ok
link_end_ins_3	host link_upper_tongue	material round it 100 % ok
pin_J2_collar_+1_ins	host collar_upper_+y	material round it 100 % ok
pin_J2_collar_-1_ins	host collar_upper_-y	material round it 100 % ok
pin_J2_turret_+1_ins	host J2_turret_p1	material round it 100 % ok
pin_J2_turret_-1_ins	host J2_turret_p1	material round it 100 % ok
pin_J3_collar_+1_ins	host collar_fore_+y	material round it 100 % ok
pin_J3_collar_-1_ins	host collar_fore_-y	material round it 100 % ok
pin_J3_fork_+1_ins	host J3_p1	material round it 100 % ok
pin_J3_fork_-1_ins	host J3_p1	material round it 100 % ok
seam_ins_fore_0	host link_fore_groove	material round it 100 % ok
seam_ins_fore_1	host link_fore_groove	material round it 100 % ok
seam_ins_fore_2	host link_fore_groove	material round it 100 % ok
seam_ins_fore_3	host link_fore_groove	material round it 100 % ok
seam_ins_fore_4	host link_fore_groove	material round it 100 % ok
seam_ins_fore_5	host link_fore_groove	material round it 100 % ok
seam_ins_upper_0	host link_upper_groove	material round it 100 % ok
seam_ins_upper_1	host link_upper_groove	material round it 100 % ok
seam_ins_upper_2	host link_upper_groove	material round it 100 % ok
seam_ins_upper_3	host link_upper_groove	material round it 100 % ok
split_ins_J2_0	host J2_turret_p1	material round it 100 % ok
split_ins_J2_1	host J2_turret_p1	material round it 100 % ok
split_ins_J2_2	host J2_turret_p1	material round it 100 % ok
split_ins_J2_3	host J2_turret_p1	material round it 100 % ok
split_ins_J3_0	host J3_p1	material round it 100 % ok
split_ins_J3_1	host J3_p1	material round it 100 % ok
split_ins_J3_2	host J3_p1	material round it 100 % ok
split_ins_J3_3	host J3_p1	material round it 100 % ok
split_ins_J5_0	host J5_p1	material round it 100 % ok
split_ins_J5_1	host J5_p1	material round it 100 % ok
split_ins_J5_2	host J5_p1	material round it 100 % ok
split_ins_J5_3	host J5_p1	material round it 100 % ok
tool_ins_135	host j6_flange	material round it 100 % ok
tool_ins_225	host j6_flange	material round it 100 % ok
tool_ins_315	host j6_flange	material round it 100 % ok
tool_ins_45	host j6_flange	material round it 100 % ok

FASTENER CHECK: 0 failure(s)

Parameter -> CAD audit (generator value vs measured file)

Parameter -> CAD audit

Generator value vs measurement of the exported file.

part	parameter	intended	measured	result
---	---	---	---	---
J3_p2	pinch channel width (y)	24.500	24.500	PASS
J3_p2	bay length (x)	46.020	46.000	PASS
J3_p2	bay depth P -> back wall	34.989	35.000	PASS
J3_p2	wiring window length (x)	7.500	7.500	PASS
J3_p2	wiring window width (y)	22.000	22.000	PASS
J3_p2	servo screw pad face = back plateau (8.30,+10.25)	32.000	32.001	PASS
J3_p2	servo screw clearance 0 (8.30,+10.25)	3.400	3.395	PASS
J3_p2	servo screw head seat above the seat face (8.30,+10.25)	37.389	37.386	PASS
J3_p2	servo screw pad face = back plateau (8.30,-10.25)	32.000	32.001	PASS
J3_p2	servo screw clearance 0 (8.30,-10.25)	3.400	3.395	PASS
J3_p2	servo screw head seat above the seat face (8.30,-10.25)	37.389	37.386	PASS
J3_p2	servo screw pad face = back plateau (32.75,+10.25)	32.000	32.001	PASS
J3_p2	servo screw clearance 0 (32.75,+10.25)	3.400	3.395	PASS
J3_p2	servo screw head seat above the seat face (32.75,+10.25)	37.389	37.386	PASS
J3_p2	servo screw pad face = back plateau (32.75,-10.25)	32.000	32.001	PASS
J3_p2	servo screw clearance 0 (32.75,-10.25)	3.400	3.395	PASS
J3_p2	servo screw head seat above the seat face (32.75,-10.25)	37.389	37.386	PASS
J3_p1	drive bearing pocket 0	42.020	42.030	PASS
J3_p1	idle bearing pocket 0	42.020	42.030	PASS
J3_p1	drive pocket open to the gap (0 at z_in+0.6)	42.020	42.030	PASS
J3_p1	floor shaft bore 0	31.000	31.026	PASS
J3_p1	servo seat plane P (floor top, x=30,y=10)	28.500	28.495	PASS
J3_p1	horn/step relief depth below P	1.400	1.400	PASS
J3_p1	split insert pocket depth at (-20,+16)	8.100	8.105	PASS
J3_p2	split bolt hole open P..back face (void fraction) (-20,+16)	1.000	1.000	PASS
J3_p1	split insert pocket depth at (-20,-16)	8.100	8.105	PASS
J3_p2	split bolt hole open P..back face (void fraction) (-20,-16)	1.000	1.000	PASS
J3_p1	split insert pocket depth at (+26,+16)	8.100	8.105	PASS
J3_p2	split bolt hole open P..back face (void fraction) (+26,+16)	1.000	1.000	PASS
J3_p1	split insert pocket depth at (+26,-16)	8.100	8.105	PASS
J3_p2	split bolt hole open P..back face (void fraction) (+26,-16)	1.000	1.000	PASS
J3_shaft	shaft OD	29.950	29.948	PASS
J3_shaft	shaft bore	22.000	22.000	PASS
J3_shaft	set-screw flat distance from axis (normal at -90 deg)	14.200	14.199	PASS
J3_shaft	set-screw flat distance from axis (normal at +90 deg)	14.200	14.199	PASS
J3_shaft	horn face = end of shaft	25.489	25.489	PASS
J3_shaft	horn hole 0 at BCD14, 0 deg	2.520	2.520	PASS
J3_shaft	horn hole 0 at BCD14, 90 deg	2.520	2.520	PASS
J3_shaft	horn hole 0 at BCD14, 180 deg	2.520	2.520	PASS
J3_shaft	horn hole 0 at BCD14, 270 deg	2.520	2.520	PASS
J5_p2	pinch channel width (y)	24.500	24.500	PASS
J5_p2	bay length (x)	46.020	46.000	PASS
J5_p2	bay depth P -> back wall	34.989	35.000	PASS
J5_p2	wiring window length (x)	7.500	7.500	PASS
J5_p2	wiring window width (y)	22.000	22.000	PASS
J5_p2	servo screw pad face = back plateau (8.30,+10.25)	32.000	32.001	PASS
J5_p2	servo screw clearance 0 (8.30,+10.25)	3.400	3.395	PASS
J5_p2	servo screw head seat above the seat face (8.30,+10.25)	37.389	37.386	PASS
J5_p2	servo screw pad face = back plateau (8.30,-10.25)	32.000	32.001	PASS
J5_p2	servo screw clearance 0 (8.30,-10.25)	3.400	3.395	PASS
J5_p2	servo screw head seat above the seat face (8.30,-10.25)	37.389	37.386	PASS
J5_p2	servo screw pad face = back plateau (32.75,+10.25)	32.000	32.001	PASS
J5_p2	servo screw clearance 0 (32.75,+10.25)	3.400	3.395	PASS
J5_p2	servo screw head seat above the seat face (32.75,+10.25)	37.389	37.386	PASS
J5_p2	servo screw pad face = back plateau (32.75,-10.25)	32.000	32.001	PASS

Parameter -> CAD audit (generator value vs measured file) (cont.)

J5_p2	servo screw head seat above the seat face (32.75,-10.25)	37.389	37.386	PASS
J5_p1	drive bearing pocket 0	37.020	37.024	PASS
J5_p1	idle bearing pocket 0	37.020	37.024	PASS
J5_p1	drive pocket open to the gap (0 at z_in+0.6)	37.020	37.024	PASS
J5_p1	floor shaft bore 0	31.000	31.026	PASS
J5_p1	servo seat plane P (floor top, x=30,y=10)	26.000	25.995	PASS
J5_p1	horn/step relief depth below P	1.400	1.405	PASS
J5_p1	split insert pocket depth at (-20,+16)	8.100	8.105	PASS
J5_p2	split bolt hole open P..back face (void fraction) (-20,+16)	1.000	1.000	PASS
J5_p1	split insert pocket depth at (-20,-16)	8.100	8.105	PASS
J5_p2	split bolt hole open P..back face (void fraction) (-20,-16)	1.000	1.000	PASS
J5_p1	split insert pocket depth at (+26,+16)	8.100	8.105	PASS
J5_p2	split bolt hole open P..back face (void fraction) (+26,+16)	1.000	1.000	PASS
J5_p1	split insert pocket depth at (+26,-16)	8.100	8.105	PASS
J5_p2	split bolt hole open P..back face (void fraction) (+26,-16)	1.000	1.000	PASS
J5_shaft	shaft OD	29.950	29.948	PASS
J5_shaft	shaft bore	22.000	22.000	PASS
J5_shaft	set-screw flat distance from axis (normal at -90 deg)	14.200	14.199	PASS
J5_shaft	horn face = end of shaft	22.989	22.989	PASS
J5_shaft	horn hole 0 at BCD14, 0 deg	2.520	2.520	PASS
J5_shaft	horn hole 0 at BCD14, 90 deg	2.520	2.520	PASS
J5_shaft	horn hole 0 at BCD14, 180 deg	2.520	2.520	PASS
J5_shaft	horn hole 0 at BCD14, 270 deg	2.520	2.520	PASS
J2_p2	pinch channel width (y)	24.500	24.500	PASS
J2_p2	bay length (x)	46.020	46.000	PASS
J2_p2	bay depth P -> back wall	34.989	35.000	PASS
J2_p2	wiring window length (x)	7.500	7.500	PASS
J2_p2	wiring window width (y)	22.000	22.000	PASS
J2_p2	servo screw pad face = back plateau (8.30,+10.25)	32.000	32.001	PASS
J2_p2	servo screw clearance 0 (8.30,+10.25)	3.400	3.395	PASS
J2_p2	servo screw head seat above the seat face (8.30,+10.25)	37.389	37.386	PASS
J2_p2	servo screw pad face = back plateau (8.30,-10.25)	32.000	32.001	PASS
J2_p2	servo screw clearance 0 (8.30,-10.25)	3.400	3.395	PASS
J2_p2	servo screw head seat above the seat face (8.30,-10.25)	37.389	37.386	PASS
J2_p2	servo screw pad face = back plateau (32.75,+10.25)	32.000	32.001	PASS
J2_p2	servo screw clearance 0 (32.75,+10.25)	3.400	3.395	PASS
J2_p2	servo screw head seat above the seat face (32.75,+10.25)	37.389	37.386	PASS
J2_p2	servo screw pad face = back plateau (32.75,-10.25)	32.000	32.001	PASS
J2_p2	servo screw clearance 0 (32.75,-10.25)	3.400	3.395	PASS
J2_p2	servo screw head seat above the seat face (32.75,-10.25)	37.389	37.386	PASS
J2_p1	drive bearing pocket 0	42.020	42.030	PASS
J2_p1	idle bearing pocket 0	42.020	42.030	PASS
J2_p1	drive pocket open to the gap (0 at z_in+0.6)	42.020	42.030	PASS
J2_p1	floor shaft bore 0	31.000	31.026	PASS
J2_p1	servo seat plane P (floor top, x=30,y=10)	28.500	28.495	PASS
J2_p1	horn/step relief depth below P	1.400	1.405	PASS
J2_p1	split insert pocket depth at (-20,+16)	8.100	8.105	PASS
J2_p2	split bolt hole open P..back face (void fraction) (-20,+16)	1.000	1.000	PASS
J2_p1	split insert pocket depth at (-20,-16)	8.100	8.105	PASS
J2_p2	split bolt hole open P..back face (void fraction) (-20,-16)	1.000	1.000	PASS
J2_p1	split insert pocket depth at (+26,+16)	8.100	8.105	PASS
J2_p2	split bolt hole open P..back face (void fraction) (+26,+16)	1.000	1.000	PASS
J2_p1	split insert pocket depth at (+26,-16)	8.100	8.105	PASS
J2_p2	split bolt hole open P..back face (void fraction) (+26,-16)	1.000	1.000	PASS
J2_shaft	shaft OD	29.950	29.940	PASS
J2_shaft	shaft bore	22.000	22.000	PASS
J2_shaft	set-screw flat distance from axis (normal at +0 deg)	14.200	14.199	PASS
J2_shaft	set-screw flat distance from axis (normal at +180 deg)	14.200	14.199	PASS
J2_shaft	horn face = end of shaft	25.489	25.489	PASS
J2_shaft	horn hole 0 at BCD14, 0 deg	2.520	2.520	PASS
J2_shaft	horn hole 0 at BCD14, 90 deg	2.520	2.520	PASS

Parameter -> CAD audit (generator value vs measured file) (cont.)

J2_shaft	horn hole 0 at BCD14, 270 deg	2.520	2.520	PASS
J5_p1	J5_FACE (mounting face to J5 axis)	28.280	28.280	PASS
j1_hub	horn recess floor z (horn face)	-11.300	-11.297	PASS
j1_hub	plug 0 (across, off the flat)	21.800	21.800	PASS
j1_hub	D-flat distance from axis (45 deg)	9.000	9.000	PASS
J2_turret_p1	spigot D-key distance (45 deg)	9.200	9.202	PASS
j1_mount	pinch channel width	24.500	24.500	PASS
j1_mount	end stop face x (case end)	-10.112	-10.115	PASS
j1_mount	lip underside z (cap face)	-14.311	-14.310	PASS
j1_mount	servo screw pad face = back plateau (8.30,+10.25)	32.000	32.001	PASS
j1_mount	servo screw clearance 0 (8.30,+10.25)	3.400	3.395	PASS
j1_mount	servo screw head seat above the seat face (8.30,+10.25)	37.389	37.389	PASS
j1_mount	servo screw head counterbore 0 (8.30,+10.25)	5.600	5.588	PASS
j1_mount	servo screw pad face = back plateau (8.30,-10.25)	32.000	32.001	PASS
j1_mount	servo screw clearance 0 (8.30,-10.25)	3.400	3.395	PASS
j1_mount	servo screw head seat above the seat face (8.30,-10.25)	37.389	37.389	PASS
j1_mount	servo screw head counterbore 0 (8.30,-10.25)	5.600	5.588	PASS
j1_mount	servo screw pad face = back plateau (32.75,+10.25)	32.000	32.001	PASS
j1_mount	servo screw clearance 0 (32.75,+10.25)	3.400	3.395	PASS
j1_mount	servo screw head seat above the seat face (32.75,+10.25)	37.389	37.389	PASS
j1_mount	servo screw head counterbore 0 (32.75,+10.25)	5.600	5.588	PASS
j1_mount	servo screw pad face = back plateau (32.75,-10.25)	32.000	32.001	PASS
j1_mount	servo screw clearance 0 (32.75,-10.25)	3.400	3.395	PASS
j1_mount	servo screw head seat above the seat face (32.75,-10.25)	37.389	37.389	PASS
j1_mount	servo screw head counterbore 0 (32.75,-10.25)	5.600	5.588	PASS
j4_base	pinch channel width	24.500	24.500	PASS
j4_base	lip underside z (cap face)	378.589	378.590	PASS
j4_base	servo screw pad face = back plateau (32.75,+10.25)	32.000	32.001	PASS
j4_base	servo screw clearance 0 (32.75,+10.25)	3.400	3.395	PASS
j4_base	servo screw head seat above the seat face (32.75,+10.25)	37.389	37.386	PASS
j4_base	servo screw head counterbore 0 (32.75,+10.25)	5.600	5.588	PASS
j4_base	servo screw pad face = back plateau (32.75,-10.25)	32.000	32.001	PASS
j4_base	servo screw clearance 0 (32.75,-10.25)	3.400	3.395	PASS
j4_base	servo screw head seat above the seat face (32.75,-10.25)	37.389	37.386	PASS
j4_base	servo screw head counterbore 0 (32.75,-10.25)	5.600	5.588	PASS
j4_base	servo seat pad face = back plateau (8.30,+10.25)	32.000	32.001	PASS
j4_base	servo seat pad face = back plateau (8.30,-10.25)	32.000	32.001	PASS
j4_cap	6706 pocket 0	37.020	37.024	PASS
j4_cap	outer-race shoulder 0	34.000	34.006	PASS
j4_cap	cap bolt counterbore depth (+14.5,-15.5)	3.800	3.800	PASS
j4_cap	cap bolt counterbore depth (+14.5,+15.5)	3.800	3.800	PASS
j4_cap	cap bolt counterbore depth (+31.0,-15.5)	3.800	3.800	PASS
j4_cap	cap bolt counterbore depth (+31.0,+15.5)	3.800	3.800	PASS
j4_hub	journal OD	29.950	29.948	PASS
j4_hub	collar OD (inner race only)	32.000	31.998	PASS
j4_hub	J5 insert depth at (-17,-7)	7.500	7.501	PASS
j4_hub	J5 insert depth at (-17,+7)	7.500	7.501	PASS
j4_hub	J5 insert depth at (+17,-7)	7.500	7.501	PASS
j4_hub	J5 insert depth at (+17,+7)	7.500	7.501	PASS
j6_body	pinch channel width	24.500	24.500	PASS
j6_body	lip underside (cap face)	64.589	64.589	PASS
j6_body	servo screw pad face = back plateau (32.75,+10.25)	32.000	32.001	PASS
j6_body	servo screw clearance 0 (32.75,+10.25)	3.400	3.395	PASS
j6_body	servo screw head seat above the seat face (32.75,+10.25)	37.389	37.386	PASS
j6_body	servo screw pad face = back plateau (32.75,-10.25)	32.000	32.001	PASS
j6_body	servo screw clearance 0 (32.75,-10.25)	3.400	3.395	PASS
j6_body	servo screw head seat above the seat face (32.75,-10.25)	37.389	37.386	PASS
j6_body	servo seat pad face = back plateau (8.30,+10.25)	32.000	32.001	PASS
j6_body	servo seat pad face = back plateau (8.30,-10.25)	32.000	32.001	PASS
J3_p1	floor bolt head counterbore depth (-17,-7)	3.200	3.202	PASS
J3_p1	floor bolt head counterbore 0 (-17,-7)	6.200	6.200	PASS

Parameter -> CAD audit (generator value vs measured file) (cont.)

J3_p1	floor bolt head counterbore 0 (-17,+7)	6.200	6.200	PASS
J3_p1	floor bolt head counterbore depth (+17,-7)	3.200	3.202	PASS
J3_p1	floor bolt head counterbore 0 (+17,-7)	6.200	6.200	PASS
J3_p1	floor bolt head counterbore depth (+17,+7)	3.200	3.202	PASS
J3_p1	floor bolt head counterbore 0 (+17,+7)	6.200	6.200	PASS
J5_p1	floor bolt head counterbore depth (-17,-7)	3.200	3.202	PASS
J5_p1	floor bolt head counterbore 0 (-17,-7)	6.200	6.200	PASS
J5_p1	floor bolt head counterbore depth (-17,+7)	3.200	3.202	PASS
J5_p1	floor bolt head counterbore 0 (-17,+7)	6.200	6.200	PASS
J5_p1	floor bolt head counterbore depth (+17,-7)	3.200	3.202	PASS
J5_p1	floor bolt head counterbore 0 (+17,-7)	6.200	6.200	PASS
J5_p1	floor bolt head counterbore depth (+17,+7)	3.200	3.202	PASS
J5_p1	floor bolt head counterbore 0 (+17,+7)	6.200	6.200	PASS
j6_body	J5 D-bore flat (a of the flat)	-14.250	-14.255	PASS
j6_body	J5 grub pilot 0	2.600	2.597	PASS
j6_cap	6706 pocket 0	37.020	37.024	PASS
j6_flange	tool insert 0 at BCD30, 45 deg	4.100	4.100	PASS
j6_flange	tool insert 0 at BCD30, 135 deg	4.100	4.100	PASS
j6_flange	tool insert 0 at BCD30, 225 deg	4.100	4.100	PASS
j6_flange	tool insert 0 at BCD30, 315 deg	4.100	4.100	PASS
base	lower 6806 pocket 0, open at the bottom	42.020	42.030	PASS
spigot_collar	bore 0	30.150	30.150	PASS
spigot_collar	inner-race boss outer r	16.750	16.746	PASS
spigot_collar	body top z (0.5 under the base)	-0.500	-0.502	PASS
J2_turret_p1	J2 spring pin z (-1 side)	125.000	125.000	PASS
J2_turret_p1	J2 spring pin z (+1 side)	125.000	125.000	PASS
collar_upper	inner width (link 50 + 0.4)	50.400	50.400	PASS
link_upper_tongue	seam head counterbore depth (28.3,-17.85)	3.200	3.202	PASS
link_upper_tongue	seam counterbore 0 (28.3,-17.85)	6.000	5.995	PASS
link_upper_tongue	seam head counterbore depth (28.3,+17.85)	3.200	3.202	PASS
link_upper_tongue	seam counterbore 0 (28.3,+17.85)	6.000	5.990	PASS
link_upper_tongue	seam head counterbore depth (56.7,-17.85)	3.200	3.202	PASS
link_upper_tongue	seam counterbore 0 (56.7,-17.85)	6.000	5.990	PASS
link_upper_tongue	seam head counterbore depth (56.7,+17.85)	3.200	3.202	PASS
link_upper_tongue	seam counterbore 0 (56.7,+17.85)	6.000	5.990	PASS
link_upper_tongue	ear head counterbore depth	2.700	2.702	PASS
link_fore_tongue	seam head counterbore depth (29.8,-18.35)	3.200	3.202	PASS
link_fore_tongue	seam counterbore 0 (29.8,-18.35)	6.000	6.000	PASS
link_fore_tongue	seam head counterbore depth (29.8,+18.35)	3.200	3.202	PASS
link_fore_tongue	seam counterbore 0 (29.8,+18.35)	6.000	6.000	PASS
link_fore_tongue	seam head counterbore depth (59.5,-18.35)	3.200	3.202	PASS
link_fore_tongue	seam counterbore 0 (59.5,-18.35)	6.000	6.000	PASS
link_fore_tongue	seam head counterbore depth (59.5,+18.35)	3.200	3.202	PASS
link_fore_tongue	seam counterbore 0 (59.5,+18.35)	6.000	6.000	PASS
link_fore_tongue	seam head counterbore depth (89.2,-18.35)	3.200	3.202	PASS
link_fore_tongue	seam counterbore 0 (89.2,-18.35)	6.000	6.000	PASS
link_fore_tongue	seam head counterbore depth (89.2,+18.35)	3.200	3.202	PASS
link_fore_tongue	seam counterbore 0 (89.2,+18.35)	6.000	5.995	PASS
link_fore_tongue	ear head counterbore depth	2.700	2.702	PASS
shaft_clamp	bore 0 (45 deg: clear of the grub pilots and the gap)	30.000	29.967	PASS
shaft_clamp	ear gap width (at r 17)	11.400	11.400	PASS
shaft_clamp	ring OD (45 deg)	38.350	38.340	PASS
shaft_clamp	M5 set-screw clearance hole 0 (90 deg)	5.400	5.396	PASS
shaft_clamp	M5 set-screw clearance hole 0 (270 deg)	5.400	5.396	PASS
link_upper_groove	M5 self-tap pilot 0 (side hole, 90 deg)	4.200	4.196	PASS
link_upper_groove	M5 self-tap pilot 0 (side hole, 270 deg)	4.200	4.196	PASS
link_upper_tongue	M5 self-tap pilot 0 (side hole, 90 deg)	4.200	4.196	PASS
link_upper_tongue	M5 self-tap pilot 0 (side hole, 270 deg)	4.200	4.196	PASS
link_fore_groove	M5 self-tap pilot 0 (side hole, 90 deg)	4.200	4.196	PASS
link_fore_groove	M5 self-tap pilot 0 (side hole, 270 deg)	4.200	4.196	PASS
link_fore_tongue	M5 self-tap pilot 0 (side hole, 90 deg)	4.200	4.192	PASS

Parameter -> CAD audit (generator value vs measured file) (cont.)

J2_turret_p1	collet slit width (x -13)	1.000	1.000	PASS
J2_turret_p1	collet slit width (x +13)	1.000	1.000	PASS
J2_turret_p1	underside land outer r (bears on the inner ring only)	16.750	16.740	PASS
spigot_collar	pinch slot width	2.000	2.000	PASS
j4_base	lip relief r (stationary lips off the 6706 inner ring)	16.750	16.750	PASS
base	foot floor thickness at r 38 (was 0.5)	3.000	2.990	PASS
j1_mount	table hole 0 at 0 deg (r 58)	5.500	5.500	PASS
j1_mount	table hole 0 at 90 deg (r 58)	5.500	5.500	PASS
j1_mount	table hole 0 at 180 deg (r 58)	5.500	5.500	PASS
j1_mount	table hole 0 at 270 deg (r 58)	5.500	5.500	PASS
j1_mount	nothing inside the rear idler horn envelope + 0.5 (void fraction)	1.000	1.000	PASS
j4_base	nothing inside the rear idler horn envelope + 0.5 (void fraction)	1.000	1.000	PASS
J5_shaft	flat runs to the idle end (surface r at the idle end)	14.200	14.199	PASS
J3_p2	pinch wall at the far end of the bay (was 0.06-0.5)	2.400	2.398	PASS
J5_p2	pinch wall at the far end of the bay (was 0.06-0.5)	2.400	2.398	PASS
J5_spacer	thickness (blade face to inner ring)	3.000	3.000	PASS
J2_p2	servo pod back wall thickness (x 25, y 8)	2.400	2.399	PASS
J2_p2	servo pod side wall present at the deep rear corner (x 32, z P+32)	1.000	1.000	PASS
J3_p1	J3 spring-lug insert pocket length (-1 side)	6.500	6.500	PASS
J3_p1	J3 spring-lug insert pocket length (+1 side)	6.500	6.500	PASS
J2_turret_p1	J2 spring-lug insert pocket length (-1 side)	6.500	6.500	PASS
J2_turret_p1	J2 spring-lug insert pocket length (+1 side)	6.500	6.500	PASS
spigot_collar	material all round the pinch insert (fraction)	1.000	1.000	PASS

Bought hardware (counted from the placed assembly)

ARM-450 rev I -- bought hardware (counted from the placed assembly)

item	qty
6706 bearing (30x37x4)	4
6806 bearing (30x42x7)	6
M2.5 heat-set insert	2
M2.5 x 16 socket head	2
M2.5 x 6 socket head (horn)	24
M3 heat-set insert	55
M3 nyloc nut	4
M3 x 10 socket head	12
M3 x 12 socket head	16
M3 x 16 socket head	10
M3 x 20 socket head	1
M3 x 4 set screw	1
M3 x 45 socket head	16
M4 heat-set insert	4
M4 x 12 socket head	4
M5 x 10 set screw (cup point)	8
ST3215 servo	6
extension spring J2 (7.4 N @ 36.0 / 12.4 N @ 58.2 mm, inside hooks)	2
extension spring J3 (2.2 N @ 43.0 / 5.0 N @ 91.4 mm, inside hooks)	2
self-tapping screw ~2.2 x 8 pan head (from the servo box)	20

Print list (FINAL_PRINT/STL_print_ready)

file	qty	orient	supports	size mm	note
01_base_x1.stl	1	+Z up	REQUIRED	120 x 120 x 50	rev I: lower 6806 pocket opens from underneath
02_spigot_collar_x1.stl	1	+Z up	none	46 x 46 x 12	split clamp + J1 collet closer, slot 2.0, 1 x M3 pinch bolt
03_j1_mount_x1.stl	1	+Z up	none	128 x 128 x 70	J1 servo mount + the arm's foot
04_j1_hub_x1.stl	1	+Z up	advised	29 x 29 x 17	J1 horn coupler, D-keys the turret
05_J2_turret_p1_x1.stl	1	+X up	REQUIRED	138 x 70 x 84	turret; spigot end slit as a collet (J1 zero play); land on the 6806 i
06_J2_turret_p2_x1.stl	1	-Y up	advised	83 x 66 x 37	J2 servo pod (2.4 shell round the whole servo), 4 x M3x45 split bolts
07_J2_shaft_x1.stl	1	+Y up	none	30 x 30 x 50	printed J2 shaft, bolts to the horn; double-D flats for the M5 set scr
08_shaft_clamp_x4.stl	4	-Z up	none	37 x 38 x 14	link centring ring (0.025 fits); links locked by 2 x M5 set screws eac
09_link_upper_groove_x1.stl	1	+Z up	advised	110 x 50 x 14	released, unchanged (reuse your printed one)
10_link_upper_tongue_x1.stl	1	+Z up	advised	110 x 50 x 16	rev I: released + head counterbores (or drill yours: 06.0x3.2 seam, 05
11_collar_upper_x2.stl	2	+Z up	none	64 x 24 x 10	J2 spring collar, print 2 (same part)
12_J3_p1_x1.stl	1	-Z up	REQUIRED	98 x 50 x 53	J3 fork; J3 spring lugs (both pockets)
13_J3_p2_x1.stl	1	-Z up	none	72 x 50 x 37	J3 servo cradle, bay 34.99 deep, 2.4 back wall
14_J3_shaft_x1.stl	1	-Z up	none	30 x 30 x 50	printed J3 shaft, double-D flats for the M5 set screws
15_link_fore_tongue_x1.stl	1	+Z up	advised	144 x 50 x 16	rev I forearm: socket end, valid solid, flush seam-screw heads
16_link_fore_groove_x1.stl	1	+Z up	advised	144 x 50 x 14	rev I forearm: socket end, valid solid
17_collar_fore_x2.stl	2	+Z up	none	64 x 24 x 10	J3 spring collar, print 2 (same part)
18_j4_base_x1.stl	1	-Z up	REQUIRED	57 x 38 x 65	J4 servo base; lip tops relieved off the 6706 inner ring
19_j4_cap_x1.stl	1	-Z up	none	58 x 45 x 10	J4 cap, holds the 6706
20_j4_hub_x1.stl	1	-Z up	none	46 x 46 x 20	J4 hub, carries the J5 fork
21_J5_p1_x1.stl	1	+X up	advised	48 x 50 x 66	J5 fork
22_J5_p2_x1.stl	1	-Z up	none	66 x 50 x 37	J5 servo cradle, bay 34.99 deep, 2.4 back wall
23_J5_shaft_x1.stl	1	-Z up	none	30 x 30 x 45	printed J5 axle
24_J5_spacer_x2.stl	2	+Z up	none	32 x 32 x 3	blade spacers, exactly 3.0 (close the J5 stack), print 2
25_j6_body_x1.stl	1	-X up	REQUIRED	43 x 54 x 87	J5 blade + J6 servo housing
26_j6_cap_x1.stl	1	-X up	none	45 x 58 x 10	J6 cap, holds the 6706
27_j6_flange_x1.stl	1	-X up	none	46 x 46 x 20	tool flange, bolts to the J6 horn

27 files, 33 pieces